



Between the Lines: Investigating Emotional Expressions and Health Beliefs in Online Mental Health Communities

Master Thesis

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Declaration of Authorship

I hereby declare that the Master's thesis entitled "**Between the Lines: Investigating Emotional Expressions and Health Beliefs in Online Mental Health Communities**" is my own research work, carried out under the academic supervision of Prof. Dr. Binh Vu and Dr. Gwendolyn Mayer, and has not been copied from any source not expressly cited.

All contents of this thesis are the result of my research, study and analysis based on the references listed in the reference list. The data, research findings and comments in the thesis are accurate, truthful and have never been published in any other research work.

I take full responsibility for the truthfulness and accuracy of the content of the thesis. I also confirm that this thesis has never been used to defend any other degree.

Signed : Nguyen Bao Khanh

Date : 13.09.2024

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Finally, I would like to thank all the individuals and organizations that provided me with documents, data and the necessary support to complete this thesis.

Acknowledgement of Sensitive Language

In this thesis, we acknowledge that the analysis of Reddit data includes explicit and distressing language, such as terms like "fuck," "die," and "kill," which reflect the raw and unfiltered expressions of users discussing mental health issues. This inclusion is essential for accurately capturing the nature of the discussions. We have carefully contextualized and interpreted these terms to ensure respectful handling and to maintain the integrity of the analysis. Readers should be aware that such language is part of the authentic user-generated content on Reddit and is addressed with the appropriate sensitivity in the study.

Abstract

This thesis investigates the application of computational techniques to analyze large-scale Reddit datasets through the lens of the Health Belief Model (HBM). The primary objective is to develop and implement a methodology that enhances the accuracy and efficiency of analyzing health-related discussions on Reddit. The study employs various models, including GPT-4o and DistilBERT Classification, to classify and interpret data on components such as Root Cause, Perceived Susceptibility, Perceived Benefits, Self-Efficacy, and Perceived Severity. While the models demonstrate strong performance with accuracy rates ranging from 75% to 84% for most components, challenges persist in predicting Perceived Severity due to its complex multi-label nature, resulting in a lower accuracy of 47%. To overcome these obstacles, the thesis proposes using GPT-4o for keyword extraction combined with human review to refine labels, and suggests an alternative method for improving accuracy to 81%.

Furthermore, the analysis reveals significant patterns in mental health discussions, such as the influence of personality on anxiety and the urgent need for targeted interventions in cases of suicidal thoughts. Despite the robustness of transformer-based models in general contexts, they face difficulties in specialized domains like health beliefs, underscoring the need for further domain-specific adaptations. Moreover, the study underscores the ethical implications and the influence of explicit language in user-generated content. Future work is recommended to enhance domain-specific model performance, improve handling of rare words, and address scalability and ethical issues. Overall, this research contributes to a deeper understanding of health-related discussions on Reddit and provides a foundation for more nuanced and effective computational analysis in this field.

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Abbreviations

AI	Artificial Intelligence
NLP	Natural Language Processing
LLMs	Large Language Models
BoW	Bag of Words
CBOW	Continuous Bag of Words
GloVe	Global Vectors for Word Representation
FNNs	Feedforward Neural Networks
CNNs	Convolutional Neural Networks
RNNs	Recurrent Neural Networks
GRU	Gated Recurrent Unit
LSTM	Long Short-Term Memory
BERT	Bidirectional Encoder Representations from Transformers
RoBERTa	Robustly Optimized BERT Approach
GPT	Generative Pre-trained Transformer
LLAMA	Language Model for Query Auto-completion
LDA	Latent Dirichlet Allocation
HBM	Health Belief Model
HBMS-E	Health Belief Model Scale for Exercise
HPV	Human Papillomavirus
GPU	Graphics Processing Unit

Chapter 1. Introduction

Mental health is a fundamental aspect of human functioning that enables individuals to cope with the challenges and demands of life, realize their potential, and contribute to their communities. However, it can be undermined by a range of factors, including poverty, violence, disability, and inequality. While many mental health conditions are treatable at low cost, health systems often lack the necessary resources, leading to significant treatment gaps and poor-quality care. Those with mental health conditions frequently face stigma, discrimination, and human rights violations. In 2019, 970 million individuals globally were affected by a mental disorder, which has a significant impact on relationships, education, and employment. Mental disorders are associated with a substantial degree of disability and can result in a reduction in life expectancy of 10 to 20 years. Furthermore, there are notable economic consequences due to productivity losses ([WHO 2019](#)).

1.1 Motivation

The prevalence of discussions on social media about mental health has increased markedly in recent years. More people have been openly sharing their emotions and experiences on social media platforms like Twitter, Instagram, Facebook, and Reddit. These conversations help create a sense of community and support, allowing individuals to connect with others facing similar challenges and easing feelings of isolation. Twitter and Instagram serve as forums for advocacy and raising awareness about mental health issues. This widespread conversation is crucial in breaking down the stigma around mental health issues, creating a supportive space where people can access resources and find comfort in shared experiences. In contrast, on platforms such as Reddit, users engage in in-depth discussions, seeking advice and sharing personal narratives.

Building on this trend, the Health Belief Model (HBM), a well-known theoretical framework in psychology as well in medicine, can be applied to analyze these discussions. The Health Belief Model (HBM) comprises five key concepts that influence behavior: perceived severity, perceived susceptibility, perceived benefits, perceived barriers, cues to action, and self-efficacy ([Siddiqui](#)

[and others 2016](#); [Janz and Becker 1984](#)). In the past, the Health Belief Model (HBM) research involved the administration of surveys to collect data, which were then manually categorized by researchers ([Jones and others 2015](#)). However, the current era's abundance of social media platforms offers a rich source of data for analysis.

By incorporating the Health Belief Model with emotional content analysis, this research aims to uncover the cognitive and emotional dimensions driving user engagement and behavior on Reddit. This combined approach offers a more nuanced understanding of the interplay between beliefs and emotions in online health discussions. This research explores emotional expressions in mental health discussions, focusing on negative sentiments. Sentiment analysis captures overall negativity, while emotional content analysis identifies specific emotions like fear or frustration, particularly regarding health actions such as seeking treatment or following medical advice. Despite the significant challenges posed by the volume of data, which requires categorization of millions of posts, this method necessitates a profound understanding of language and emotion to interpret the five distinct HBM concepts effectively.

1.2 Problem Statement

The main challenge lies in the methodological approach required to analyze large datasets with long texts, such as those found on Reddit. While traditional qualitative methods are accurate, they are labor-intensive and therefore not feasible for extensive datasets. As previously stated in the paper, despite the voluminous nature of the data set, the authors were able to categorize 800 rows ([Rani and others 2024](#)).

The rapid and exponential rise of large language models (LLMs) recently has created an opportunity to utilize them for language understanding, categorization and the application of the Health Belief Model to social media data. Models such as GPT4, GPT4o, LLAMA3, etc. have demonstrated their ability to understand deep context and label with zero shot learning ([Minaee and others 2024](#)). One of the papers showed their ability to do sentiment labeling of the tweets ([Krugmann and Hartmann 2024](#)). However, the implementation of these models have not been proven to annotate more complex data in other social media platforms.

To address this, we need a method that can maintain accuracy without being overly labor-intensive. This necessitates the integration of advanced computational techniques, such as natural language processing (NLP) and machine learning, to process and analyze vast amounts of textual data efficiently.

The overarching objectives of this thesis can be summarized as follows:

- The objective is to formulate and apply a methodology that leverages computational techniques for accurate and efficient analysis of large Reddit datasets within the framework of the Health Belief Model.
- The objective is to explore the combined effect of health beliefs and emotional content on public health discussions and behaviors on Reddit.

1.3 Research Questions

The first objective is to formulate and apply a methodology that leverages computational techniques for accurate and efficient analysis of large Reddit datasets within the framework of the Health Belief Model. This raises the following questions:

- RQ1: What computational techniques can be utilized to effectively analyze large-scale Reddit datasets within the context of the Health Belief Model?
- RQ2: How can these techniques be applied to ensure accuracy and efficiency in the analysis of health-related discussions on Reddit?
- RQ3: What are the key challenges in applying computational methods to Reddit data, and how can they be addressed?

The second objective is to explore the combined effect of health beliefs and emotional content in shaping public health discussions and behaviors on Reddit. This leads to the following questions:

- RQ4: How do health beliefs influence the nature and content of public health discussions on Reddit?

- RQ5: What role does emotional content play in shaping these discussions and subsequent public health behaviors?
- RQ6: Can patterns be identified in how different health beliefs and emotional tones correlate with engagement levels and behavior changes on Reddit?

1.4 Approach

To accomplish the objectives outlined in this thesis, a systematic approach will be undertaken, comprising several key steps:

1. Literature Review:

Conduct an extensive review of existing literature on the Health Belief Model, computational techniques for data analysis, and the influence of emotional content on public health discussions. This will establish a theoretical framework and highlight gaps in the existing research.

2. Data Collection:

Gather large-scale datasets from Reddit, focusing on health-related discussions. The dataset is accessible via Kaggle for the purposes of public research.

3. Data Preprocessing:

Clean and preprocess the gathered data to filter out noise and irrelevant details. This step will include tasks such as text normalization, tokenization, and filtering of non-health-related content.

4. Methodology Development:

Develop a computational methodology leveraging techniques such as natural language processing (NLP), machine learning, and sentiment analysis. This methodology will be designed to annotate a large dataset, then analyze the data within the framework of the Health Belief Model, focusing on the identification of health beliefs and emotional content.

5. Implementation:

Implement the developed methodology using appropriate computational tools and libraries. Ensure that the implementation is optimized for accuracy and efficiency in handling large datasets.

6. Analysis:

Apply the implemented methodology to the preprocessed Reddit data. Analyze the results to understand how health beliefs and emotional content influence public health discussions and behaviors.

7. Validation:

Validate the findings by comparing them with established theories and models in the literature. Perform statistical tests to ensure the robustness and reliability of the results.

8. Interpretation and Reporting:

Interpret the analysis results in the context of the Health Belief Model and provide insights into the combined effect of health beliefs and emotional content. Present the findings in an organized format, emphasizing key patterns and their implications for public health discussions on Reddit.

1.5 Thesis Contribution

This thesis makes several significant contributions to both academic research and practical applications in the fields of health communication, behavioral science, and computational linguistics.

Firstly, the thesis introduces a novel computational methodology specifically designed to analyze large-scale health-related discussions on Reddit through the lens of the Health Belief Model (HBM). By combining advanced natural language processing (NLP) techniques with the HBM framework, this research provides a systematic approach to extracting and analyzing the six components of the model from user-generated content. This methodological advancement not only enhances the accuracy and efficiency of analyzing vast amounts of social media data but

also sets a foundation for future studies in health communication research, particularly those involving big data.

Secondly, the thesis offers new theoretical insights into the complex interplay between health beliefs and emotional content in shaping public health discussions and behaviors on Reddit. By exploring how these two factors influence user engagement and behavior, the research extends current knowledge in the domains of health communication and behavioral science. It reveals how specific health beliefs, when combined with certain emotional tones, can significantly impact the nature of public discourse and drive behavior changes. This contributes to a deeper understanding of how online communities process and react to health information, providing a nuanced view of the social dynamics at play.

Thirdly, this research has substantial practical implications for public health professionals, policymakers, and digital health communicators. The findings can inform the design of more effective health communication strategies that leverage the power of social media platforms like Reddit. By understanding the factors that drive engagement and behavior change, health campaigns can be more precisely targeted and tailored to address the specific beliefs and emotions of different audience segments. This can lead to more impactful public health initiatives, better resource allocation, and ultimately, improved health outcomes.

In addition to these contributions, the thesis also provides valuable resources for the research community. The development of computational tools, such as NLP models and automated labeling systems, represents a significant step forward in the analysis of social media data. These tools, along with the comprehensive dataset generated throughout the study, will be accessible for future research and applications, thereby supporting ongoing advancements in the field.

Overall, this thesis not only advances academic understanding but also offers practical solutions and resources that can be utilized in real-world health communication efforts, making it a valuable contribution to both research and practice.

1.6 Thesis Outline

The remainder of this thesis is organized as follows. Chapter 2 provides a comprehensive review of the Health Belief Model and natural language processing techniques, along with relevant literature on emotional content analysis and the application of social media data in health research. Chapter 3 outlines the methodology, addressing the model design and description of the datasets used. Chapter 4 elaborates on the methodological approach, covering data collection, preprocessing, and the computational techniques applied. Chapter 5 presents the results, focusing on the performance of the data labeling model, as well as the analyses of the Health Belief Model and emotional content. Finally, Chapter 6 concludes the thesis with a summary of the key findings, answers to the research questions, discussion of limitations, and recommendations for future research.

Chapter 2. State of the Art

This chapter provides an exhaustive review of the theoretical underpinnings and contemporary techniques pertinent to this research. We commence with an in-depth exploration of the Health Belief Model, elucidating its components and its applicability to health communication and behavioral analysis. Following this, we characterize the text dataset derived from Reddit, detailing the data collection methodologies and preprocessing techniques employed to prepare the text for subsequent analysis.

Subsequently, we delve into the fundamentals of text representation and word embedding, which are pivotal for transforming raw text into a machine-readable format. This section encompasses a spectrum of techniques, ranging from traditional methods to sophisticated embeddings. An introduction to neural networks in NLP is then presented, covering essential architectures such as Feedforward Neural Networks (FNNs), Convolutional Neural Networks (CNNs), and Recurrent Neural Networks (RNNs). Each subsection explicates the architecture, operational principles, and specific applications in NLP tasks.

The discourse then transitions to the transformative impact of transformer models on NLP. We provide an intricate examination of the mechanics of transformer models, highlighting their expertise in managing complex language tasks with unmatched accuracy and efficiency. Key transformer models, including BERT, DistilBERT, and RoBERTa, are analyzed comprehensively, highlighting their architectural nuances, training methodologies, and performance metrics.

The chapter further explores the practical applications of these transformer models across various NLP tasks, demonstrating their versatility and efficacy in real-world scenarios. This segment also includes a critical evaluation and comparative analysis of different transformer models, offering insights into their respective strengths and limitations. In addition, we examine generative AI models in NLP, such as GPT-4o and LLAMA3, discussing their capabilities, particularly in generating human-like text and their role in augmenting data labeling and classification processes.

Considering the crucial role of preprocessing in improving model performance, we introduce various preprocessing techniques, including resampling, data augmentation, and label combination strategies. These methodologies are indispensable for optimizing the training data and improving model accuracy.

Finally, we address the prevailing challenges and potential future directions in NLP research, particularly in the context of health-related text analysis. The chapter concludes with a synthesis of the key points discussed, thereby laying the groundwork for the subsequent chapters, which will elaborate on the implementation and outcomes of our categorization and analysis process.

This comprehensive review of the theoretical foundations and state-of-the-art techniques establishes a robust framework for understanding the methodologies employed in this research, thus providing a scholarly context for the innovative application of NLP models in health belief analysis.

2.1 Background

The Background section offers a detailed overview of the key concepts and methodologies that underpin this study. It begins with an exploration of the Health Belief Model, followed by a discussion on Text Data and Representation techniques essential for processing and analyzing textual data. The section then delves into the role of Neural Networks and Deep Learning in modern data analysis, as well as the impact of Generative AI Models like LLAMA and GPT. Finally, the section addresses the Preprocessing and Training Techniques necessary to prepare and refine the dataset for effective model training and analysis.

2.1.1 Health Belief Model

The Health Belief Model (HBM) was developed in the 1950s by social psychologists at the U.S. Public Health Service, including Irwin M. Rosenstock, Godfrey M. Hochbaum, S. Stephen Kegeles, and Howard Leventhal. It was initially created to understand why people were not participating in tuberculosis screening programs, but has since been extended to a variety of health behaviors ([Rosenstock 1974](#)).

As shown in Figure 2.1 below, the HBM consists of five key components: perceived susceptibility, perceived severity, perceived benefits, perceived barriers, cues to action. These constructs work together to explain and predict an individual's likelihood of engaging in health-promoting behaviors. The model proposes that individuals are more inclined to take health-related actions if they believe they are susceptible to a condition, perceive the condition as severe, recognize the benefits of taking action, perceive fewer barriers to action, are exposed to factors that prompt action, and are confident in their ability to successfully perform an action ([Rosenstock 1974](#)).

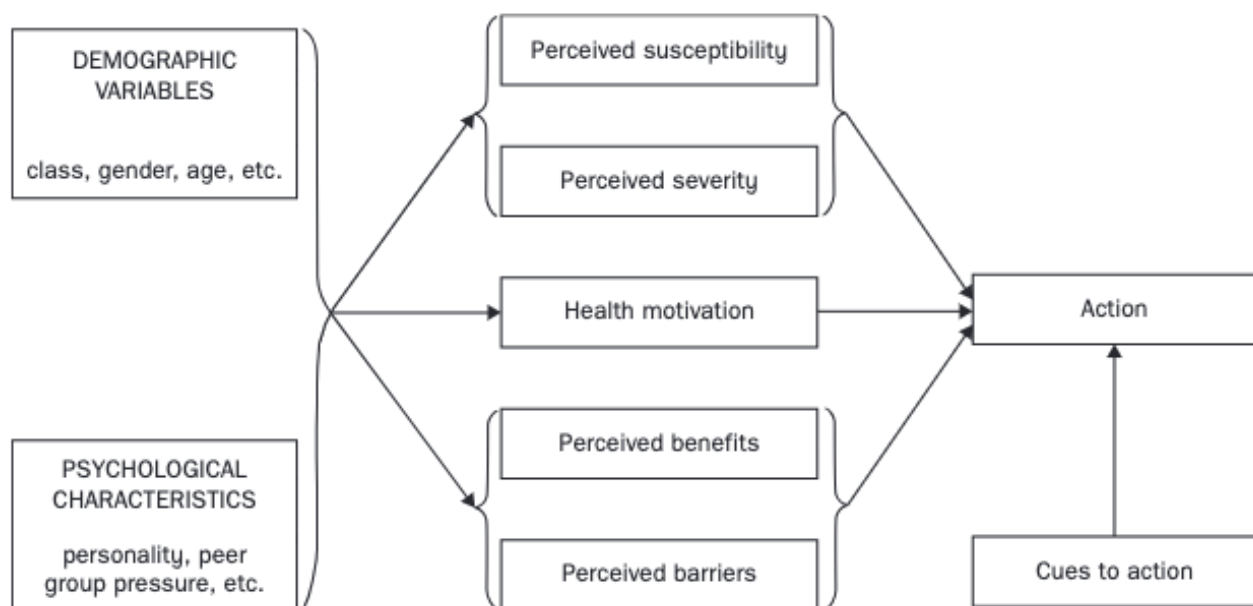


Figure 2.1. The health belief model (Rosenstock 1974)

In 1988, Rosenstock, along with Strecher and Becker, added self-efficacy as the sixth main construct to the model, as illustrated in Figure 2.2. This addition was made to better explain and predict health behaviors, especially in cases of long-term behavior change. The inclusion of self-efficacy was based on emerging evidence in psychology about its crucial role in decision-making and behavior. Researchers recognized that even if individuals perceived a health threat as serious and believed that taking action would be beneficial, they might not take action if they did not feel confident in their ability to effectively carry out the necessary behavior ([Rosenstock 1988](#)).

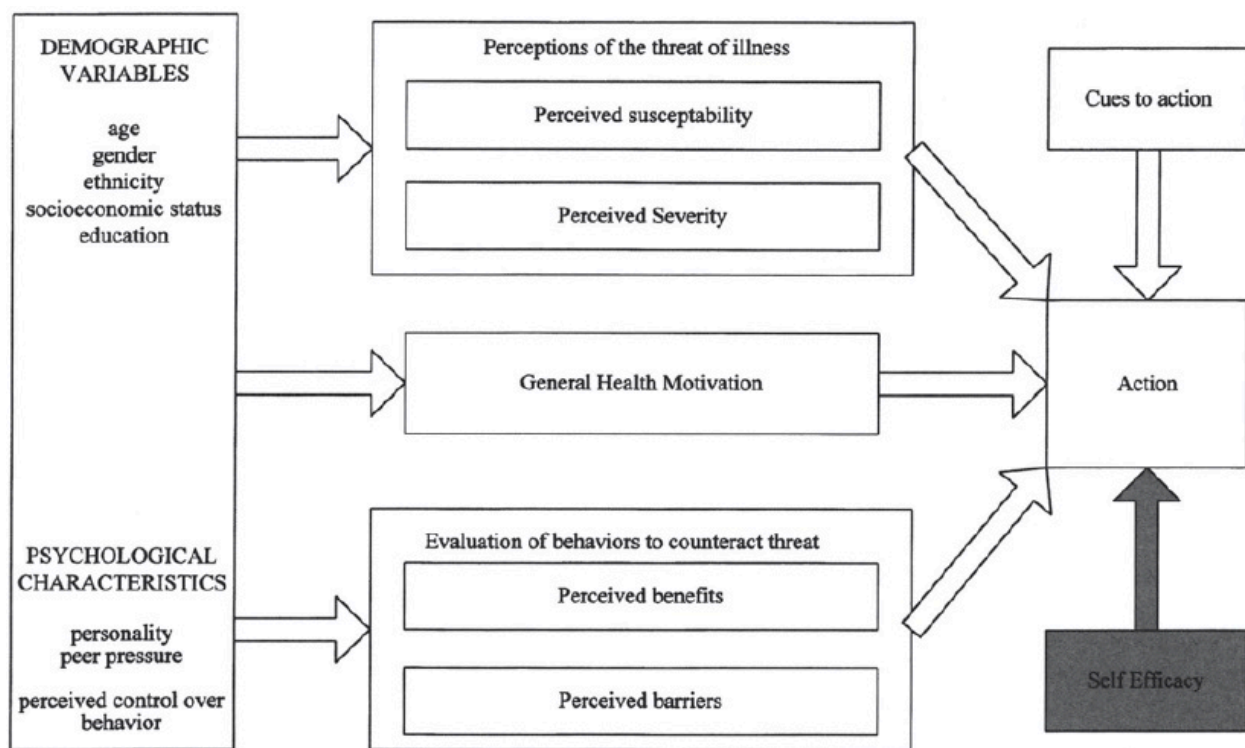


Figure 2.2. The modified health belief model suggested by Rosenstock (1988)

The HBM has been widely applied in various health contexts, including disease prevention, health promotion, and adherence to medical treatments. It has been employed to create effective interventions by targeting these key constructs, such as providing education about disease risks, addressing perceived barriers, and enhancing self-efficacy through training in specific health behaviors ([Jones and others 2015](#)). Furthermore, during the COVID-19 pandemic, the HBM was

used to understand vaccine hesitancy and promote vaccination. [Wong and others \(2021\)](#) applied the HBM to examine the public's acceptance of COVID-19 vaccines in Hong Kong, addressing perceived susceptibility to infection, benefits of vaccination, and barriers such as misinformation. Despite these efforts, challenges persisted as vaccine resistance continued in some communities due to entrenched beliefs and perceived risks.

2.1.2 Text Data and Representation

This section explores the methods and strategies used for handling and representing text data in this study. It begins with a description of the Text Dataset utilized, outlining its sources and characteristics. Following this, the section explores various Text Representation techniques, such as embeddings and vectorization, which are essential for converting raw text into formats suitable for machine learning and analysis.

Text Dataset

Text datasets are crucial in numerous fields, especially in natural language processing (NLP) and social media research. Platforms like Reddit, Twitter, and Facebook provide rich sources of textual data, including posts, comments, and user interactions. Current methodologies leverage advanced computational techniques, such as sentiment analysis, topic modeling, and text classification, to derive insights from these datasets. Researchers frequently apply the Health Belief Model (HBM) to analyze health-related surveys, illuminating the factors influencing public health behaviors ([N K Janz and Becker 1984](#)).

Additionally, sentiment analysis has gained prominence, enabling the detection and exploration of emotional nuances in text data. Despite the abundance of information, challenge remains in analyzing large-scale text datasets, including data volume, computational efficiency, and ensuring data quality. Recent studies address these challenges through innovative approaches, advancing the field and providing deeper insights into online discourse ([Sesagiri Raamkumar and others 2020](#)).

Text Representation & Word Embedding

In the field of natural language processing, the representation of text has evolved from the use of simple methods, such as the Bag of Words (BoW) model and n-grams, to the implementation of sophisticated word embeddings. BoW models text as high-dimensional, sparse vectors based on word counts, while n-grams capture some local context by considering sequences of words. The Continuous Bag of Words (CBOW) model, a component of the Word2Vec system, represents a significant advancement over previous methods. Rather than simply predicting a target word based on its context, CBOW generates dense, low-dimensional embeddings that capture semantic relationships. The Skip-gram model, also included in Word2Vec, predicts context words based on a target word, thereby enriching the embeddings, particularly for infrequent terms. As illustrated in the Figure 2.3, Word2Vec employs a shallow neural network to learn distributed representations of words based on their co-occurrence patterns in a corpus of text ([Mikolov and others 2013](#)).

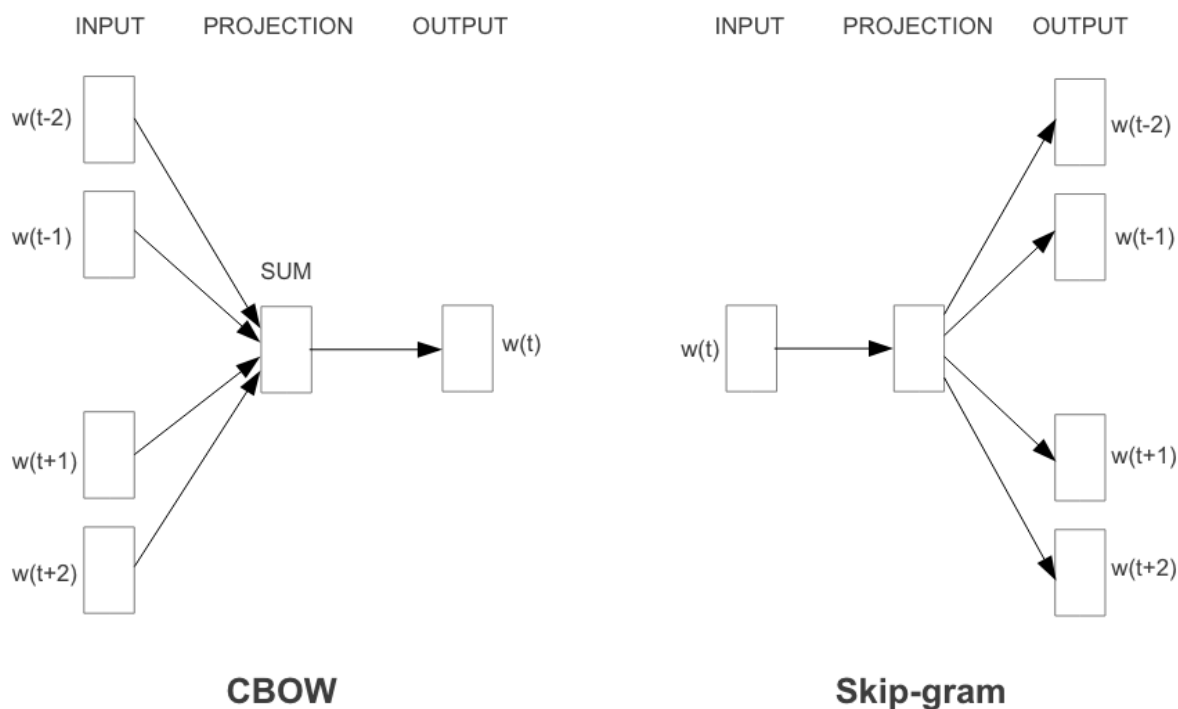


Figure 2.3. Continuous Bag of Words (CBOW) and Skip-gram

Moreover, FastText, developed by Facebook's AI Research lab, builds upon Word2Vec by representing words as bags of character n-grams. This enables FastText to capture subword information, rendering it particularly efficacious for languages with complex morphology and for rare or out-of-vocabulary words. By considering character-level information, FastText embeddings are robust to misspellings and variations, enhancing the model's ability to understand and process text ([Joulin and others 2017](#)).

Similarly, GloVe (Global Vectors for Word Representation) develops word embeddings by leveraging global word-word co-occurrence statistics across the entire corpus. These word embeddings have played a pivotal role in various NLP applications, such as sentiment analysis, named entity recognition, and machine translation, by enabling models to generalize better and infer semantic relationships between words ([Pennington and others 2014](#)). As shown in Figure 2.4, the linear substructures of words in the vector space of GloVe demonstrate how semantic relationships, such as analogies, can be captured and represented mathematically.

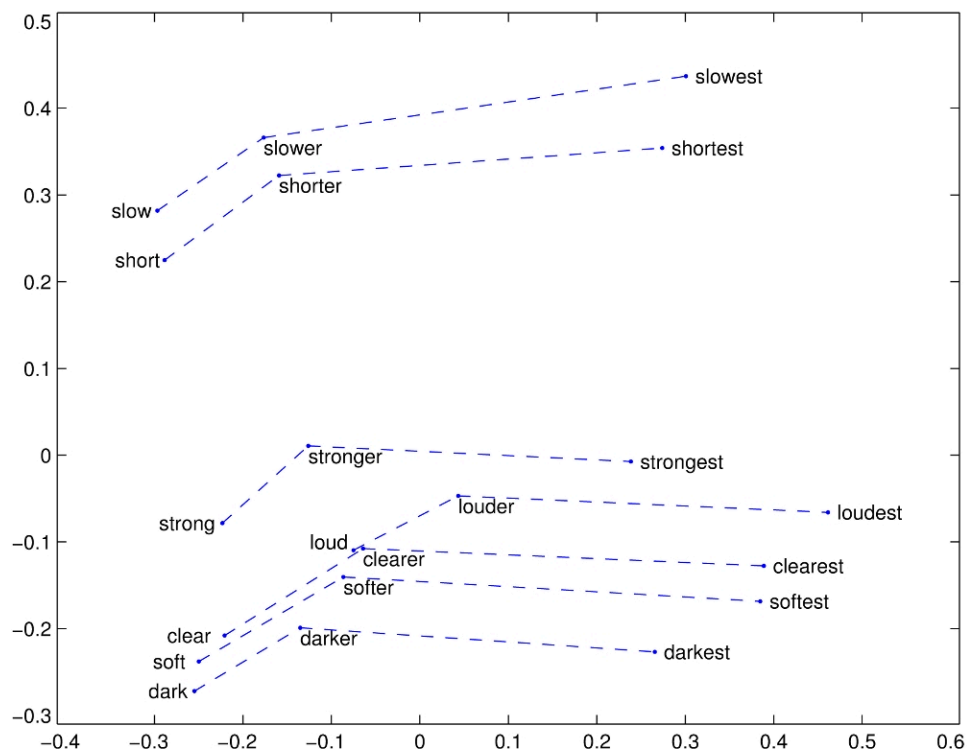


Figure 2.4. Linear Substructures of Words in the Vector Space of GloVe

Traditional natural language processing (NLP) models, such as bag-of-words and n-gram models, have played a foundational role in text analysis. However, they have inherent limitations that affect their effectiveness in capturing the complexity of language ([Mikolov and others 2013](#)). One of the primary challenges with these traditional approaches is their inability to capture long-range dependencies in sequential data. Bag-of-words models, for example, treat each word in isolation, ignoring the sequential nature of language and the relationships between words in a sentence. Similarly, n-gram models consider only local contexts of adjacent words, limiting their ability to capture broader semantic associations ([Mikolov and others 2013](#)).

2.1.3 Neural Networks and Transformer Models in NLP

Neural networks are foundational tools in natural language processing (NLP), designed to mimic the human brain's ability to learn from and understand language patterns. These models are composed of interconnected nodes arranged in layers, where each node represents a neuron that processes and transmits information. In the context of NLP, neural networks excel at tasks ranging from text classification and sentiment analysis to machine translation and natural language understanding ([Goldberg 2017](#)).

Neural Networks in NLP

The foundations of neural networks provide a framework for understanding their evolution and widespread application in various domains, including natural language processing (NLP). Historically, neural network models have undergone significant development, tracing back to the pioneering work of Warren McCulloch and Walter Pitts in the 1940s (McCulloch and Pitts 1943). This foundational work established the conceptual framework for early neural network architectures, inspiring subsequent research that culminated in the development of perceptrons and feedforward networks in the 1950s and 1960s ([Ahamed and Akthar 2016](#)).

These early architectures provided the basis for understanding how neural networks could be trained to learn from data and make predictions, thereby laying the groundwork for further progress in the field. These networks are effective for tasks requiring hierarchical feature

extraction and classification, such as sentiment analysis, document categorization, and text generation based on learned patterns ([Goodfellow and others 2017](#)).

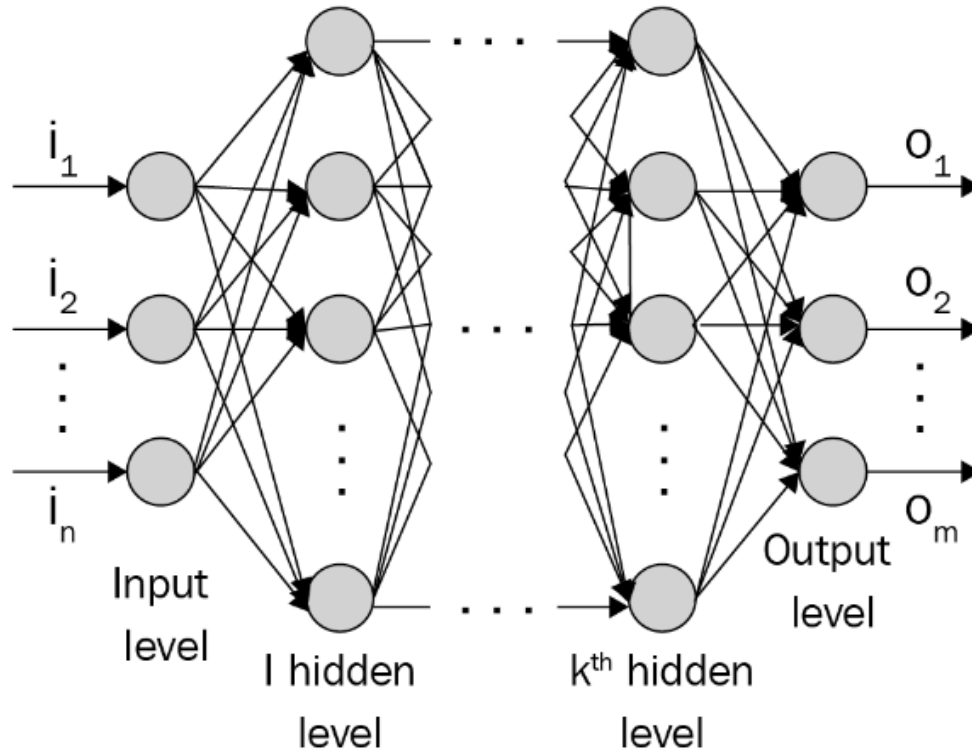


Figure 2.5. Feed-forward Neural Network (Ayyadevara 2019)

The architecture of a simple feed-forward neural network is depicted in Figure 2.5. The network comprises an input layer, multiple hidden layers, and an output layer. The training algorithms, activation functions, and optimization techniques that constitute the key components of neural networks have been instrumental in determining their effectiveness and efficiency. Training algorithms, including backpropagation, stochastic gradient descent, and their variations, constitute the fundamental basis of how neural networks learn from data. Activation functions, such as rectified linear unit (ReLU), sigmoid, and tanh, determine how neurons in the network respond to input stimuli, thereby influencing its capacity to capture complex patterns in data. Optimization techniques, which encompass a range of approaches, including basic gradient descent and more advanced methods such as RMSprop and Adam, are instrumental in

fine-tuning the model parameters and enhancing its performance. An understanding of these foundational elements is essential for the effective utilization of neural networks in NLP and other domains, driving innovation and expanding the boundaries of what is possible in artificial intelligence ([Goodfellow and others 2017](#)).

Convolutional neural networks (CNNs) have proven to be a versatile architecture, initially developed for image processing tasks, which has now been successfully adapted for natural language processing (NLP). In NLP, the ability to capture local dependencies and patterns in text is crucial. In a convolutional neural network (CNN), layers consist of filters (or kernels) that convolve across input sequences, extracting local features such as word sequences or n-grams that contribute to the understanding of the meaning of text. Subsequently, pooling layers aggregate these features (e.g., max-pooling or average-pooling), reducing the dimensionality while retaining the essential information. Subsequently, fully connected layers handle the extracted features for tasks such as text classification, sentiment analysis, and named entity recognition (NER) ([LeCun 2015](#)).

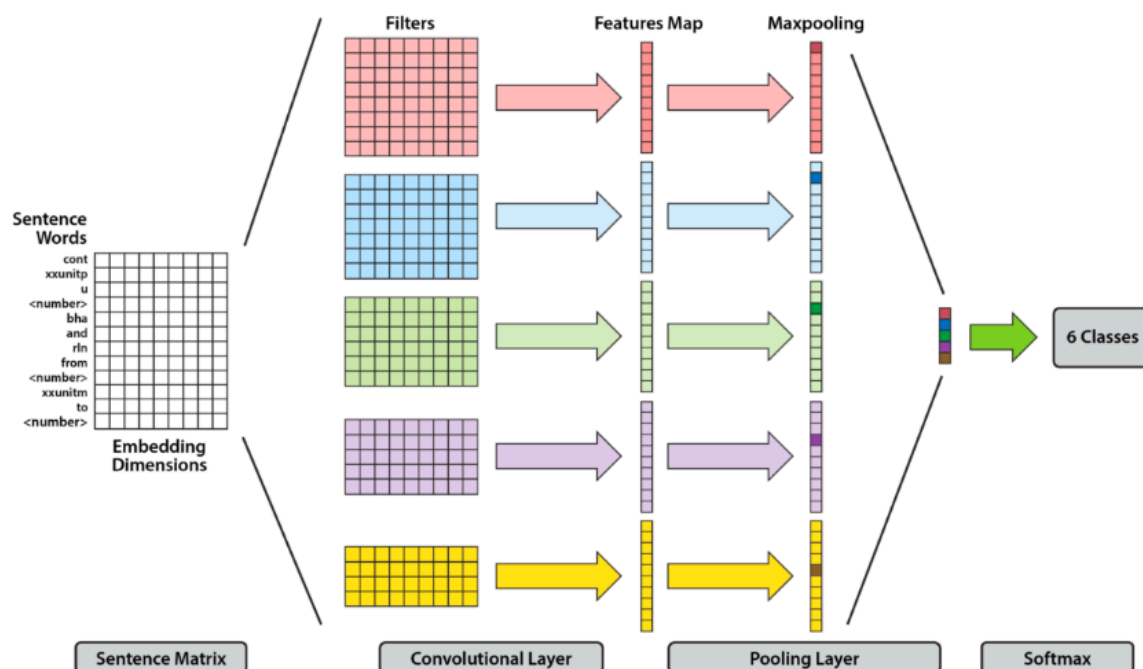


Figure 2.6. Convolutional Neural Network (DeepLearning.AI 2022)

Convolutional neural networks (CNNs), as shown in Figure 2.6, are particularly adept at tasks where spatial relationships and local context play a significant role, offering efficiency in capturing nuanced textual patterns. By leveraging parameter sharing and hierarchical feature extraction, convolutional neural networks (CNNs) optimize computational efficiency and generalize well across different natural language processing (NLP) applications ([Goldberg 2017](#)).

Recurrent Neural Networks (RNNs), as demonstrated in Figure 2.7, are specialized architectures built to handle sequential data, making them ideal for tasks where the order and context of words or tokens are crucial. Unlike feedforward networks, RNNs retain a hidden state that evolves dynamically with each input token, capturing temporal dependencies over time. This recurrent nature allows RNNs to model sequences effectively, which is essential for tasks like language modeling, machine translation, and speech recognition ([Goodfellow and others 2017](#)).

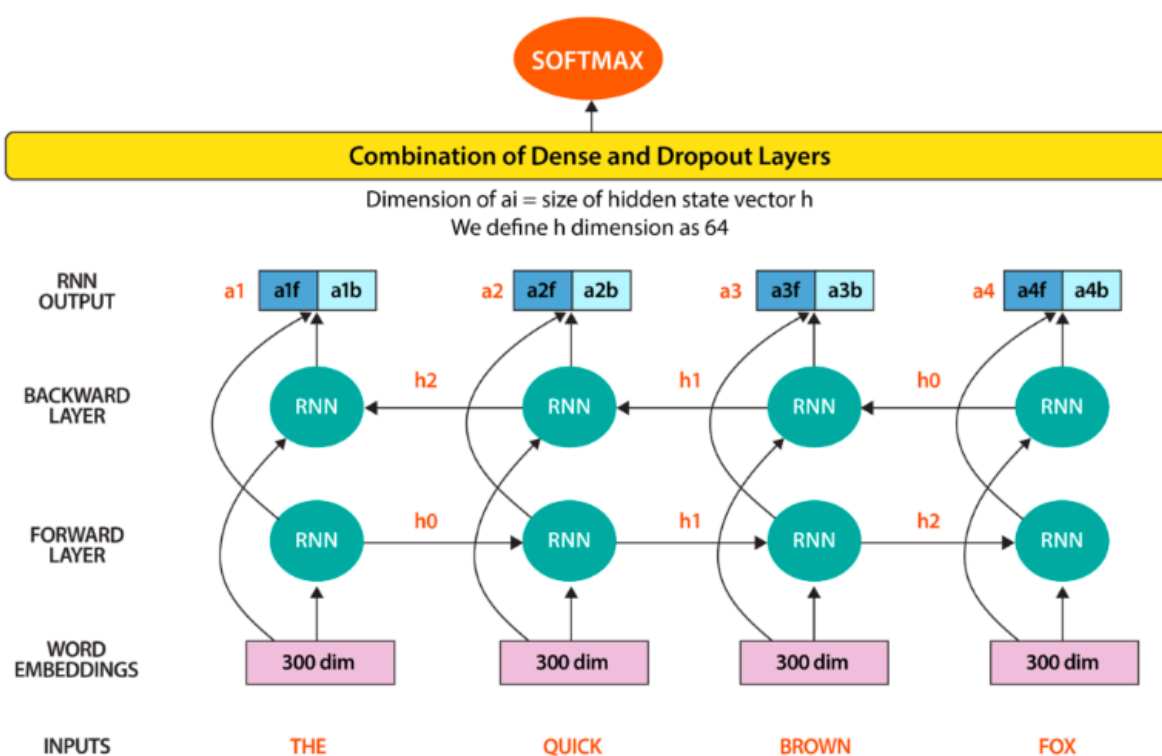


Figure 2.7. Recurrent Neural Network (DeepLearning.AI 2022)

RNN variants such as Long Short-Term Memory (LSTM) and Gated Recurrent Unit (GRU) address inherent challenges like vanishing gradients, enabling them to capture long-term dependencies and improve performance on complex NLP tasks. Their flexibility and ability to learn from sequential patterns make them indispensable for a wide range of NLP applications, from sentiment analysis to sequence labeling tasks such as named entity recognition and part-of-speech tagging ([Hochreiter and Schmidhuber 1997](#)).

However, RNNs also have their limitations, particularly in handling long sequences and capturing dependencies over long distances. This is due to the vanishing gradient problem, where gradients diminish exponentially as they propagate through the network during training, leading to difficulties in learning long-range dependencies. Additionally, RNNs suffer from issues with retaining relevant information over long sequences, known as the short-term memory problem, which can affect their performance in tasks requiring the modeling of complex linguistic structures ([Bengio and others 2013](#)).

Revolutionizing NLP with Transformers

The introduction of transformer architecture has sparked a revolution in natural language processing (NLP), significantly advancing the ability of models to understand and generate human-like text. At the heart of transformers lie self-attention mechanisms and multi-head attention mechanisms, which allow the model to focus on different parts of the input sequence simultaneously and capture complex dependencies between words. The attention mechanism, as introduced in the groundbreaking paper "Attention is All You Need", revolutionized sequence processing tasks by enabling models to focus selectively on relevant parts of input sequences ([Vaswani and others 2017](#)).

Unlike traditional sequence-to-sequence models that rely on fixed-length representations and suffer from information loss, attention mechanisms dynamically compute weighted sums of context vectors based on their relevance to each output word, as shown in Figure 2.8. This allows models to capture long-range dependencies more effectively and adaptively allocate attention to

different parts of the input sequence during both encoding and decoding phases ([Vaswani and others 2017](#)).

By incorporating attention mechanisms, models like transformers have achieved state-of-the-art performance in various natural language processing tasks, including machine translation, text summarization, and language modeling. This architecture has played a key role in the evolution of advanced models like BERT and GPT, setting new benchmarks in tasks ranging from question answering to text generation and understanding ([Devlin and others 2018](#); [Brown and others 2020](#)).

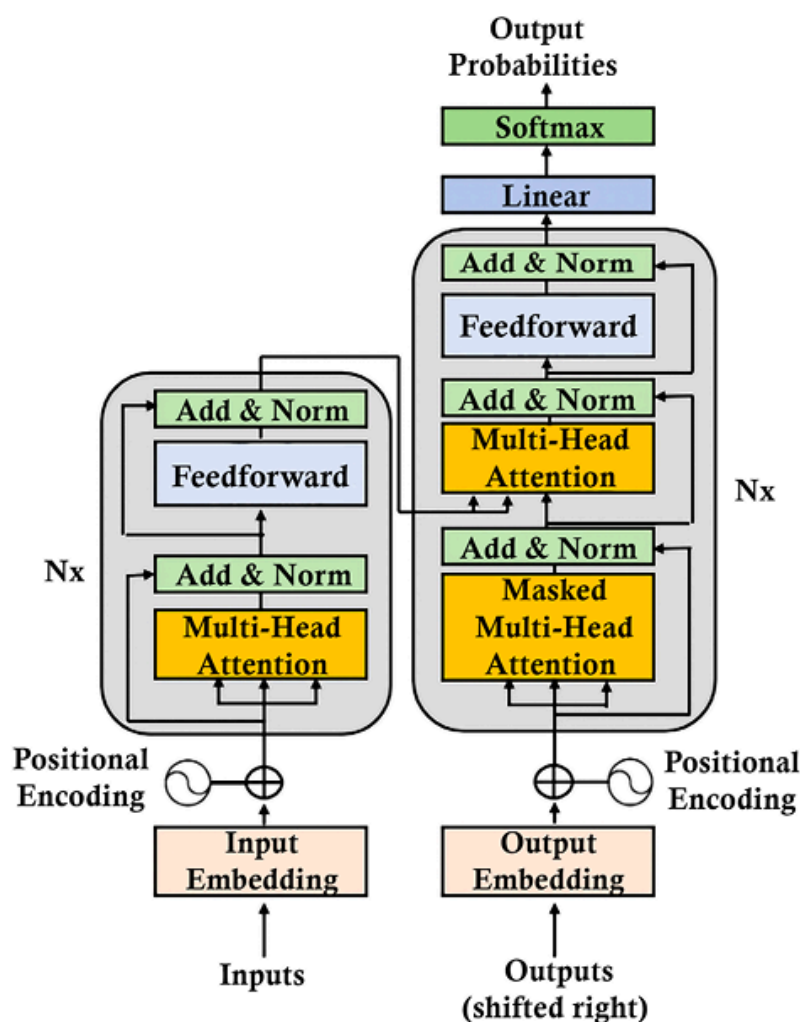


Figure 2.8. Encoder-Decoder Architecture with Attention Mechanism (Vaswani and others 2017)

A major advantage of transformers over RNNs and CNNs lies in their ability to capture long-range dependencies in sequential data. Unlike RNNs, which experience the vanishing gradient problem and have difficulty retaining information across long sequences, transformers can effectively capture contextual information across the entire input sequence. Similarly, transformers outperform CNNs in NLP tasks by leveraging self-attention mechanisms to capture global relationships between words, allowing them to better understand the contextual meaning of text. Furthermore, transformers are highly parallelizable, making them ideal for training on large-scale text corpora and achieving state-of-the-art performance on a wide range of NLP tasks ([Vaswani and others 2017](#)).

Applications of Transformer Models in NLP

Transformers, such as BERT (Bidirectional Encoder Representations from Transformers), GPT (Generative Pre-trained Transformer), and LLAMA, have transformed the field of Natural Language Processing (NLP) with their sophisticated capabilities in language understanding and generation.

The BERT model, as illustrated in Figure 2.9, employs a bidirectional context representation approach, enabling it to capture intricate semantic relationships between words and achieve state-of-the-art results across a range of Natural Language Processing (NLP) tasks ([Devlin and others 2018](#)). GPT, developed by OpenAI, demonstrates remarkable proficiency in natural language generation tasks through a combination of pre-training on vast text corpora and fine-tuning for specific tasks ([Brown and others 2020](#)).

These transformer-based architectures represent a departure from traditional static word embeddings, as they generate contextualized representations of words and sentences. In contrast to methods such as Word2Vec or GloVe, which assign fixed vectors to each word, BERT and GPT generate embeddings that vary based on the context in which words appear within sentences or paragraphs. This contextualization enhances their ability to understand and process natural language with greater accuracy and flexibility ([Devlin and others 2018](#); [Brown and others 2020](#)).

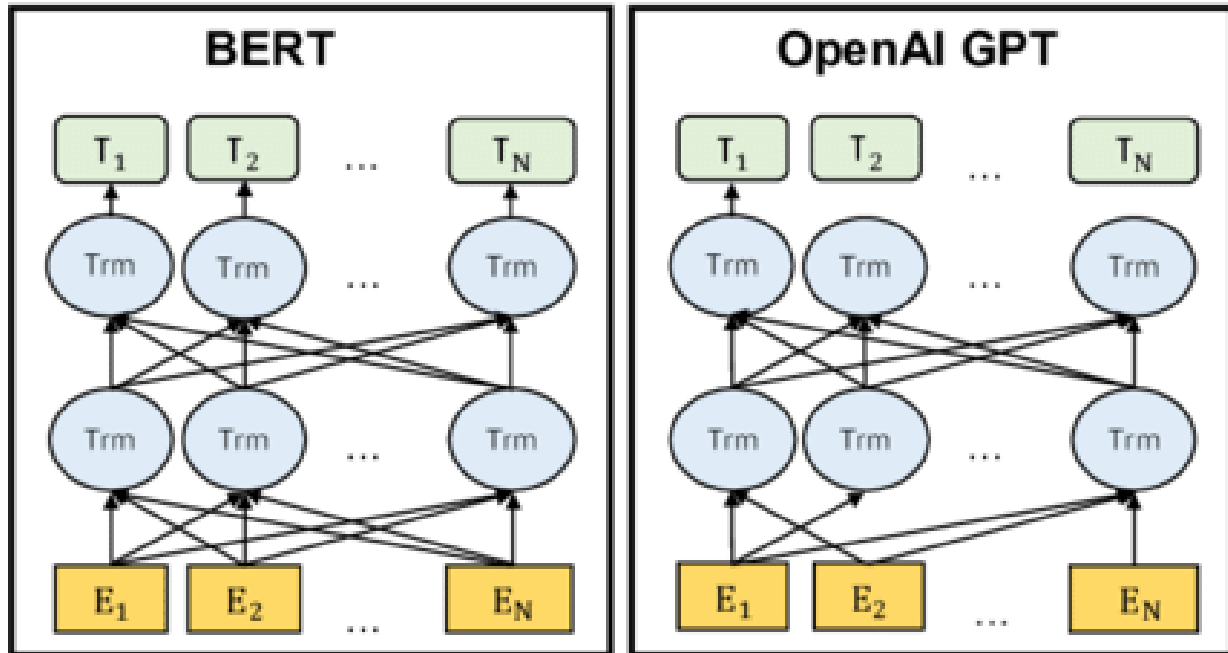


Figure 2.9. BERT and GPT Architectures (Huang and others 2020)

The adoption of transformer models in NLP is further facilitated by their effective use of transfer learning techniques. Initially, these models are pre-trained on extensive text datasets through self-supervised learning tasks, such as masked language modeling and next sentence prediction ([Devlin and others 2018](#)). This pre-training phase enables the models to learn general language representations that capture syntactic and semantic structures. Subsequently, the models are fine-tuned on specific downstream tasks with labeled data, adjusting their learned representations to optimize performance for tasks such as named entity recognition, sentiment analysis, machine translation, and other related tasks ([Howard and Ruder 2018](#)).

BERT has inspired several variations that aim to improve its performance, efficiency, and applicability to different tasks. Among these, DistilBERT and RoBERTa stand out for their unique enhancements and optimizations, as illustrated in Figure 2.10. DistilBERT is a smaller and more efficient adaptation of BERT that maintains impressive performance while significantly reducing model size and computational requirements. It was developed by Hugging Face as a smaller, faster alternative to BERT, achieving 95% of BERT's performance with 40%

fewer parameters. DistilBERT uses knowledge distillation during the pre-training phase to compress the knowledge from the more extensive BERT model into a smaller one. This results in a model that is about 40% smaller, 60% faster, and still retains much of BERT's language understanding capabilities, making it well-suited to applications that require minimal computational resources or those requiring faster inference times ([Sanh and others 2019](#)).

RoBERTa (Robustly Optimized BERT Approach) represents an enhanced iteration of BERT that improves upon the original model's performance through modified training techniques. Developed by Facebook AI, RoBERTa uses dynamic masking, larger batch sizes, and longer training times to achieve state-of-the-art results on a wide range of natural language processing tasks. It also removes BERT's next sentence prediction objective and uses a larger dataset for pre-training. These optimizations allow RoBERTa to outperform BERT on benchmark tasks while maintaining a similar model architecture, demonstrating the importance of training methodology in addition to model design ([Liu and others 2019](#)).

Comparison	BERT October 11, 2018	RoBERTa July 26, 2019	DistilBERT October 2, 2019
Parameters	Base: 110M Large: 340M	Base: 125 Large: 355	Base: 66
Layers / Hidden Dimensions / Self-Attention Heads	Base: 12 / 768 / 12 Large: 24 / 1024 / 16	Base: 12 / 768 / 12 Large: 24 / 1024 / 16	Base: 6 / 768 / 12
Training Time	Base: 8 x V100 x 12d Large: 280 x V100 x 1d	1024 x V100 x 1 day (4-5x more than BERT)	Base: 8 x V100 x 3.5d (4 times less than BERT)
Performance	Outperforming SOTA in Oct 2018	88.5 on GLUE	97% of BERT-base's performance on GLUE
Pre-Training Data	BooksCorpus + English Wikipedia = 16 GB	BERT + CCNews + OpenWebText + Stories = 160 GB	BooksCorpus + English Wikipedia = 16 GB
Method	Bidirectional Transformer, MLM & NSP	BERT without NSP, Using Dynamic Masking	BERT Distillation

Figure 2.10. Comparison between BERT, RoBERTa and DistilBERT (Godbole and others 2020)

Evaluation and Comparison of Transformer Models

Evaluating and comparing transformer models in natural language processing (NLP) involves the use of various performance evaluation metrics tailored to specific tasks. For language understanding tasks, metrics such as accuracy, recall, precision, and F1 score are frequently utilized to assess the model's ability to correctly classify text data ([Rogers and others 2021](#)). Benchmarking transformer models against traditional approaches is another important aspect of evaluation and comparison in NLP. Traditional approaches, such as rule-based systems, statistical models, and shallow learning algorithms, have been widely used in NLP tasks for many years.

Comparing the performance of transformer models against these traditional approaches helps to demonstrate the advancements and improvements achieved by transformer architectures ([Houlsby and others 2019](#)). By evaluating metrics such as accuracy, efficiency, and scalability, researchers can assess the relative strengths and weaknesses of transformer models compared to traditional methods, providing valuable insights into their utility and effectiveness in real-world applications.

A comparative analysis of different transformer architectures and their performance is essential for understanding the strengths and weaknesses of each model. Transformer architectures vary in terms of model size, complexity, and computational requirements, which can impact their performance on different NLP tasks. By systematically comparing the performance of different transformer models, researchers determine the optimal architectural solution for a specific task or application. Factors including model size, training data, and hyperparameters can significantly influence performance, and conducting rigorous comparative analyses helps to identify the optimal configuration for achieving the desired outcomes ([Pipalia and others 2020](#)). Additionally, analyzing the trade-offs between model performance and computational resources provides valuable insights into the practical considerations of deploying transformer models in real-world scenarios.

2.1.4 Generative AI Models in NLP

Generative AI models have emerged as powerful tools in natural language processing (NLP), revolutionizing the way machines understand and generate human-like text. These models, unlike traditional rule-based or statistical approaches, are capable of autonomously generating text that is coherent, contextually relevant, and linguistically accurate. One of the most significant benefits of generative models is their ability to capture complex linguistic patterns and generate text that closely resembles natural language, making them highly versatile for a wide range of NLP tasks.

LLAMA

Several key generative AI models have made significant strides in the field of NLP, each offering unique approaches and capabilities. LLAMA (Language Model for Query Auto-completion), developed by Meta (formerly Facebook), is a specialized generative model designed for query auto-completion, providing accurate and contextually relevant suggestions for partial user queries. Meta's LLAMA3 is the latest generation of large language models designed to excel in a range of natural language processing tasks.

Building on the success of its predecessors, LLAMA3 introduces several key improvements in model architecture, training data, and performance optimization. The system employs a standard decoder-only transformer architecture and utilizes a tokenizer with a vocabulary of 128,000 tokens, thereby enhancing the efficiency of language encoding. LLAMA 3 is trained on an extensive dataset comprising over 15 trillion tokens, which is seven times larger than that used for LLAMA 2, including a significant portion of high-quality non-English data covering more than 30 languages. This extensive training data and advanced filtering techniques ensure high-quality inputs, resulting in a model that performs exceptionally well across various use cases, including coding, reasoning, and creative writing. LLAMA 3 is available in two parameter versions, 8B and 70B, and represents a significant advance in the field of open-source large language models, offers robust performance and versatility for a wide range of applications ([Touvron and others 2023](#)).

GPT

Among the most notable are the Generative Pre-trained Transformer (GPT) series, developed by OpenAI. GPT models employ transformer-based architectures pre-trained on vast amounts of text data, allowing them to generate contextually relevant and coherent text by predicting the next word in a sequence given its context ([Marr 2023](#)).

GPT-3.5 represents a significant advancement in the field of language modeling, building upon the success of its predecessors in the GPT (Generative Pre-trained Transformer) series. It represents a significant advancement in natural language processing, capable of generating human-like text, answering questions, and performing a wide range of language tasks with remarkable fluency and coherence. GPT-3.5 excels in understanding context and generating appropriate responses across various domains, making it useful for applications such as chatbots, content creation, and language translation ([Marr 2023](#)). GPT-4 is the latest iteration in OpenAI's GPT series, marking a major leap forward in artificial intelligence and natural language processing. It demonstrates enhanced capabilities in understanding context, generating more nuanced and accurate responses, and handling complex tasks across multiple domains. GPT-4 shows improved performance in areas such as reasoning, problem-solving, and creative writing compared to its predecessors. Additionally, its ability to process and generate both text and image inputs opens up new possibilities for multimodal applications, making it a versatile tool for various industries and research fields ([Open AI 2023](#)).

OpenAI has introduced two new AI models: GPT-4o and GPT-4o mini. GPT-4o is a powerful multimodal model designed to handle text, audio, and video inputs, with the ability to generate outputs in text, audio, and image formats. GPT-4o mini, conversely, is a more cost-efficient and compact version of GPT-4o, offering superior performance compared to GPT-3.5 Turbo while being significantly more affordable. Both models feature a 128,000 token context window and knowledge up to October 2023 ([Open AI 2024c](#)). The GPT-4o mini is currently equipped to accept both text and vision inputs in its current API release, with future plans to expand to include audio and video capabilities ([Open AI 2024b](#)).

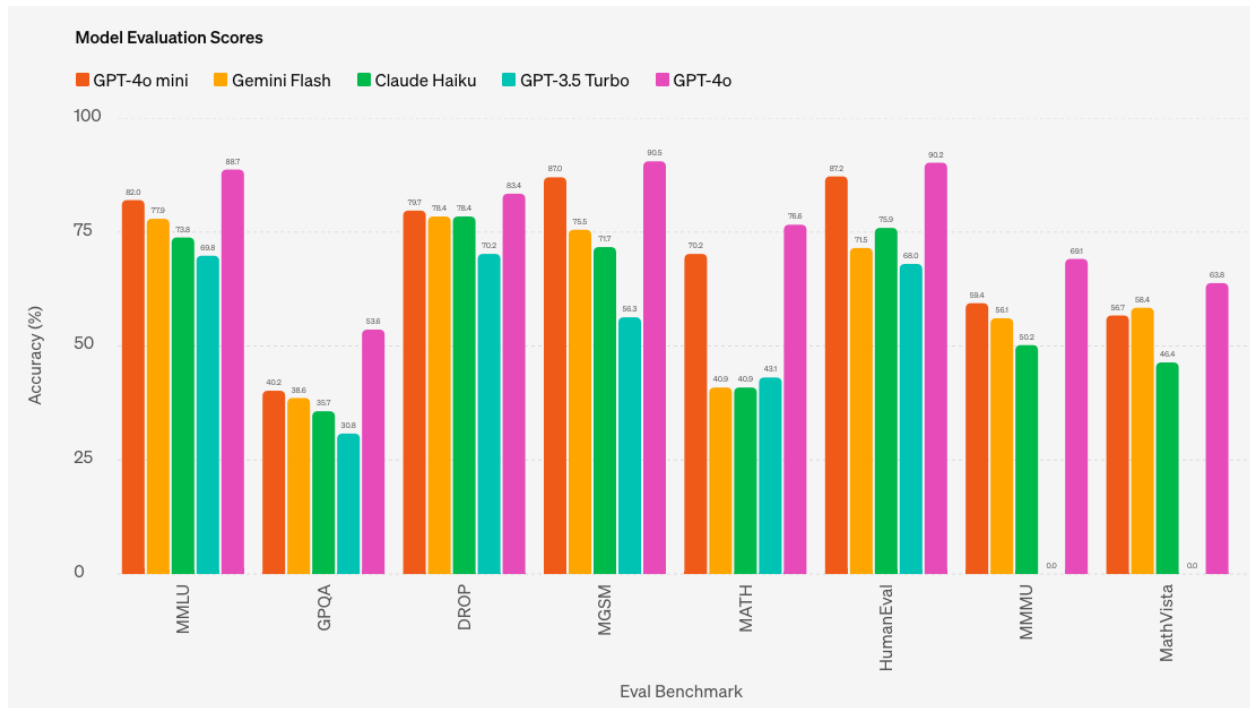


Figure 2.11. Model Evaluation Scores (Open AI 2024b)

As shown in Figure 2.11, GPT-4o mini demonstrates impressive performance for its size and cost. It outperforms GPT-3.5 Turbo on textual intelligence and multimodal reasoning benchmarks while being significantly more affordable. On the MMLU benchmark, GPT-4o mini scores 82.0%, which is lower than GPT-4o's 88.7% but still competitive. It demonstrates optimal performance in tasks that require low latency and high throughput, such as real-time customer interactions or the processing of large data volumes. The model generates output at a rate of 94.3 tokens per second, which is faster than GPT-4o but slower than some competitors like Llama 3 8B and Gemini 1.5 Flash. GPT-4o mini's performance, combined with its cost-effectiveness, makes it an attractive option for many applications, especially those requiring a balance between capability and affordability ([Open AI 2024b](#)). Notably, OpenAI has made fine-tuning available for GPT-4o mini on July 18, 2024 and GPT-4o on August 03, 2024, allowing developers to customize the model for specific use cases and potentially enhance its performance for targeted applications ([Open AI 2024a](#)).

Comparison of Generative Models

LLAMA 3 and GPT-4o represent significant advancements in large language models, each with distinct strengths. LLAMA 3, developed by Meta AI, offers impressive performance, offering enhanced efficiency and cost-effectiveness compared to its predecessors. It excels in tasks like translation and dialogue generation, leveraging techniques such as Grouped-Query Attention for improved focus. GPT-4o, on the other hand, builds upon OpenAI's GPT-4 architecture with enhanced multimodal capabilities, allowing it to process both text and visual inputs. As shown in Figure 2.12 and 2.13, while LLAMA 3 shines in efficiency and specific applications, GPT-4o demonstrates superior performance in complex reasoning tasks and offers a more versatile, multimodal interaction. The choice between these models ultimately depends on the specific requirements of the application, with LLAMA 3 being ideal for streamlined, cost-sensitive projects and GPT-4o better suited for advanced, multimodal AI assistants ([Vellum.ai 2024](#)).

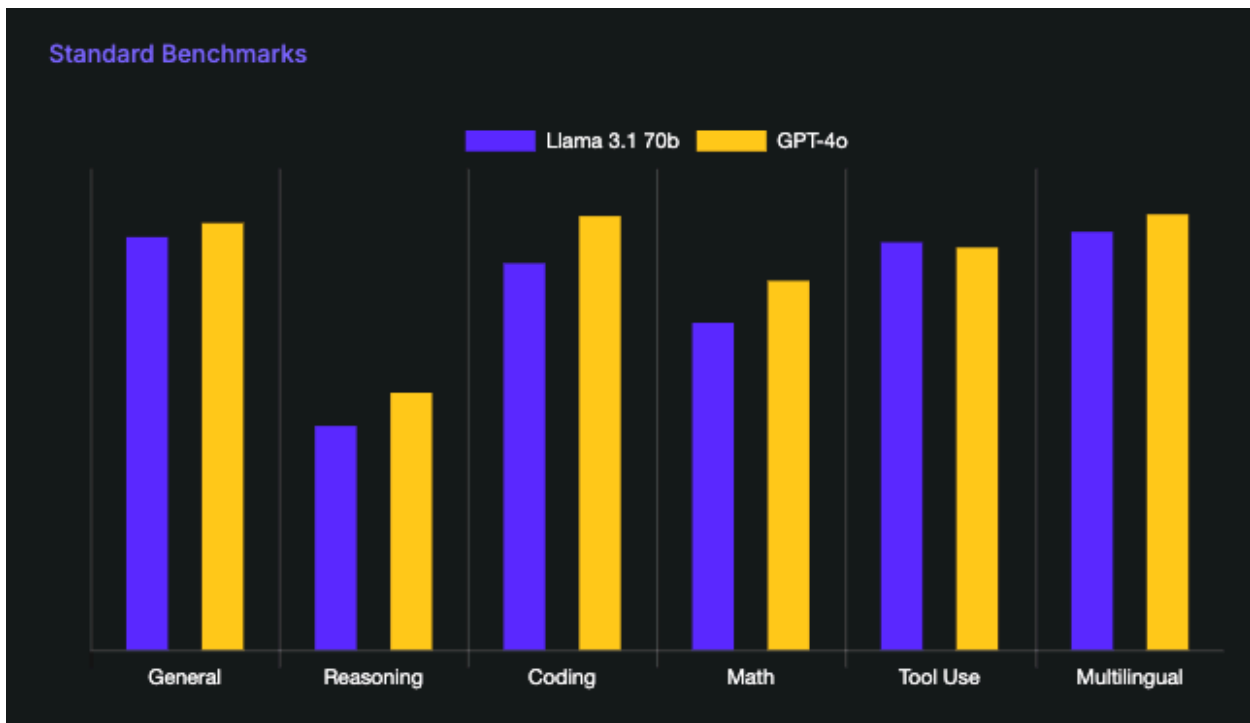


Figure 2.12. LLAMA 3.1 70b and GPT-4o Task Performance Comparison

Model	 Llama 3.1 70b	 GPT-4o
Context size	128,000	128,000
Cutoff date	Dec 2023	Oct 2023
Input/Output cost	\$0.52/\$0.75	\$5/\$15
Max output tokens	4096	4096
Latency (TTFT)	0.43s	0.48s
Throughput	330 t/s (Groq)	79 t/s

Figure 2.13. LLAMA 3.1 70b and GPT-4o Computing Performance Comparison

In summary, Generative AI models have a wide range of capabilities and applications in NLP tasks, including text summarization, generation, and dialogue systems. These models can generate coherent and contextually relevant text on a variety of topics, making them valuable tools for content creation, storytelling, and creative writing. They can also distill lengthy documents by extracting essential information and presenting it in a clear, concise format, facilitating efficient information retrieval and comprehension. Additionally, generative models can interact with users by producing responses that are both contextually relevant and linguistically precise, making them suitable for applications such as virtual assistants, chatbots, and customer service systems. As these models continue to evolve and improve, they hold the potential to revolutionize the way we interact with and utilize natural language in various domains.

2.1.5 Preprocessing and Training Strategies

Preprocessing techniques are essential for preparing text data from Reddit for classification based on the Health Belief Model. These techniques ensure that the data is balanced, augmented, and appropriately labeled to enhance the performance of classification models like DistilBERT and RoBERTa. By tackling challenges like class imbalance and data scarcity, preprocessing sets the foundation for effective model training and accurate categorization.

Preprocessing Techniques

In natural language processing (NLP) and machine learning, the preprocessing of textual data is a vital step that significantly impacts the performance of various tasks. For user-generated content, several standard preprocessing techniques are commonly employed to clean and standardize the text, ensuring it is ready for analysis. These methods are well-established and widely recognized in the field of text analytics and are supported by extensive research and best practices ([Houlsby and others 2019](#)).

One of the foundational steps in preprocessing is text normalization. Converting all text to lowercase is a widely adopted practice that reduces the dimensionality of the data, ensuring that words are treated consistently regardless of their case. This is particularly significant when working with user-generated content, where capitalization can be inconsistent and may lead to variations in word recognition ([Uysal and Gunal 2014](#)).

Handling special characters and whitespace is another critical aspect of preprocessing. The removal of tabulations, newlines, and excess whitespace, often referred to as "whitespace normalization," helps create a uniform text structure. This standardization improves the performance of subsequent NLP tasks by ensuring that the text format is consistent across the dataset ([Uysal and Gunal 2014](#)).

Character-level preprocessing also plays a significant role, especially when dealing with text that may include inputs from various languages. Removing accented characters through processes like "accent stripping" or "unicode normalization" standardizes characters across different

encodings, enhancing text consistency and ensuring uniformity in character representation ([Denny and Spirling 2017](#)).

Expanding contractions, such as converting "don't" to "do not," is another valuable preprocessing technique. This approach standardizes text and can improve the performance of NLP models, particularly those that rely on word-level analysis. By treating contracted forms consistently with their expanded counterparts, this step ensures that the text is processed in a uniform manner, aiding in accurate analysis ([Denny and Spirling 2017](#)).

In addition to these core techniques, other common preprocessing methods often used in text analytics include removing punctuation and special characters, tokenization (breaking text into individual words or subwords), eliminating stop words, and stemming or lemmatization. While not explicitly mentioned, these additional steps can further refine the text data, depending on the specific requirements of the annotation and classification models ([Uysal and Gunal 2014](#)).

By applying these preprocessing techniques, user-generated content is transformed into a cleaner, more standardized format. This processed text is better suited for input into various NLP models, leading to improved performance in tasks such as text classification, sentiment analysis, and topic modeling.

Data Augmentation and Sample Generation

Data augmentation and sample generation are crucial for expanding the dataset and improving model robustness. Contextual word embeddings from models like BERT or RoBERTa can replace words in the text with semantically similar alternatives, generating diverse samples ([Devlin and others 2018](#)). Additionally, generative models like GPT-4 and LLAMA 3 can be employed to create synthetic data that mimics the linguistic characteristics of Reddit posts. These models can generate new text samples by conditioning on existing data, thereby increasing the dataset size and diversity, which is particularly useful for addressing class imbalance and enriching small datasets ([Liu and others 2019](#); [OpenAI and others 2023](#)).

Sample Size Considerations

When determining the appropriate sample size for a large dataset, such as one with 1 million rows, researchers often use Cohran's (1977) formula, which is widely accepted in survey research and other fields. The formula is shown in Figure 2.14 below.

$$\text{Sample size} = \frac{\frac{z^2 \times p(1-p)}{e^2}}{1 + \left(\frac{z^2 \times p(1-p)}{e^2 N} \right)}$$

Figure 2.14. Cohran's formula for sample size (1977)

Where n is the sample size, Z is the z-score for the desired confidence level, p is the estimated proportion of the population (0.5 is used for maximum variability), e is the desired margin of error, and N is the population size. For a 95% confidence level ($Z = 1.96$) and 5% margin of error, this formula yields a sample size of approximately 384 for a population of 1 million or more. For a 99% confidence level ($Z = 2.576$), the required sample size increases to about 664.

These sample sizes are consistent with the guidance provided by [Krejcie and Morgan \(1970\)](#), who developed widely-used tables for determining sample size based on population size. As [Israel \(1992\)](#) notes, the sample size doesn't increase much for populations larger than 100,000, making these calculations applicable to very large datasets. However, [Bartlett \(2001\)](#) caution that while these formulas provide a useful starting point, researchers should also factor in elements like the variability of the characteristic being measured and the specific sampling design being used. Additionally, as [Taherdoost \(2017\)](#) points out, in some fields or for certain research questions, even larger sample sizes might be preferred to increase precision or to allow for more detailed subgroup analyses.

When dealing with large text datasets, traditional sample size formulas like Cohran's are often not directly applicable because of the distinctive nature of textual information. Instead, researchers typically rely on the concept of "saturation" - the point at which additional data does not lead to new information or themes ([Morse 1995](#)). For many linguistic features, samples of 1,000 to 2,000 words can be sufficient to represent the full text population, though this can differ depending on the diversity and complexity of the corpus ([Biber 1993](#)).

In topic modeling or sentiment analysis tasks, larger samples are often needed, with [Qiu and others \(2019\)](#) suggesting that sample sizes of 5,000 to 10,000 documents often provide stable results for large corpora. A practical approach, as recommended by [Hovy and Lavid \(2010\)](#), is to use an iterative sampling method, starting with a smaller sample (e.g., 1,000 texts) and gradually increasing it until the results stabilize. This method enables researchers to identify an optimal sample size for their particular dataset and analysis task, rather than adhering to a universal, one-size-fits-all approach. Additionally, ensuring that the sampling method captures the diversity of the dataset, possibly through stratified sampling based on metadata, is crucial for representative analysis of text data ([Biber 1993](#)).

In the realm of large-scale text analysis, determining optimal sample sizes remains a critical consideration. For overall analysis, [Cohran's formula \(1977\)](#) is still widely used, suggesting that for populations over 1 million, a sample size of 384 is sufficient for a 95% confidence level with a 5% margin of error. However, recent work by [Singh and Masuku \(2014\)](#) argues for larger samples in complex datasets, recommending up to 1,000 samples for more precise estimates. For training data in machine learning models, the rule of thumb proposed by [Beleites and others \(2013\)](#) of 50-100 samples per class still holds, but this can vary based on model complexity.

[Figuerola and others \(2012\)](#) suggest using power analysis to determine sample size, particularly for imbalanced datasets. Their approach considers effect size and desired power, often resulting in larger sample sizes than traditional methods. For test sets, [Varoquaux \(2018\)](#) emphasizes the importance of larger samples, recommending at least 1,000 instances to ensure stable performance estimates, especially in cross-validation scenarios. This aligns with [Sogaard and others \(2021\)](#), who argue for test sets of at least 1,000 samples to reliably distinguish between

high-performing models. These researchers also propose using bootstrap sampling to estimate confidence intervals on test set performance, providing a more nuanced view of model accuracy. Collectively, these approaches suggest a trend towards larger sample sizes than traditionally used, particularly for test sets, to ensure robust and reliable results in large-scale text analysis.

2.2 Related Work

The Related Work section reviews existing research and methodologies relevant to this study. It covers key areas such as Data Labeling practices, the application of the Health Belief Model Analysis in various contexts, and approaches to Emotional Content Analysis. This section provides a foundation by connecting the current study with established knowledge, highlighting gaps that this research aims to address.

2.2.1 Data Labeling

The Health Belief Model is composed of six key components: perceived susceptibility, perceived severity, perceived benefits, perceived barriers, cues to action, and self-efficacy. Accurate labeling of these components within text data is essential for developing effective health communication strategies and interventions. As highlighted by Jones and others (2015), correctly identifying these components helps in understanding the mechanisms through which health messages influence behavioral change. However, the process of labeling health belief model components becomes increasingly complex when dealing with large datasets. Manual labeling, while precise, can be extremely time-consuming and prone to inconsistencies, particularly when labeling vast amounts of text data. To address these challenges, automated or semi-automated approaches have been explored in recent studies to handle large volumes of data more efficiently ([Shorten and others 2021](#)).

Machine learning techniques demonstrate potential for further development in the identification of health belief model components within text. Natural language processing (NLP) methods can be implemented to automatically extract and classify text segments according to the health belief model components. Supervised learning approaches involve training classifiers on a subset of manually labeled data, which can then be used to predict labels for the larger dataset.

Furthermore, unsupervised learning techniques can be applied to group similar text segments, which can then be mapped to corresponding health belief model components ([Jun Yan and others 2006](#)).

Hybrid approaches that combine machine learning with expert review offer another effective strategy. Active learning, for instance, involves an iterative process where the model identifies uncertain cases for human experts to label, thereby improving the model's accuracy over time. Rule-based systems, developed through expert linguistic knowledge, can complement machine learning models by providing a set of predefined rules for labeling ([Sahin and others 2012](#)).

It is crucial to guarantee data quality through preprocessing when working with large datasets. Preprocessing steps may include text normalization, removal of irrelevant information, and handling of missing data ([Manning and others 2008](#)). For manually labeled subsets, assessing inter-rater reliability is crucial to ensure consistency in labeling. Techniques like Cohen's Kappa can be used to measure agreement between different raters, ensuring that the labeling process is reliable ([Cohen 1960](#); [Artstein and Poesio 2008](#)).

Validation is another critical aspect of the labeling process. Techniques such as cross-validation should be employed to evaluate the performance of the automated labeling methods. This helps ensure that the model generalizes well to unseen data and maintains accuracy ([Nordhausen 2009](#)).

Several examples from the literature demonstrate successful approaches to labeling HBM components in text. [Jones and others \(2015\)](#) utilized a combination of manual coding and statistical analysis to identify HBM components in health communication messages. [Haller and others \(2007\)](#) applied the HBM framework to analyze illness beliefs among young people, offering insights into labeling health-related behaviors. Additionally, the development of the Health Belief Model Scale for Exercise (HBMS-E) by researchers in China illustrates how HBM components can be operationalized and measured in specific health contexts ([Sher and others 2016](#)).

In conclusion, labeling the components of the Health Belief Model in large text datasets requires a multifaceted approach that combines machine learning techniques with expert knowledge. The process should be iterative, with the labeling methodology continuously refined based on the results of the validation. Ensuring consistency and accuracy in labeling is essential for the effective application of the HBM in health communication research.

2.2.2 Health Belief Model Analysis

The Health Belief Model has been increasingly applied to analyze health-related content and behaviors on social media platforms. [Du and others \(2019\)](#) used deep learning techniques to analyze Twitter data and understand health beliefs about the Human Papillomavirus (HPV) vaccine, classifying tweets into different HBM constructs. This approach demonstrated how social media data can be leveraged to understand public perceptions of health issues.

[Guidry and others \(2019\)](#) applied the HBM to analyze Instagram posts about the Zika virus, examining how different HBM constructs were represented in social media content related to Zika. [Wang and others \(2021\)](#) explored COVID-19-related health beliefs using Twitter data, analyzing how people's perceptions of susceptibility, severity, benefits, and barriers related to COVID-19 changed over time. [Meadows and others \(2019\)](#) examined Twitter messages during the 2015 measles outbreak in California, using the HBM to categorize tweets and analyze how different health beliefs were associated with vaccine attitudes.

More recently, [Mao and others \(2023\)](#) conducted a cross-national study of health promotion strategies against COVID-19 on Twitter, analyzing tweets from national health departments of six countries using the HBM framework. They explored how different HBM constructs were used in public health messaging and their impact on social media engagement. The collective findings of these studies illustrate the adaptability of the Health Belief Model in analyzing social media data across various health contexts, from vaccine attitudes to pandemic response strategies, showcasing how the HBM can be applied to understand public health communication and perception in the digital age.

2.2.3 Emotional Content Analysis

Emotional content analysis is a multifaceted approach to examining text or other media to identify and categorize emotional expressions. This field has grown significantly in recent years, driven by advances in natural language processing, machine learning, and the increasing availability of digital text data ([Liu 2020](#)). A number of analytical techniques have been developed to facilitate this process, each with its own strengths and limitations, catering to different research needs and contexts.

Lexicon-based methods, such as Text Emotion and NRCLex, rely on predefined dictionaries of emotion-associated words to classify text into emotional categories. These methods are relatively simple to implement and can be computationally efficient, making them suitable for large-scale analyses. However, they may struggle with context and nuance, as they typically don't consider word order or more complex linguistic structures. Text2emotion, for instance, focuses on five basic emotions (Happy, Angry, Sad, Surprise, and Fear), while NRCLex covers a broader range of eight emotions and two sentiments. These tools can be particularly useful for quick, broad-stroke analyses of emotional content in large text corpora ([Mohammad and Turney 2012](#)).

Machine learning-based approaches, like IBM's Tone Analyzer, use trained algorithms to detect emotions in text, offering more sophisticated analysis that can consider sentence structure and context. These methods can often yield more accurate results as they can learn complex patterns from data, potentially capturing subtleties that lexicon-based methods might miss. However, these solutions require substantial computational resources and extensive training data sets to perform effectively. The key benefit of these approaches is their ability to adapt to specific domains or types of text, potentially offering more nuanced and accurate emotional analysis ([Cambria 2016](#)).

Topic modeling techniques, such as Latent Dirichlet Allocation (LDA) and its variants, can be valuable tools in emotional content analysis. These methods identify latent topics within a corpus of documents, which can then be analyzed for emotional content. By combining topic modeling with emotion detection methods, researchers can uncover the emotional associations of different topics and how they evolve over time or across different text sources. For instance, [Bao and](#)

[others \(2009\)](#) proposed an emotion-topic model that jointly models topics and emotions in text, facilitating a deeper understanding of how emotions are expressed in relation to specific themes or subjects. This method can be especially effective for analyzing large-scale social media data, where emotions are often closely tied to current events or trending topics. Additionally, topic modeling can help in preprocessing steps by grouping similar texts together, potentially improving the accuracy of subsequent emotion classification tasks ([Rao and others 2016](#)).

Natural language processing (NLP) APIs, such as those offered by Google Cloud Natural Language API and Microsoft Azure Text Analytics, can perform entity-level sentiment analysis and emotion detection. These services often provide comprehensive analysis, including not just emotion detection but also sentiment analysis, entity recognition, and other linguistic features. They can be particularly useful for researchers or businesses looking for ready-to-use solutions without the need to develop their own models. However, these services may come with usage costs and privacy considerations, as data typically needs to be sent to external servers for processing ([Poria and others 2019](#)).

Qualitative content analysis methods, as described by [Mayring \(2023\)](#), involve systematic, rule-guided categorization of text, which can be adapted to focus on emotional content. This approach allows for more nuanced interpretation and can be particularly useful when dealing with complex or ambiguous emotional expressions. Researchers using this method typically develop a coding framework based on theoretical considerations and iteratively refine it through application to the data. While this approach can yield rich, context-sensitive insights, it can be time-consuming and may be subject to researcher bias. To mitigate this, researchers often employ multiple coders and calculate inter-rater reliability to ensure consistency.

Manual coding techniques, like emotion coding in qualitative research, involve researchers directly identifying and labeling emotional content in data sources such as interviews or field notes. This method allows for deep, context-aware analysis of emotional expressions in qualitative data. Researchers can capture subtle nuances, implicit emotions, and cultural-specific expressions that automated methods might miss. However, this approach requires a significant investment of time and resources, making it less suitable for large datasets. It also requires

well-trained coders with a strong understanding of emotional expression in the context being studied. Despite these limitations, manual coding remains valuable, especially in exploratory research or when dealing with complex, nuanced emotional expressions ([Saldaña 2021](#)).

2.3 Related Work Comparison

This section compares previous research in three key areas: Data Labeling, Health Belief Model Analysis, and Emotional Content Analysis. Each of these areas contributes to the understanding and processing of health-related data, offering insights that inform our approach.

2.3.1 Data Labeling

[Jones and others \(2015\)](#) demonstrated the effectiveness of manual labeling for accurately identifying Health Belief Model (HBM) components, particularly in understanding the mechanisms of health messages. However, this approach has limitations in terms of scalability; it is time-consuming and prone to inconsistencies when applied to large datasets, making it less practical for extensive text analysis ([Artstein and Poesio 2008](#)).

Furthermore, [Shorten and others \(2021\)](#) highlighted the potential of machine learning and NLP for automating the identification of HBM components, offering scalability and consistency. Nevertheless, these methods are limited by their dependence on high-quality, manually labeled training data, which may not always be available. Moreover, automated approaches often struggle with contextual nuances, leading to less accurate labeling in complex or ambiguous texts ([Manning and others 2008](#)).

In addition, [Sahin and others \(2012\)](#) showed that hybrid approaches, such as combining rule-based systems with active learning, can improve labeling accuracy by integrating human expertise with automation. However, these methods are resource-intensive and complex to implement, particularly for large datasets. Their effectiveness is heavily dependent on the quality of the rules and human input, which can introduce errors if not properly managed. [Haller and others \(2007\)](#) and [Sher and others \(2016\)](#) offered valuable insights into the use of the HBM

framework in specific contexts. However, these studies primarily relied on manual methods, which limits their scalability and generalizability across different health contexts.

Given the limitations of existing methods, a combined approach that integrates expert labeling on a small, high-quality dataset with machine learning for broader application is proposed. This method leverages the precision and contextual understanding of human experts in the initial stages, followed by the efficiency and scalability of machine learning to handle larger datasets. Continuous validation and iteration, incorporating human expertise as needed, can further enhance the model's accuracy and applicability across diverse health communication contexts. This approach provides a balanced solution to the difficulties of labeling complex constructs like the Health Belief Model components in large datasets.

2.3.2 Health Belief Model Analysis

[Du and others \(2019\)](#) effectively utilized deep learning to analyze Twitter data and classify tweets into various Health Belief Model constructs related to the HPV vaccine. This study highlighted how social media data can be leveraged to gain insights into public perceptions of health issues. However, the deep learning approach, while powerful, may have limitations in interpreting the nuanced meanings within tweets, potentially leading to misclassification of complex health beliefs.

[Guidry and others \(2019\)](#) successfully applied the HBM to analyze Instagram posts about the Zika virus, demonstrating how HBM constructs are represented in social media content. This approach provided valuable insights into how health-related information is shared and perceived on visual platforms. However, the study's focus on Instagram's visual content might limit its applicability to text-heavy platforms, and the variability in post content could challenge consistent HBM construct identification.

[Wang and others \(2021\)](#) explored COVID-19-related health beliefs on Twitter, offering a dynamic analysis of how public perceptions of susceptibility, severity, benefits, and barriers evolved over time. This temporal analysis is a key strength, showing how public health messages can shift in response to ongoing events. Nevertheless, the reliance on keyword-based analysis

could miss out on more subtle or evolving language trends that do not match predefined categories, limiting the study's comprehensiveness.

[Meadows and others \(2019\)](#) examined Twitter messages during the 2015 measles outbreak in California, using the HBM framework to categorize tweets and analyze their association with vaccine attitudes. The study effectively linked social media content to public health outcomes, showcasing the relevance of HBM in real-world health crises. However, its retrospective design might not capture the full scope of real-time shifts in public sentiment, and the focus on a single event could limit generalizability to other health contexts.

[Castilla-Puentes and others \(2022\)](#) applied the Health Belief Model (HBM) to analyze digital conversations about depression across various racial and ethnic groups before and during the COVID-19 pandemic. Utilizing advanced data mining techniques, the study mapped online discussions to HBM constructs such as perceived susceptibility, severity, benefits, and barriers. This innovative approach provided insights into cultural differences in mental health attitudes and highlighted shifts in discussions related to depression due to the pandemic. While the study effectively demonstrates the applicability of the HBM to social media data, it also faces limitations, such as potential selection bias in underrepresenting older adults and immigrant populations.

[Mao and others \(2023\)](#) conducted a cross-national study of COVID-19 health promotion strategies on Twitter, using the HBM framework to analyze tweets from six countries. The study's strength lies in its comparative approach, highlighting differences in public health messaging and engagement across nations. However, the cross-national focus could lead to challenges in standardizing HBM construct interpretations across different cultural contexts, potentially affecting the consistency of the findings.

The collective results of these studies demonstrate the versatility of the Health Belief Model in analyzing social media content across various health issues, from vaccines to pandemics. However, they also highlight key limitations, such as potential misclassification of nuanced content, platform-specific challenges, and difficulties in capturing real-time or cross-cultural dynamics. Considering these strengths and limitations, the proposed solution would involve a

combined approach that leverages both machine learning for scalability and expert input for nuanced interpretation, ensuring accurate and comprehensive analysis across diverse social media platforms and health contexts.

2.3.3 Emotional Content Analysis

Lexicon-based methods, such as Text2emotion and NRCLex, offer a straightforward approach to emotional content analysis by classifying text into predefined emotional categories using dictionaries of emotion-associated words. [Liu \(2020\)](#) notes that these methods are computationally efficient and simple to deploy, making them well-suited for large-scale analyses. However, their reliance on fixed word lists means they often miss context and nuance, as they do not account for word order or complex linguistic structures. This limitation can lead to oversimplified or inaccurate emotional classifications, especially in texts with subtle or context-dependent emotions ([Mohammad and Turney 2012](#)).

Machine learning-based approaches, like IBM's Tone Analyzer, provide more sophisticated emotional analysis by using trained algorithms that consider sentence structure and context. These methods can capture complex patterns and subtleties in emotional expression, offering higher accuracy compared to lexicon-based methods ([Cambria 2016](#)). However, these solutions require significant computational resources and substantial amounts of training data to function effectively, which can be a significant barrier, particularly for smaller research teams or projects with limited data availability.

Topic modeling techniques, such as Latent Dirichlet Allocation (LDA), allow researchers to identify latent topics within a text corpus and analyze their associated emotional content. [Bao and others \(2009\)](#) highlighted the utility of combining topic modeling with emotion detection to uncover how emotions are linked to specific themes or subjects, making this approach valuable for analyzing large-scale social media data. However, while these methods provide nuanced insights, they can be intricate to implement and interpret, necessitating a deep understanding of both the modeling techniques and the domain-specific context of the text ([Rao and others 2016](#)).

Natural Language Processing (NLP) APIs, such as those from Google Cloud and Microsoft Azure, offer comprehensive emotional content analysis tools, including entity-level sentiment analysis and emotion detection. These ready-to-use solutions are convenient for researchers or businesses needing robust analysis without developing custom models ([Poria and others 2019](#)). However, these services come with usage costs and potential privacy concerns, as they typically require data to be processed on external servers, which may not be suitable for sensitive or confidential information.

Qualitative content analysis methods, as described by [Mayring \(2023\)](#), involve systematic, rule-guided categorization of text, allowing for a nuanced interpretation of emotional content. This approach is particularly useful for analyzing complex or ambiguous emotional expressions. However, it is time-consuming and may be prone to researcher bias, necessitating the use of multiple coders and inter-rater reliability measures to ensure consistency and validity.

Manual coding techniques, such as emotion coding in qualitative research, enable researchers to directly identify and label emotional content, capturing subtle nuances and culturally specific expressions that automated methods might overlook ([Saldaña, 2021](#)). This method is valuable for deep, context-aware analysis, especially in exploratory research. However, it is highly labor-intensive and time-consuming, making it impractical for large datasets. Additionally, the effectiveness of this approach depends on the coders' expertise and understanding of the emotional context within the data.

Each method for emotional content analysis has distinct strengths and limitations. Lexicon-based methods are efficient for large-scale analyses but lack nuance, while machine learning approaches offer more accuracy but require significant resources. Topic modeling provides nuanced insights but is complex, and NLP APIs are convenient but costly and raise privacy concerns. Qualitative content analysis and manual coding offer depth and context sensitivity but are labor-intensive and susceptible to bias. Based on these considerations, a combined approach that leverages the strengths of these methods while mitigating their limitations is proposed, balancing efficiency, accuracy, and contextual understanding in emotional content analysis.

2.4 Summary

In analyzing the strengths and limitations of various methods for Health Belief Model (HBM) component labeling, social media health belief analysis, and emotional content analysis, it becomes clear that no single approach is universally optimal. For HBM component labeling, manual methods, as demonstrated by [Jones and others \(2015\)](#), provide precise identification but lack scalability. Automated approaches, such as those discussed by [Shorten and others \(2021\)](#), offer efficiency for large datasets but often struggle with the contextual nuances essential for accurate HBM labeling. Hybrid methods, like those proposed by [Sahin and others \(2012\)](#), strike a balance by combining machine learning with expert input, though they are resource-intensive and complex to implement.

In the context of social media health belief analysis, studies by [Du and others \(2019\)](#), [Guidry and others \(2019\)](#), and [Mao and others \(2023\)](#) illustrate the versatility of the HBM in understanding public health perceptions across platforms like Twitter and Instagram. However, these studies also highlight limitations such as misclassification risks in deep learning, platform-specific challenges, and difficulties in standardizing cross-national data, underscoring the need for more adaptive and nuanced methods. Regarding emotional content analysis, lexicon-based methods, like Text2emotion and NRCLEx, offer efficiency but lack depth in capturing emotional subtleties. Machine learning approaches, such as IBM's Tone Analyzer, provide more accurate analysis but require significant resources. Topic modeling and NLP APIs offer sophisticated, context-aware analysis but come with complexity and potential privacy concerns. Qualitative methods, including manual coding, allow for deep, context-sensitive insights but are labor-intensive and prone to bias.

Across these domains, a combined approach is recommended, leveraging the strengths of both manual expertise and automated techniques. This strategy involves using manual labeling for small datasets to ensure accuracy, training machine learning models on this data, and then applying these models to larger datasets. Hybrid methods that integrate expert validation with automated analysis provide a balanced solution, ensuring scalability without sacrificing contextual understanding or accuracy.

Chapter 3. Dataset and Methodology

This chapter provides a comprehensive overview of the dataset and the computational techniques employed in this research. It begins by detailing the dataset, including its source, characteristics, and preprocessing steps, which are critical for comprehending the context and quality of the data used in the study. Following the dataset description, the chapter delves into the design and deployment of the machine learning model. This section outlines the methodologies adopted for analyzing the data, including the model architecture, training procedures, and evaluation strategies. By integrating these elements, the chapter aims to offer a clear and structured approach to how the data supports the research objectives and how the model was created in order to address the research questions effectively.

3.1 Dataset

This section provides an overview of the dataset used in this study, covering the key steps from data acquisition to preliminary analysis. It begins with a description of the data collection process, followed by data profiling to understand the structure and quality of the dataset. The feature description subpart details the key variables and attributes, while the data test subpart addresses the initial testing and validation of the dataset.

3.1.1 Data Collection

The raw dataset was sourced from Reddit, a social media platform with millions of posts across five subreddits: anxiety, loneliness, mental health, suicide watch, and depression. The data was collected in 2022 and encompasses posts from 2019 to 2022. In the original paper, the authors concentrated on the underlying causes of mental health as perceived by the individuals in question, as evidenced by their posts. Eight hundred posts were annotated by experts, and the results were compared to determine the final outcomes ([Rani and others 2024](#)). Nevertheless, the further approach for health belief models has not been applied, nor has an emotional analysis been conducted. Consequently, a comprehensive analysis of the health belief model and emotional content analysis would be a valuable addition to the existing literature.

In addition to the raw dataset, there was an annotated dataset with an additional column defining the mental health root cause that users believed based on their posts. This column was annotated by two annotators based on the mental health root cause categories, and the results were then compared using Cohen's kappa ([Cohen 1960](#)). This could produce highly accurate results, but due to the labor intensive nature of the process, only 800 rows were categorized.

3.1.2 Data Profiling

The raw dataset is organized into folders, each corresponding to a specific month. Within each folder, there are five files representing data from five subreddits. The detailed data tables contain millions of rows with seven columns. The detailed data in each table can be categorized as below:

- Qualitative features: Title, Author, Selftext, Subreddit
- Quantitative (Discrete) features: Score
- Quantitative (Continuous) features: Created_utc, Timestamp

The root cause labeled dataset comprises one table with 800 rows and six columns. The detailed data in each table can be categorized as below:

- Qualitative features: Title, Author, Selftext, Subreddit, Label
- Quantitative (Discrete) features: Score

3.1.3 Feature Description

A detailed description of each feature included in the dataset is provided in Table 3.1 and 3.2 below.

Number	Feature Name	Description
1	Title	The title of the Reddit post
2	Author	The author of the post
3	Selftext	The main text content of the post
4	Subreddit	The subreddit from which the post was sourced
5	Score	The net score (upvotes minus downvotes) of the post
6	Created_utc	The UTC timestamp of when the post was created
7	Timestamp	The local date and time when the post was created, converted from the UTC timestamp

Table 3.1. Columns of Raw Dataset (Rani and others 2024)

Number	Feature Name	Description
1	Title	The title of the Reddit post
2	Selftext	The main text content of the post
3	Subreddit	The subreddit from which the post was sourced
4	Score	The net score (upvotes minus downvotes) of the post
5	Label (Root Cause)	The assigned label indicates the identified root cause of mental health issues, based on the annotation process: Drug and Alcohol , Early Life, Personality, Trauma and Stress.
6	CAT 1	The assigned sub-label indicates the identified root cause of mental health issues in Trauma and Stress, based on the annotation process: Relationship, Loneliness, Work Stress, Financial, DV.

Table 3.2. Columns of Labeled Dataset (Rani and others 2024)

3.1.4 Data Test

Table 3.3 and 3.4 below shows a concise overview of the data types associated with each feature in the dataset.

Number	Column	Non-Null	Count	Dtype
1	author	142901	non-null	object
2	created_utc	142901	non-null	int64
3	score	142901	non-null	int64
4	selftext	138907	non-null	object
5	subreddit	142901	non-null	object
6	title	142901	non-null	object
7	timestamp	142901	non-null	object

Table 3.3 Type of each column of Raw Dataset

Number	Column	Non-Null	Count	Dtype
1	score	800	non-null	int64
2	selftext	800	non-null	object
3	subreddit	800	non-null	object
4	title	800	non-null	object
5	Label (Root cause)	800	non-null	object
6	CAT 1	200	non-null	object

Table 3.4 Type of each column of Root Cause Labeled Dataset

3.2 Methodology

Section 3.2 outlines the methodology used in the study, detailing the process from model design to data analysis. It includes steps such as data preprocessing, tokenization, and label combination, followed by the training and evaluation of text classification and generation models. The section also details the approach to data annotation and the specific analyses conducted, including the Health Belief Model and Emotional Content Analysis.

3.2.1 Model Design

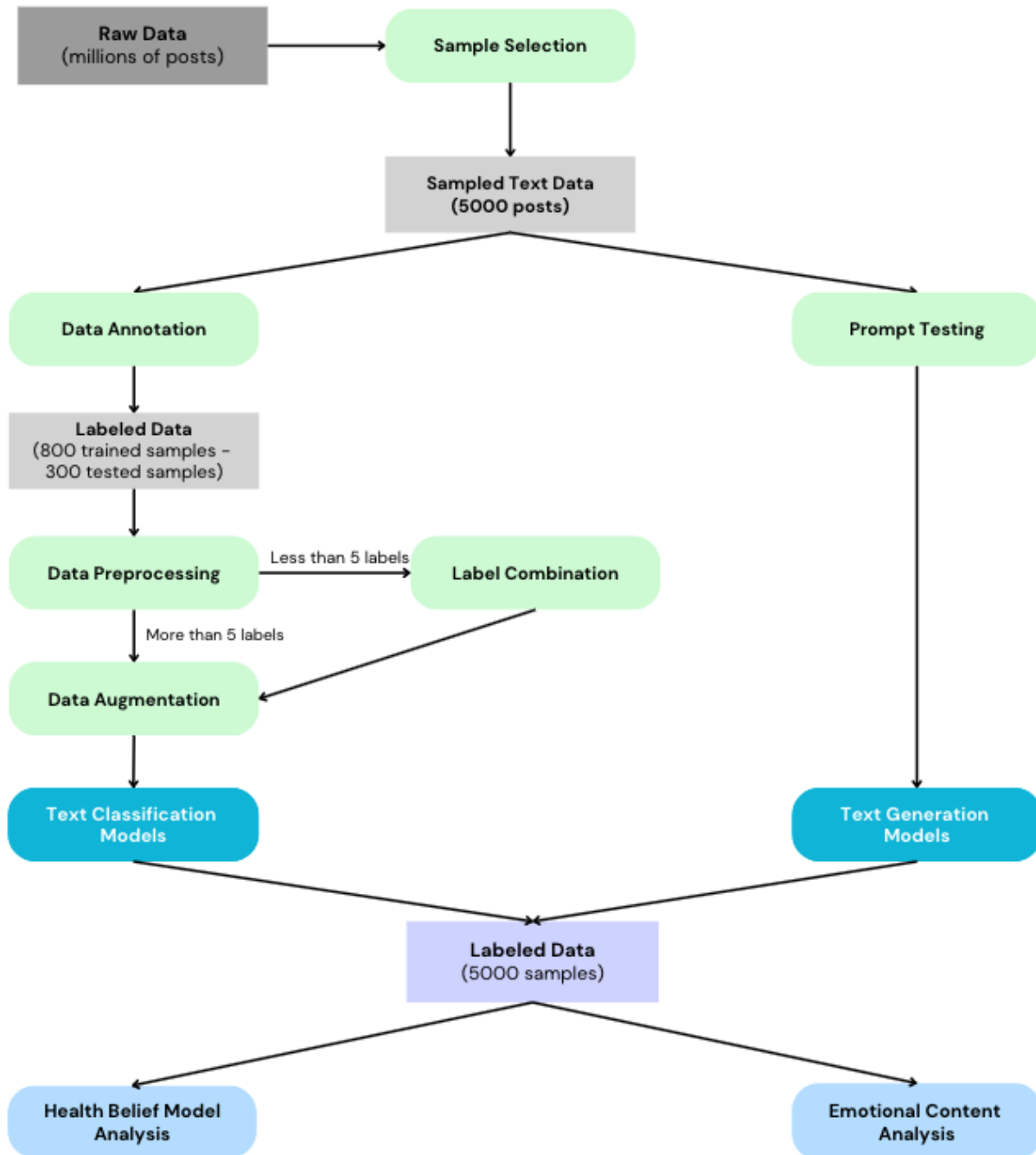


Figure 3.1. Overall Flow From Data Annotation to Data Analysis

The original dataset comprises millions of posts from 2019 to 2022. For the purposes of this report, however, our focus is on the period following the onset of the pandemic, specifically from May to July 2022. Original posts with insufficient content (i.e., less than 50 characters) were excluded from the analysis, as it proved challenging to comprehend the context and label the data rows.

Subsequently, 5,000 posts were randomly selected for analysis, which is a sufficient number to represent a text dataset. These posts were distributed between two different flows. One flow involved two annotators labeling the posts, followed by preprocessing and feeding them to a text classification model to label based on Health Belief Model components. The other flow involved feeding text to a text generation model to extract information, particularly regarding perceived severity, as this column varied significantly and was challenging to train with a text classification model.

The outcomes of training and evaluating these models demonstrated their efficacy, prompting the application of the models to a corpus of 5,000 posts. Subsequently, 288 to 296 posts were selected for further evaluation based on the outcomes of the sample size test. Should the evaluation yield favorable outcomes, specifically an accuracy rate exceeding 75%, the subsequent phase would commence, entailing the utilization of these labels to examine the Health Belief Model and the emotional content of the aforementioned posts.

3.2.2 Data Preprocessing

The preprocessing phase was designed to ensure the dataset's consistency and readiness for analysis. Essential tasks included handling missing values, removing duplicates, and normalizing text. This stage aimed to clean and standardize the dataset, which is crucial for the reliability of subsequent analyses. By addressing these preprocessing steps, we prepared the data for effective tokenization and model training.

3.2.3 Tokenization

The tokenization process involves converting text into a format that is compatible with machine learning models. This process included breaking down text into token IDs using advanced tokenizers compatible with models like RoBERTa and DistilBERT. Given the input length constraints of these models, texts were truncated to retain the most relevant portions. This approach ensured that the data fed into the models was both efficient and representative, optimizing model performance.

3.2.4 Label Combination and Data Augmentation

To address class imbalance, labels that constituted less than seven percent of the dataset were consolidated into an "Others" category. The objective of this strategy was to achieve a more balanced dataset and enhance model accuracy by minimizing the influence of underrepresented classes. Additionally, data augmentation techniques were applied to enhance the dataset, particularly for minority labels. By generating synthetic samples, the dataset's diversity and robustness were increased, which contributed to improved performance of the predictive models.

3.2.5 Dataset Splitting and Evaluation

The dataset was divided into training and testing subsets using an 80/20 ratio. This splitting ensured that the models were trained on a substantial portion of the data while retaining a portion for unbiased evaluation. Model performance was assessed using metrics such as accuracy, precision, recall, and F1-score, which provided a detailed evaluation of the models' effectiveness in both classification and text generation tasks. These metrics were computed to gauge how well the models performed and to identify areas for improvement.

3.2.6 Data Annotation

Data annotation is a critical step in preparing the dataset for subsequent analysis, ensuring that the textual data is accurately labeled and categorized. This section outlines the procedures followed to annotate the data, concentrating on the creation and use of machine learning models for text classification and generation. The process begins with the training of text classification

models to categorize data according to predefined labels, followed by the use of text generation models to enhance and expand the annotated dataset. Each of these steps is essential for refining the dataset and enhancing the overall quality of the analysis.

Human-Led Sample Annotation

For the annotation process, the training data and testing data were selected. These datasets were annotated based on the principles of the Health Belief Model by two annotators, one from the Data Team and one from the Psychology Team. Each label could encompass multiple values. After initial annotation, the results were compared to identify any discrepancies. The annotators then engaged in discussions to resolve these differences and finalize the labels, ensuring consistency and accuracy in the dataset.

Text Classification Model Training

For text classification, pre-trained models such as RoBERTa and DistilBERT were utilized and fine-tuned using established training practices. This involved optimizing hyperparameters such as learning rate and batch size to enhance model performance. Early stopping techniques were implemented to prevent overfitting and ensure that the models are capable of generalizing to new, unseen data. This step was crucial for developing robust and accurate classifiers.

Text Generation Models

Text generation capabilities were explored using advanced models such as LLAMA3 and GPT-4o. These models were evaluated for their capacity to generate coherent and contextually relevant text based on the input data. Integration and fine-tuning of these models were carried out using appropriate libraries. Generated text was processed and structured into JSON objects, which were then converted into a format suitable for analysis. This allowed for the comparison of text generation outputs with classification results and facilitated comprehensive analysis and interpretation.

3.2.7 Data Analysis

The data analysis process is critical for extracting meaningful insights and validating the hypotheses outlined in this study. This section details the analytical approaches used to explore the relationships and patterns within the collected data. Specifically, the analysis is divided into two primary areas of focus: the Health Belief Model Analysis and the Emotional Content Analysis. Each of these analyses fulfills a distinct role in understanding the underlying factors influencing health-related behaviors and emotional responses

Health Belief Model Analysis

Once the data had been labeled, a variety of data visualizations were employed for the purpose of analyzing the components of the health belief models. In the case of columns with multiple labels, such as perceived severity, a word cloud was employed for visualizing the distribution of key terms. For other columns, we employed donut charts and bar charts to illustrate the distribution and relationships between different categories in a more effective manner. In addition to these visualizations, correlation analysis was conducted to analyze the relationships between various pairs of factors.

Specifically, we examined the correlation between sentiment scores and self-efficacy levels to determine whether more positive sentiments correspond with higher self-efficacy. We also evaluated how perceived benefits correlate with perceived barriers, assessing whether individuals who see high benefits towards taking health actions also report lower barriers. Finally, we analyzed how cues to action relate to other factors, considering their potential roles as mediators or moderators in belief formation or behavior change.

Emotional Content Analysis

Subsequently, we conducted a sentiment analysis for each word and compared it across different perceived severity levels to determine if users employed more negative language in certain cases. Additionally, we identified the top 10 negative words in each group to understand the particular language patterns employed.

3.3 Summary

This chapter explained the methodological framework implemented to analyze and process the Reddit dataset for this research. The methodology was designed to ensure that the dataset was thoroughly prepared and effectively utilized for both classification and generation tasks.

Data Preprocessing involves essential steps such as removing duplicates, handling missing values, and normalizing text to standardize and clean the dataset. This prepared the data for the next stages of analysis. Tokenization converted the text into a format compatible with machine learning models, ensuring that the data input was efficient and representative despite model constraints.

To address class imbalance, Label Combination was implemented by consolidating underrepresented labels into an "Others" category. Data Augmentation further enhanced the dataset by generating synthetic samples, which increased its diversity and robustness. Dataset Splitting was performed to create training and testing subsets, with performance evaluated using metrics such as accuracy, precision, recall, and F1-score to assess model effectiveness.

The Text Classification Model Training involved fine-tuning pre-trained models like RoBERTa and DistilBERT, optimizing their hyperparameters, and employing early stopping techniques to prevent overfitting. In parallel, Text Generation Models such as LLAMA3 and GPT-4o were explored for their ability to generate coherent and contextually relevant text. The generated outputs were processed and structured for detailed analysis.

These methodological steps were crucial in preparing the data, training the models, and assessing their performance. The following chapter will delve into the implementation details, focusing on the practical application of these methods and the specific results achieved through their application.

Chapter 4. Implementation

Implementation details the practical steps taken to apply the methodologies outlined in the previous chapter. This includes the technical setup, the application of data preprocessing techniques, and the training of machine learning models. The chapter describes the execution of the methods, including any adjustments made throughout the process and the challenges encountered.

4.1 Tools

This section outlines an overview of the tools and software libraries utilized for data analysis, data pre-processing, and machine learning construction.

4.1.1 Data Preprocessing

Data preprocessing involves using Python libraries “tqdm” and “contractions” to perform essential tasks such as removing duplicates, handling missing values, and normalizing text. These steps ensured that the Reddit posts dataset was consistent and well-prepared for subsequent analysis.

4.1.2 Tokenization

Tokenization was carried out using the Hugging Face Transformers library, employing tokenizers designed for models like RoBERTa and DistilBERT. This process converted text into token IDs, making it compatible with model input requirements. Due to the inherent input length constraints of certain models, text inputs were truncated using the Hugging Face library's features. This ensured that the most critical portions of the text were retained, allowing the models to make more accurate predictions.

4.1.3 Label Combination and Data Augmentation

To mitigate class imbalance, labels that represented less than seven percent of the dataset were grouped into an "Others" category. This approach was critical in reducing data imbalance and enhancing the accuracy of model predictions.

Data augmentation was performed using the “nlpaug.augmentation” package to enhance the dataset, particularly for minority labels. This tool generated synthetic samples, improving the dataset’s diversity and robustness, ultimately leading to better model performance.

4.1.4 Dataset Splitting and Evaluation

The dataset was split into training and testing sets using the `train_test_split` function from `scikit-learn`, maintaining an 80/20 ratio. Model performance was evaluated using metrics such as accuracy, precision, recall, and F1-score, calculated through `scikit-learn`’s `classification_report`. These metrics provided a comprehensive assessment of the models' effectiveness in both classification and text generation tasks.

4.1.5 Data Annotation

The data was imported into a Google Sheet and distributed to the annotators for parallel labeling of the data. For text classification, TensorFlow and the Hugging Face Transformers library were used to train and evaluate pre-trained models such as RoBERTa and DistilBERT. These tools allowed for efficient model training with configurations like optimized batch size and learning rate. Early stopping was used to avoid overfitting and ensure the models generalized well to unseen data.

In addition to classification, text generation was explored using state-of-the-art models, including LLAMA3 and GPT-4o. These models were evaluated for their ability to generate coherent and contextually relevant text based on the input data. The Hugging Face Transformers library provided support for integrating and fine-tuning these models within the implementation. After generating text, the results were returned in a JSON object. The `json` package was then employed to extract the relevant information from these JSON objects, converting the generated text into

structured data. This data was subsequently mapped into predicted columns, facilitating further analysis and interpretation of the text generation output. This process ensured that the predictions were stored in a format that could be easily analyzed alongside the classification results.

4.1.6 Data Analysis

This section outlines the tools used for analyzing the dataset, focusing on both Health Belief Model and emotional content analyses. The analysis leveraged a range of Python packages to systematically explore and visualize patterns and sentiment within the data, ensuring robust and insightful results.

In order to conduct a health belief model analysis, a number of Python packages were employed. These included “pandas” for data manipulation and management, “wordcloud” for the generation of visual representations of word frequency, “matplotlib” for the creation of various charts and visualizations, and “Counter” from the “collections” module for the efficient counting and analysis of word occurrences. These tools provided a means for systematically exploring and visualizing the patterns and trends in the dataset.

In order to perform an emotional content analysis, a set of Python packages was employed for the assessment of sentiment within the dataset. The data handling was conducted using “pandas”, while the counting of sentiment categories was performed with the “Counter” function from the “collections” module. The emotional tone of the text was classified using the “nltk.sentiment.SentimentIntensityAnalyzer” module, and the visualization of the results was illustrated by “matplotlib”. These tools enabled a systematic analysis and comparison of the emotional content across different perceived severity groups.

4.1.7 Computational Resources

The training and testing processes were conducted in a GPU-enabled cloud-based environment, providing the necessary computational power to handle large datasets and optimize model performance effectively.

4.2 Data Preprocessing

This section outlines the preprocessing steps applied to prepare the dataset for analysis. Initially, discrepancies in data values were corrected, followed by a series of cleaning procedures to standardize and normalize the text, ensuring the data was accurate and ready for annotation and classification.

4.2.1 Raw Dataset

The initial column to be examined and preprocessed was "subreddit." At certain data points, there was a discrepancy between the "subreddit" and "selftext" columns, with the values of these two columns being interchanged. Consequently, following the application of lowercase and the removal of white space from these two columns, the five subreddits were defined in accordance with the original paper.

The following terms were identified as relevant: "suicidewatch," "depression," "mentalhealth," "anxiety," and "lonely." Subsequently, for each value in the "subreddit" column that was not included in the aforementioned five subreddits, the value was switched between the "subreddit" and "selftext" columns. The code for the preprocessing steps is illustrated in Figure 4.1.

```

!pip install contractions
import contractions
from bs4 import BeautifulSoup
import re
import tqdm
import unicodedata

from nltk.stem import SnowballStemmer
stemmer = SnowballStemmer('english')

def strip_html_tags(text):
    soup = BeautifulSoup(text, "html.parser")
    [s.extract() for s in soup(['iframe', 'script'])]
    stripped_text = soup.get_text()
    stripped_text = re.sub(r'[\r|\n|\r\n|'+, '\n', stripped_text)
    return stripped_text

def pre_process_corpus(docs, stem = False):
    norm_docs = []
    # tqdm to display a progress bar while looping
    for doc in tqdm.tqdm(docs):
        # remove HTML tags
        doc = strip_html_tags(doc)

        # convert tab, new lines to empty spaces
        doc = doc.translate(doc.maketrans("\n\t\r", " "))

        # lowercase
        doc = doc.lower()

        # remove accented chars
        doc = unicodedata.normalize('NFKD', doc).encode('ascii', 'ignore').decode('utf-8', 'ignore')

        # expand shortened words, e.g. don't to do not
        doc = contractions.fix(doc)

        # remove white spaces
        doc = re.sub(' +', ' ', doc)
        doc = doc.strip()

        norm_docs.append(doc)
    return norm_docs

```

Figure 4.1. The code for the preprocessing steps

The subsequent column to be subjected to preprocessing was the "selftext" column, which represents posts from users. A number of steps were taken to ensure that the data was free from any contamination and was ready to be fed into annotation and classification models. These included the conversion of tabulated data into empty spaces, the conversion of new lines into empty spaces, the conversion of uppercase letters into lowercase, the removal of accented characters, the expansion of shortened words, such as "don't" to "do not", and the removal of white spaces. The Table 4.1 below shows the sample of text before and after preprocessing.

Raw Text	Preprocessed Text
No matter which way, whether sadness, hate, fear, pain, tiredness, anger, despair, clueless,..... No matter where there is only depth and abyss... No matter how often I build a bridge to go further (feel good) it's always a "matter" of time until I feel the fall... Everytime I think that I am as down as I could ever be that I could not be anymore broken or whatsoever time goes until I fall deeper and again and always and forever. No matter how deep/much I feel to be cut, there is more meat that gets cut even deeper..... Nothing ever cuts through. I never get true cut of (die). It only cuts near the edge...	no matter which way, whether sadness, hate, fear, pain, tiredness, anger, despair, clueless,..... no matter where there is only depth and abyss... no matter how often i build a bridge to go further (feel good) it is always a "matter" of time until i feel the fall... everytime i think that i am as down as i could ever be that i could not be anymore broken or whatsoever time goes until i fall deeper and again and always and forever. no matter how deep/much i feel to be cut, there is more meat that gets cut even deeper..... nothing ever cuts through. i never get true cut of (die). it only cuts near the edge...

Table 4.1. Sample of text before and after preprocessing

4.2.2 Labeled Dataset

The same methodology was employed for the raw data set. The label data was extracted using the pandas package. The first column to be subjected to preprocessing was the "selftext" column, which followed the same procedure as the raw dataset. Subsequently, the 'label' column was converted to lowercase and spaces were removed to maintain consistency with the mentioned procedure.

4.3. Sample Size Selection

For the analysis of a large dataset comprising 1 million Reddit posts, we adopt a strategic sampling approach to balance statistical validity with practical constraints:

1. Overall analysis sample: 5,000 posts

This sample size is sufficient for stable results in large corpora analysis, as suggested by [Qiu and others \(2019\)](#).

2. Manual labeling for training: 500 to 750 posts

This range ensures adequate representation for each class, following the guideline of at least 100 samples per class ([Beleites and others 2013](#)).

3. Testing sample: 125 to 225 posts

This maintains a 70/30 to 80/20 split between training and testing data, a common practice in machine learning.

4. Validation of machine-labeled data: 288 to 300 posts

Given the labor-intensive nature of manual review, we adjusted the confidence level to 95% and maintained a 5.7% margin of error. This sample size was determined using the formula shown in Figure 4.2.

$$E = Z \times \sqrt{\frac{p \times (1 - p)}{n}} \times \sqrt{\frac{N - n}{N - 1}}$$

Figure 4.2. Sample size calculation based on Cohran's formula 1977

Where:

$Z = 1.96$ (95% confidence level)

$p = 0.5$ (assumed proportion)

$e = 0.057$ (5.7% margin of error)

$N = 5000$ (population of machine-labeled posts)

This approach allows for a statistically valid assessment of the model's performance while minimizing the manual labeling effort. With this sample size, we can be 95% confident that the

true accuracy of our model lies within $\pm 5.7\%$ of what we observe in our sample ([Bartlett and others 2001](#)).

4.4 Data Annotation

Although the existing annotated dataset comprises 800 rows of root causes, the objective of this research is to conduct a more in-depth examination of the Health Belief Model. Consequently, the scope of data annotation has been expanded to encompass six additional categories: perceived susceptibility, perceived severity, perceived benefits, perceived barriers, cue to action, and self-efficacy.

4.4.1 Human-Led Sample Annotation

The 800 rows of data designated for training and the 296 rows of data designated for testing were annotated in accordance with the Health Belief Model by two annotators from the Data Team and the Psychology Team. Subsequently, the annotations were compared to identify discrepancies, discussed, and aligned on the final labels.

4.4.2 Text Classification Model

In light of the aforementioned considerations, our preliminary recommendation is to utilize text classification models for the categorization of data. This approach is favored because such models represent the cutting edge of the field, are available at no cost, and offer a high degree of flexibility.

The three-step process was subjected to testing in this context. In cases where more than five labels are involved, the label combination entails the aggregation of all labels with a value count of less than seven percent into the "Others" category, while preserving the original categorization of the remaining labels. This process ensures a reduction in data imbalance and enhances the accuracy of model predictions. The subsequent step is data augmentation, which generates additional data, particularly for minority labels. The code for these steps is presented in Figure 4.3 below.

```
[ ] # Calculate the value counts
value_counts = df['Perceived severity'].value_counts()

# Calculate the threshold for 10%
threshold = 0.10 * len(df)
print(f"Threshold (10% of total): {threshold}")
```

Threshold (10% of total): 80.0

```
[ ] # Create a mask for labels to keep
labels_to_keep = value_counts[value_counts > threshold].index

# Modify the DataFrame
df['Perceived severity'] = df['Perceived severity'].apply(lambda x: x if x in labels_to_keep else 'Others')

# Display the modified value counts
print("\nModified value counts:")
print(df['Perceived severity'].value_counts())
```

```
[ ] #Augment only for minority classes
!pip install nlpaug
import pandas as pd
import nlpaug.augmenter.word as naw

# Identify the minority classes
class_counts = df['Perceived severity'].value_counts()
minority_classes = class_counts[class_counts < class_counts.max()].index.tolist()

# Define the augmenter
aug = naw.SynonymAug(aug_src='wordnet')

# Function to augment text
def augment_text(text):
    augmented = aug.augment(text)
    return augmented[0] if isinstance(augmented, list) else augmented

# Create a new DataFrame to store augmented data
augmented_df = pd.DataFrame(columns=df.columns)

# Augment only the minority classes
for label in minority_classes:
    minority_samples = df[df['Perceived severity'] == label]

    # Calculate how many new samples to generate
    if class_counts.max() < len(minority_samples)*3:
        samples_to_generate = class_counts.max() - len(minority_samples)
    else:
        samples_to_generate = len(minority_samples)*2

    # Augment the minority class
    for _ in range(samples_to_generate):
        sample = minority_samples.sample(1)
        sample['preprocessed_selftext'] = sample['preprocessed_selftext'].apply(augment_text)
        augmented_df = pd.concat([augmented_df, sample], ignore_index=True)

# Combine original and augmented data
balanced_df = pd.concat([df, augmented_df], ignore_index=True)

print(balanced_df["Perceived severity"].value_counts())
```

Figure 4.3. Data Preprocessing and Label Aggregation Code

This guarantees that the models can not only predict with high accuracy but also high precision, recall, and F1, which is essential for imbalanced datasets. In consideration of the constraints imposed by the model's input length, two approaches were devised to ensure the retention of the initial 512 characters, or alternatively, the initial 256 characters and the remaining 256 characters at the conclusion. Additionally, models such as RoBERTa and DistilBERT have demonstrated efficacy in processing text datasets, as shown in Figure 4.4 and 4.5.

```
# Convert labels to numerical format
label_dict = {label: i for i, label in enumerate(set(labels))}
labels = [label_dict[label] for label in labels]

# Perform label encoding
label_encoder = LabelEncoder()
encoded_labels = label_encoder.fit_transform(labels)

# Split the data
train_texts, test_texts, train_labels, test_labels = train_test_split(texts, labels, test_size=0.2, random_state=42)

# Tokenize the data
tokenizer = DistilBertTokenizer.from_pretrained('distilbert-base-uncased')

train_encodings = tokenizer(train_texts, truncation=True, padding=True, max_length=512)
test_encodings = tokenizer(test_texts, truncation=True, padding=True, max_length=512)

# Convert to TensorFlow datasets
train_dataset = tf.data.Dataset.from_tensor_slices((
    dict(train_encodings),
    train_labels
))
test_dataset = tf.data.Dataset.from_tensor_slices((
    dict(test_encodings),
    test_labels
))

# Define and compile the model
model = TFDistilBertForSequenceClassification.from_pretrained('distilbert-base-uncased', num_labels=len(label_dict))

optimizer = tf.keras.optimizers.Adam(learning_rate=5e-5)
loss = tf.keras.losses.SparseCategoricalCrossentropy(from_logits=True)
metric = tf.keras.metrics.SparseCategoricalAccuracy('accuracy')

model.compile(optimizer=optimizer, loss=loss, metrics=[metric])

# Train the model
history = model.fit(
    train_dataset.shuffle(1000).batch(16),
    epochs=10,
    batch_size=16,
    validation_data=test_dataset.batch(16)
)

# Evaluate the model
predictions = model.predict(test_dataset.batch(16))
pred_labels = np.argmax(predictions.logits, axis=1)

print(classification_report(test_labels, pred_labels, target_names=list(label_dict.keys())))

# Save the model
model.save_pretrained('./my_distilbert_model')
tokenizer.save_pretrained('./my_distilbert_model')
```

Figure 4.4. Code for DistilBERT Classification Model

```

# Initialize tokenizer and model
tokenizer = RobertaTokenizer.from_pretrained('roberta-base')
num_labels = len(label_encoder.classes_)
model = TFRobertaForSequenceClassification.from_pretrained('roberta-base', num_labels=num_labels)

# Prepare datasets
def encode_batch(texts, labels):
    texts = texts.numpy().decode('utf-8')
    encoded = tokenizer(texts, padding='max_length', truncation=True, max_length=256, return_tensors='tf')
    return encoded['input_ids'][0], encoded['attention_mask'][0], tf.cast(labels, tf.int32)

train_dataset = tf.data.Dataset.from_tensor_slices((train_texts, train_labels))
val_dataset = tf.data.Dataset.from_tensor_slices((val_texts, val_labels))

train_dataset = train_dataset.map(lambda x, y: tf.py_function(
    func=encode_batch, inp=[x, y], Tout=[tf.int32, tf.int32, tf.int32]
))
val_dataset = val_dataset.map(lambda x, y: tf.py_function(
    func=encode_batch, inp=[x, y], Tout=[tf.int32, tf.int32, tf.int32]
))

# Set the shapes of the tensors
def set_shapes(input_ids, attention_mask, labels):
    input_ids.set_shape([256])
    attention_mask.set_shape([256])
    labels.set_shape([])
    return ('input_ids': input_ids, 'attention_mask': attention_mask), labels

train_dataset = train_dataset.map(set_shapes)
val_dataset = val_dataset.map(set_shapes)

# Prepare TF datasets
batch_size = 8
train_dataset = train_dataset.shuffle(1000).batch(batch_size).prefetch(tf.data.AUTOTUNE)
val_dataset = val_dataset.batch(batch_size).prefetch(tf.data.AUTOTUNE)

# Define the model
num_train_steps = len(train_texts) // batch_size * 3
warmup_steps = int(0.1 * num_train_steps)

initial_learning_rate = 2e-5
lr_schedule = tf.keras.optimizers.schedules.PolynomialDecay(
    initial_learning_rate,
    decay_steps=num_train_steps,
    end_learning_rate=0,
    power=1.0,
    cycle=False
)

optimizer = tf.keras.optimizers.Adam(learning_rate=lr_schedule)

loss = tf.keras.losses.SparseCategoricalCrossentropy(from_logits=True)
model.compile(optimizer=optimizer, loss=loss, metrics=['accuracy'])

# Train the model
history = model.fit(
    train_dataset,
    validation_data=val_dataset,
    epochs=10
)

# Evaluate the model
def compute_metrics(y_true, y_pred):
    y_pred = np.argmax(y_pred, axis=1)
    return classification_report(y_true, y_pred, target_names=label_encoder.classes_, output_dict=True)

val_predictions = model.predict(val_dataset)['logits']
metrics = compute_metrics(val_labels, val_predictions)

print("Classification Report:")
print(classification_report(val_labels, np.argmax(val_predictions, axis=1), target_names=label_encoder.classes_))

```

Figure 4.5. Code for RoBERTa Classification Model

Subsequently, we evaluated the model with the highest accuracy on a large dataset (5,000 samples) and manually checked the accuracy of 296 samples from this test set. To prepare the dataset, we began by defining the parent directory containing the raw data files and specifying the months of interest. We then iterated through the directories for May, June, and July 2022, loading and combining the CSV files into a single DataFrame. After combining the data, we filtered out entries with missing or insufficient content, retaining only those with non-null 'selftext' values of 150 characters or more. Finally, to ensure a manageable yet representative dataset, we randomly sampled 5,000 rows from the filtered data for further analysis. The whole process is presented in Figure 4.6 below.

```
# Define the parent directory containing the data
base_dir = '/content/drive/My Drive/Thesis 2024/Original Reddit Data/Raw Data/2022'

# List of months to include
months = ['May 22', 'June 22', 'Jul 22']

raw_dfs = []

# Iterate over the specified months
for month in months:
    raw_data_dir = os.path.join(base_dir, month)

    # Iterate over the files in the current month's directory
    for file in os.listdir(raw_data_dir):
        if file.endswith('.csv'):
            file_path = os.path.join(raw_data_dir, file)

            # Read the CSV file into a DataFrame and append to the list
            raw_df = pd.read_csv(file_path)
            raw_dfs.append(raw_df)

# Concatenate all DataFrames in the list into one
combined_raw_df = pd.concat(raw_dfs, ignore_index=True)

# Print the combined DataFrame
combined_raw_df.head()

# Filter out rows where selftext is '[removed]' or has fewer than 20 characters
filtered_df = combined_raw_df[
    ...combined_raw_df['selftext'].notna() &
    ... (combined_raw_df['selftext'] != '[removed']) &
    ... (combined_raw_df['selftext'].str.len() >= 150)
]

# Display the first few rows of the filtered DataFrame
print(filtered_df.head())

# Take a random sample of 5,000 rows
sampled_df = filtered_df.sample(n=5000, random_state=42)
```

Figure 4.6. Code for preparing dataset with 5000 samples from original dataset

The following code in Figure 4.7 provides an outline of the procedure for generating predictions using a loop of seven models, each corresponding to one of the seven components. Each model processes the dataset to predict its respective component. The predictions are then collected and organized into columns, with each column representing the predicted outcomes for one of the components. This structured format allows for straightforward integration and analysis of the model outputs.

```
import tensorflow as tf
from transformers import DistilBertTokenizer, TFDistilBertForSequenceClassification
import pandas as pd

# Define the path to the models and their label mappings
models_info = {
    'rootcause_distilbert_model': {0: "Early life", 1: "Personality", 2: "Trauma and Stress", 3: "Drug and Alcohol"},
    'sentiment_distilbert_model': {0: "Neutral", 1: "Negative", 2: "Positive"},
    'perceived_severity_distilbert_model': {0: "Others", 1: "Depression", 2: "Anxiety", 3: "Loneliness"},
    'perceived_benefits_distilbert_model': {0: "Not Mentioned", 1: "Getting access to treatment", 2: "Finding support", 3: "Others"},
    'perceived_barriers_distilbert_model': {0: "Others", 1: "Feeling unheard and misunderstood", 2: "Feeling hopeless", 3: "Feeling helpless", 4: "Not Mentioned"},
    'cutoaction_distilbert_model': {0: "Looking for information and resources", 1: "Seeking support and treatment", 2: "Sharing their situation", 3: "Others", 4: "No Action"},
    'selfefficacy_distilbert_model': {0: "Denial Mindset", 1: "Empowered Mindset", 2: "Overcome Mindset", 3: "Troubled Mindset"},
}

# Define a function to load model and tokenizer, and predict labels
def load_model_and_predict(model_name, label_map, text):
    # Load tokenizer and model
    tokenizer = DistilBertTokenizer.from_pretrained(model_name)
    model = TFDistilBertForSequenceClassification.from_pretrained(model_name)

    # Tokenize input text
    inputs = tokenizer(text, return_tensors='tf', truncation=True, padding=True, max_length=512)

    # Get model predictions
    outputs = model(inputs)
    logits = outputs.logits
    predicted_label = tf.argmax(logits, axis=-1).numpy()[0]

    # Map numerical label to text label
    return label_map[predicted_label]

# Apply predictions for each model and add results to the DataFrame
for model_name, label_map in models_info.items():
    column_name = f"predicted_{model_name.split('_distilbert_model')[0]}" # Create a column name from the model name
    final_df[column_name] = final_df['preprocessed_selftext'].apply(
        lambda text: load_model_and_predict(model_name, label_map, text)
    )

# Print the DataFrame with the new columns
print(final_df.head())
```

Figure 4.7. Code for predicting dataset with 5000 samples from original dataset

4.4.3 Text Generation Model

Initially, we tested the models on a dataset of 40 rows, refining the prompt to enhance the models' contextual understanding. The first prompt was designed for a more open-ended analysis, where the GPT model generated responses freely based on the Health Belief Model. The categories are broadly defined, allowing the model to interpret and categorize the input text with a degree of flexibility. This approach led to less predictable labeling, as the model was not provided with specific examples or detailed guidance for each category. The output was concise, limited to 5 words per category, but since the model has more freedom in its analysis, the labels varied widely depending on the text's content.

```
system_prompt = """You are an experienced psychiatrist. Analyze the following
text based on the Health Belief Model. Categorize it into these 7 categories:
    1. Sentiment: Overall sentiment (Positive, Neutral, or Negative)
    2. Perceived severity: Health issue or concern perceived as severe
    3. Perceived benefits: Benefits perceived in taking health-seeking action
    4. Perceived barriers: Barriers perceived in taking health-seeking action
    5. Cue to Action: Prompts or reasons for taking steps to improve mental
well-being
    6. Self-efficacy: Mindset (empowered, overcome, denial, trouble)
    7. Root cause: Drug and Alcohol, Trauma and Stress, Personality, Early Life

Respond with a JSON object containing these 7 categories as keys and your
analysis as values. Keep each value concise, preferably not more than 5
words."""
```

The second prompt provided a more controlled and structured approach to analyzing the text. It included specific examples and more detailed descriptions for each category within the Health Belief Model. This guidance helped the model generate more consistent and focused labels, reducing the variability seen in the first prompt. The inclusion of specific terms and scenarios for each category made the output more predictable and aligned with the desired analysis. As with the first prompt, responses were kept concise, not exceeding 5 words per category, but the model was likely to produce more accurate and relevant labels due to the clearer instructions.

```
system_prompt = """You are an experienced psychiatrist. Analyze the following text based on the Health Belief Model. Categorize it into these 7 categories:
```

```
    Root cause: Root cause of mental health problems (Drug and Alcohol, Trauma and Stress, Personality, Early Life)
```

```
    Sentiment: Overall sentiment (Positive, Neutral, or Negative)
```

```
    Perceived severity: Health issue or concern perceived as severe (Depression, Anxiety, Suicide Attempt, Hallucination, etc.)
```

```
    Perceived benefits: Benefits perceived in taking health-seeking action (Not Mentioned, Finding support, Feeling heard and understood, Getting access to treatment, etc.)
```

```
    Perceived barriers: Barriers perceived in taking health-seeking action (Not Mentioned, Feeling helpless, Feeling unheard and misunderstood, Lack of resources, etc.)
```

```
    Cue to Action: Actions or reasons for taking steps to improve mental well-being (No Action, Sharing their situation, Seeking for support and treatment, Looking for resources, etc.)
```

```
    Self-efficacy: Mindset (Empowered, Overcome, Denial, Trouble)
```

```
Respond with a valid JSON object containing these 7 categories as keys and your analysis as values. Keep each value concise, preferably not more than 5 words."""
```

Once optimized, we applied the updated prompt to a larger dataset of 296 rows and compared the results across different models: LLAMA3, OpenAI GPT-3.5, OpenAI GPT-4, and OpenAI GPT-4 mini, as illustrated in Figure 4.8 below.


```

# !pip install together

import os
from together import Together
import pandas as pd
from tqdm import tqdm

# Replace 'your_api_key_here' with your actual Together API key
client = Together(api_key=together_api_key)

def analyze_text(text):
    prompt = f"""You are an experienced psychiatrist. Analyze the following text based on the Health Belief Model. Categorize it into these 7 categories:
    1. Sentiment: Overall sentiment (Positive, Neutral, or Negative)
    2. Perceived severity: Health issue or concern perceived as severe
    3. Perceived benefits: Benefits perceived in taking health-seeking action
    4. Perceived barriers: Barriers perceived in taking health-seeking action
    5. Cue to Action: Prompts or reasons for taking steps to improve mental well-being
    6. Self-efficacy: Mindset (empowered, overcome, denial, trouble)
    7. Root cause: Drug and Alcohol, Trauma and Stress, Personality, Early Life

    Respond with a JSON object containing these 7 categories as keys and your analysis as values. Keep each value concise, preferably not more than 5 words."""

    response = client.chat.completions.create(
        model="meta-llama/Llama-3-70b-chat-hf",
        messages=[
            {"role": "system", "content": "system_prompt"},
            {"role": "user", "content": text}],
    )
    return response.choices[0].message.content

# Assuming sampled_df is your DataFrame and 'selftext' is the column with the text to analyze
texts = sampled_df['selftext']

# Process the DataFrame
results = []

for text in tqdm(texts, desc="Processing texts"):
    results.append(analyze_text(text))

# Add the raw results to the DataFrame
sampled_df['analysis'] = results

# Save the updated DataFrame
sampled_df.to_csv('analyzed_data_llama3.csv', index=False)

print("Analysis complete. Results saved to 'analyzed_data_llama3.csv'")

```

```

# !pip install openai
from openai import OpenAI
import time
from tqdm import tqdm

client = OpenAI(api_key=openai_key)

def analyze_text(text):
    prompt = """You are an experienced psychiatrist. Analyze the following text based on the Health Belief Model. Categorize it into these 7 categories:

    Root cause: Root cause of mental health problems (Drug and Alcohol, Trauma and Stress, Personality, Early Life)
    Sentiment: Overall sentiment (Positive, Neutral, or Negative)
    Perceived severity: Health issue or concern perceived as severe (Depression, Anxiety, Suicide Attempt, Hallucination, etc.)
    Perceived benefits: Benefits perceived in taking health-seeking action (Not Mentioned, Finding support, Feeling heard and understood, Getting access to treatment, etc.)
    Perceived barriers: Barriers perceived in taking health-seeking action (Not Mentioned, Feeling helpless, Feeling unheard and misunderstood, Lack of resources, etc.)
    Cue to Action: Actions or reasons for taking steps to improve mental well-being (No Action, Sharing their situation, Seeking for support and treatment, Looking for resources, etc.)
    Self-efficacy: Mindset (Empowered, Overcome, Denial, Trouble)

    Respond with a valid JSON object containing these 7 categories as keys and your analysis as values. Keep each value concise, preferably not more than 5 words."""

    try:
        response = client.chat.completions.create(
            model="gpt-4o",
            messages=[
                {"role": "system", "content": prompt},
                {"role": "user", "content": text}
            ],
            max_tokens=200
        )
        return response.choices[0].message.content
    except Exception as e:
        print(f"Error processing text: {text[:50]}...\nError: {e}")
        return None

texts = test_df['selftext']

# Process the DataFrame
batch_size = 50
results = []

for i in tqdm(range(0, len(texts), batch_size), desc="Processing texts"):
    batch_texts = texts[i:i+batch_size]
    batch_results = [analyze_text(text) for text in batch_texts]
    results.extend(batch_results)
    time.sleep(1) # To avoid hitting rate limits

# Add the raw results to the DataFrame
test_df['analysis'] = results

# Save the updated DataFrame
test_df.to_csv('ssr_gpt4o_analyzed_data.csv', index=False)

```

Figure 4.8. Code for LLAMA 3 and GPT-4o Models

Furthermore, we fine-tuned GPT-4o and GPT-4o mini using 800 rows of training data, carefully adjusting the model to enhance its accuracy, as shown in Figure 4.9. To accomplish this, we first needed to nest seven labeled columns into a JSON object, which served as the model's response for training purposes. We then created a JSON file specifically for fine-tuning the model, as presented in Figure 4.10.

```
[ ] train_df['selftext'] = train_df['selftext'].astype(str)

# Create json object as response
train_df['json_object'] = train_df.apply(lambda row: row[['Root cause', 'Sentiment', 'Perceived severity', 'Perceived benefits', 'Perceived barriers', 'Cue to Action', 'Self-efficacy']].to_dict(), axis=1)
train_df['json_string'] = train_df['json_object'].apply(lambda x: str(x))

# Define system prompt
system_prompt = """Analyze the following text based on the Health Belief Model. Categorize it into these 7 categories:

Root cause: Root cause of mental health problems (Drug and Alcohol, Trauma and Stress, Personality, Early Life)
Sentiment: Overall sentiment (Positive, Neutral, or Negative)
Perceived severity: Health issue or concern perceived as severe (Depression, Anxiety, Suicide Attempt, Hallucination, etc.)
Perceived benefits: Benefits perceived in taking health-seeking action (Not Mentioned, Finding support, Feeling heard and understood, Getting access to treatment, etc.)
Perceived barriers: Barriers perceived in taking health-seeking action (Not Mentioned, Feeling helpless, Feeling unheard and misunderstood, Lack of resources, etc.)
Cue to Action: Actions or reasons for taking steps to improve mental well-being (No Action, Sharing their situation, Seeking for support and treatment, Looking for resources, etc.)
Self-efficacy: Mindset (Empowered, Overcome, Denial, Trouble)

Respond with a valid JSON object containing these 7 categories as keys and your analysis as values. Keep each value concise, preferably not more than 5 words."""

# Mapping to fine-tuning format
train_df['fine_tune_data'] = train_df.apply(
    lambda row: {
        'messages': [
            {"role": "system", "content": system_prompt},
            {"role": "user", "content": row['selftext']},
            {"role": "assistant", "content": row['json_string']}
        ]
    }, axis=1
)

# Display the DataFrame with fine-tune data
print(train_df[['selftext', 'fine_tune_data']])
```

Figure 4.9. Code for preparing data to train GPT-4o and GPT-4o mini Fine-Tuning Models

```
() fine_tune_data (1).jsonl ×
Users > admin > Desktop > () fine_tune_data (1).jsonl
1 {"messages": [{"role": "system", "content": "Analyze the following text based on the Health Belief Model. Categorize it into these 7 categories:\n\n Root cause: Root cause of
2 {"messages": [{"role": "system", "content": "Analyze the following text based on the Health Belief Model. Categorize it into these 7 categories:\n\n Root cause: Root cause of
3 {"messages": [{"role": "system", "content": "Analyze the following text based on the Health Belief Model. Categorize it into these 7 categories:\n\n Root cause: Root cause of
4 {"messages": [{"role": "system", "content": "Analyze the following text based on the Health Belief Model. Categorize it into these 7 categories:\n\n Root cause: Root cause of
5 {"messages": [{"role": "system", "content": "Analyze the following text based on the Health Belief Model. Categorize it into these 7 categories:\n\n Root cause: Root cause of
6 {"messages": [{"role": "system", "content": "Analyze the following text based on the Health Belief Model. Categorize it into these 7 categories:\n\n Root cause: Root cause of
7 {"messages": [{"role": "system", "content": "Analyze the following text based on the Health Belief Model. Categorize it into these 7 categories:\n\n Root cause: Root cause of
8 {"messages": [{"role": "system", "content": "Analyze the following text based on the Health Belief Model. Categorize it into these 7 categories:\n\n Root cause: Root cause of
9 {"messages": [{"role": "system", "content": "Analyze the following text based on the Health Belief Model. Categorize it into these 7 categories:\n\n Root cause: Root cause of
10 {"messages": [{"role": "system", "content": "Analyze the following text based on the Health Belief Model. Categorize it into these 7 categories:\n\n Root cause: Root cause of
11 {"messages": [{"role": "system", "content": "Analyze the following text based on the Health Belief Model. Categorize it into these 7 categories:\n\n Root cause: Root cause of
12 {"messages": [{"role": "system", "content": "Analyze the following text based on the Health Belief Model. Categorize it into these 7 categories:\n\n Root cause: Root cause of
13 {"messages": [{"role": "system", "content": "Analyze the following text based on the Health Belief Model. Categorize it into these 7 categories:\n\n Root cause: Root cause of
14 {"messages": [{"role": "system", "content": "Analyze the following text based on the Health Belief Model. Categorize it into these 7 categories:\n\n Root cause: Root cause of
15 {"messages": [{"role": "system", "content": "Analyze the following text based on the Health Belief Model. Categorize it into these 7 categories:\n\n Root cause: Root cause of
16 {"messages": [{"role": "system", "content": "Analyze the following text based on the Health Belief Model. Categorize it into these 7 categories:\n\n Root cause: Root cause of
17 {"messages": [{"role": "system", "content": "Analyze the following text based on the Health Belief Model. Categorize it into these 7 categories:\n\n Root cause: Root cause of
18 {"messages": [{"role": "system", "content": "Analyze the following text based on the Health Belief Model. Categorize it into these 7 categories:\n\n Root cause: Root cause of
19 {"messages": [{"role": "system", "content": "Analyze the following text based on the Health Belief Model. Categorize it into these 7 categories:\n\n Root cause: Root cause of
20 {"messages": [{"role": "system", "content": "Analyze the following text based on the Health Belief Model. Categorize it into these 7 categories:\n\n Root cause: Root cause of
21 {"messages": [{"role": "system", "content": "Analyze the following text based on the Health Belief Model. Categorize it into these 7 categories:\n\n Root cause: Root cause of
22 {"messages": [{"role": "system", "content": "Analyze the following text based on the Health Belief Model. Categorize it into these 7 categories:\n\n Root cause: Root cause of
23 {"messages": [{"role": "system", "content": "Analyze the following text based on the Health Belief Model. Categorize it into these 7 categories:\n\n Root cause: Root cause of
24 {"messages": [{"role": "system", "content": "Analyze the following text based on the Health Belief Model. Categorize it into these 7 categories:\n\n Root cause: Root cause of
25 {"messages": [{"role": "system", "content": "Analyze the following text based on the Health Belief Model. Categorize it into these 7 categories:\n\n Root cause: Root cause of
26 {"messages": [{"role": "system", "content": "Analyze the following text based on the Health Belief Model. Categorize it into these 7 categories:\n\n Root cause: Root cause of
27 {"messages": [{"role": "system", "content": "Analyze the following text based on the Health Belief Model. Categorize it into these 7 categories:\n\n Root cause: Root cause of
28 {"messages": [{"role": "system", "content": "Analyze the following text based on the Health Belief Model. Categorize it into these 7 categories:\n\n Root cause: Root cause of
29 {"messages": [{"role": "system", "content": "Analyze the following text based on the Health Belief Model. Categorize it into these 7 categories:\n\n Root cause: Root cause of
30 {"messages": [{"role": "system", "content": "Analyze the following text based on the Health Belief Model. Categorize it into these 7 categories:\n\n Root cause: Root cause of
```

Figure 4.10. JSON File for fine tuning GPT-4o and GPT-4o mini Models

Following this preparation, we trained the model and conducted predictions on a test dataset consisting of 296 rows to evaluate its performance, as shown in Figure 4.11 and 4.12.

```
[ ] # Convert fine-tune data to a list of dictionaries
    fine_tune_list = train_df['fine_tune_data'].tolist()

    # Export to JSONL
    import json

    with open('fine_tune_data.jsonl', 'w') as outfile:
        for entry in fine_tune_list:
            json.dump(entry, outfile)
            outfile.write('\n')
```

```
▶ from openai import OpenAI

    # Set the API key and model name
    client = OpenAI(api_key=openAI_key)

    vFile=client.files.create(
        file=open("fine_tune_data.jsonl", "rb"),
        purpose="fine-tune"
    )

    vJob=client.fine_tuning.jobs.create(
        training_file=vFile.id,
        model="gpt-4o-mini-2024-07-18"
    )
```

```
[ ] print(client.fine_tuning.jobs.list(limit=10))
    print(client.fine_tuning.jobs.retrieve(vJob.id))
    print(client.fine_tuning.jobs.list_events(fine_tuning_job_id=vJob.id, limit=10))
```

➡ Show hidden output

```
[ ] client.fine_tuning.jobs.retrieve("ftjob-t9U9EPx8gPOHtCe3q4uckIC8")
```

➡ Show hidden output

```
[ ] client.fine_tuning.jobs.list_events(fine_tuning_job_id="ftjob-t9U9EPx8gPOHtCe3q4uckIC8", limit=10)
```

Figure 4.11. Code for fine tuning GPT-4o and GPT-4o mini Models

4.5 Data Analysis

This section outlines the methodologies used for analyzing the dataset. It includes an examination of Health Belief Model components and a detailed assessment of emotional content, using various analytical techniques to derive meaningful insights from the data.

4.5.1 Health Belief Model Analysis

The Health Belief Model analysis begins with an exploration of Perceived Severity and its root causes (Figure 4.13), followed by an investigation into Perceived Susceptibility and Self-Efficacy (Figure 4.14). The analysis then covers Perceived Benefits, Perceived Barriers, and Cues to Action, offering insights into how these elements influence health behaviors. Subsequently, we mapped related conditions to create more manageable categories: alcohol and drug addiction were grouped together, and trauma and stress were combined, using a threshold of 59, as presented in Figure 4.15. This grouping resulted in eight distinct categories, facilitating a more nuanced analysis.

```
[ ] import pandas as pd
    from wordcloud import WordCloud
    import matplotlib.pyplot as plt
    from collections import Counter

    # Split the words in 'perceived_severity' column by commas
    final_labeled_df['perceived_severity_split'] = final_labeled_df['predicted_perceived_severity'].apply(
        lambda x: [word.strip() for word in x.split(',') if isinstance(x, str)] if isinstance(x, str) else []
    )

    # Flatten the list of words
    all_words = [word for sublist in final_labeled_df['perceived_severity_split'] for word in sublist]

    # Count the occurrences of each word
    word_counts = Counter(all_words)

    print(word_counts)

Counter({'Depression': 1471, 'Anxiety': 1414, 'Suicidal Thoughts': 1123, 'Loneliness': 799, 'Panic Attacks': 150, 'Trauma': 133, 'Bipolar': 106, 'Stress':

# Generate a word cloud with adjusted parameters
wordcloud = WordCloud(
    width=800,
    height=400,
    background_color='black',
    max_font_size=100,      # Set maximum font size
    min_font_size=3,       # Set minimum font size
    relative_scaling=0.5,  # Adjust scaling to give more prominence to less frequent words
    max_words=200,
    normalize_plurals=False, # Avoid combining singular and plural forms
    colormap='PuBuGn_r'
).generate_from_frequencies(word_counts)

# Plot the word cloud
plt.figure(figsize=(13, 8))
plt.imshow(wordcloud, interpolation='bilinear')
plt.axis('off')
plt.show()
```

Figure 4.13. Code for Wordcloud presented Perceived Severity

```

▶ # Create a pivot table
pivot_table = pd.pivot_table(
    final_labeled_df,
    index='predicted_selfefficacy', # Rows
    columns='predicted_sentiment', # Columns
    aggfunc='size', # Count occurrences
    fill_value=0 # Replace NaN with 0 for missing values
)

# Display the pivot table
print(pivot_table)

```

→ Show hidden output

```

▶ import seaborn as sns
# Plotting the heatmap
plt.figure(figsize=(10, 6)) # Adjust the size as needed
sns.heatmap(
    pivot_table,
    annot=True, # Annotate each cell with the numeric value
    cmap='YlGnBu', # Color map
    linewidths=.5, # Line width between cells
    fmt='d' # Format for the annotation (integer)
)
plt.title('Heatmap of Counts by Predicted Self-Efficacy and Sentiment')
plt.xlabel('Predicted Sentiment')
plt.ylabel('Predicted Self-Efficacy')
plt.show()

```

Figure 4.14. Code for examining relationship between Perceived Susceptibility and Self-Efficacy

```

# Define the threshold
threshold = 59

# Filter severities with count <= 500 and group them as 'Others'
filtered_severity_counts = {severity: count for severity, count in severity_counts.items() if count > threshold}
other_count = sum(count for severity, count in severity_counts.items() if count <= threshold)
if other_count > 0:
    filtered_severity_counts['Others'] = other_count

print("Filtered Severity Counts:")
print(filtered_severity_counts)

# Map severities to 'Others' if their count is below the threshold
def map_severities(severity):
    return severity if filtered_severity_counts.get(severity, 0) > threshold else 'Others'

final_labeled_df['predicted_perceived_severity_mapped'] = final_labeled_df['predicted_perceived_severity_combined'].apply(map_severities)

# Debug: Check the mapping result
print("Mapped Perceived Severity:")
print(final_labeled_df[['predicted_perceived_severity_combined', 'predicted_perceived_severity_mapped']].head())

```

Figure 4.15. Code for mapping Perceived Severity

The final aspect of this analysis reviews the HBM components across major mental health issues, providing a broader understanding of their application and impact, as shown in Figure 4.16.

```
import matplotlib.pyplot as plt

# Define perceived_severity_levels
perceived_severity_levels = ["Trauma or Stress", "Alcoholism or Drug Addiction", "Others"]

# Get all unique values across the columns for consistent coloring
unique_values = pd.concat([
    final_labeled_df['predicted_sentiment'],
    final_labeled_df['predicted_perceived_benefits'],
    final_labeled_df['predicted_perceived_barriers'],
    final_labeled_df['predicted_cueaction'],
    final_labeled_df['predicted_selfefficacy'],
    final_labeled_df['predicted_rootcause'],
]).unique()

# Helper function to create a donut chart
def plot_donut_chart(data, column_name, ax, title):
    # Count the occurrences of each value in the column
    value_counts = data[column_name].value_counts()

    # Plot donut chart
    wedges, texts, autotexts = ax.pie(
        value_counts,
        autopct='%1.1f%%',
        startangle=140,
        wedgeprops=dict(width=0.45), # Width of the donut
        pctdistance=1.12
    )

    ax.set_title(title, fontsize=11)

    # Add a legend below the chart, with more space for the chart
    ax.legend(
        wedges, # Use the wedge objects for legend
        labels=value_counts.index, # Labels for the legend
        loc='center',
        bbox_to_anchor=(1.05, 0), # Position legend closer to the chart
        fontsize='small', # Small font size for the legend
        title_fontsize='small'
    )

# Set up the plotting area
fig, axs = plt.subplots(
    len(perceived_severity_levels) * 2, # Two rows for each severity level
    3, # Three columns (plus an extra column for root cause)
    figsize=(20, 10 * len(perceived_severity_levels))
)

# Loop through each self-efficacy level and plot charts
for i, level in enumerate(perceived_severity_levels):
    print(level)
    # Filter DataFrame based on the current self-efficacy level
    filtered_df = final_labeled_df[final_labeled_df['predicted_perceived_severity_mapped'] == level]

    # Plot each donut chart for the current self-efficacy
    plot_donut_chart(filtered_df, 'predicted_rootcause', axs[2*i, 0], 'Root Cause')
    plot_donut_chart(filtered_df, 'predicted_sentiment', axs[2*i, 1], 'Sentiment')
    plot_donut_chart(filtered_df, 'predicted_perceived_benefits', axs[2*i, 2], 'Perceived Benefits')
    plot_donut_chart(filtered_df, 'predicted_perceived_barriers', axs[2*i+1, 0], 'Perceived Barriers')
    plot_donut_chart(filtered_df, 'predicted_cueaction', axs[2*i+1, 1], 'Cue to Action')
    plot_donut_chart(filtered_df, 'predicted_selfefficacy', axs[2*i+1, 2], 'Self-Efficacy')

# Adjust layout
plt.tight_layout()
plt.show()
```

Figure 4.16. Code for HBM Components Analysis Across Major Mental Health Issues

4.5.2 Emotional Content Analysis

In terms of emotional content, the analysis first presents Word Sentiment Counts by perceived severity groups, offering a quantitative view of emotional tones, as depicted in Figure 4.17. It then identifies the Top Negative Words within these severity groups, highlighting specific terms associated with negative sentiments, as shown in Figure 4.18.

```
# Initialize SentimentIntensityAnalyzer
sia = SentimentIntensityAnalyzer()

# Function to classify words based on sentiment
def classify_word(word):
    sentiment = sia.polarity_scores(word)['compound']
    if sentiment > 0.1:
        return 'Positive'
    elif sentiment < -0.1:
        return 'Negative'
    else:
        return 'Neutral'

# Classify each word in the 'selftext' column
final_labeled_df['selftext_split'] = final_labeled_df['selftext'].apply(
    lambda x: [word.strip() for word in x.split() if word.strip()]
)

final_labeled_df['selftext_classified'] = final_labeled_df['selftext_split'].apply(
    lambda words: [classify_word(word) for word in words]
)

# Step 3: Count positive, negative, and neutral words for each mapped severity
def count_by_category(group_df):
    classified_words = [word for sublist in group_df['selftext_classified'] for word in sublist]

    counts = Counter(classified_words)
    return pd.Series({
        'Positive': counts.get('Positive', 0),
        'Negative': counts.get('Negative', 0),
        'Neutral': counts.get('Neutral', 0),
    })

# Apply the counting function and aggregate by mapped perceived severity group
severity_group_counts = final_labeled_df.groupby('predicted_perceived_severity_mapped').apply(count_by_category)

print("Severity Group Counts:")
print(severity_group_counts)

[ ] import seaborn as sns
import matplotlib.pyplot as plt

# Convert counts to percentages by row
severity_group_percentages = severity_group_counts.div(severity_group_counts.sum(axis=1), axis=0) * 100

# Plotting the heatmap with percentages
plt.figure(figsize=(10, 6)) # Adjust the size as needed
sns.heatmap(
    severity_group_percentages,
    annot=True, # Annotate each cell with the percentage value
    cmap='OrRd', # Color map
    linewidths=.5, # Line width between cells
    fmt='.2f', # Format for the annotation (2 decimal places)
    cbar_kws={'label': 'Percentage (%)'} # Color bar label
)
plt.title('Heatmap of Percentages by Perceived Severity and Sentiment')
plt.xlabel('Sentiment')
plt.ylabel('Perceived Severity')
plt.show()
```

Figure 4.17. Code for presenting presents Word Sentiment Counts by perceived severity groups


```

# Initialize SentimentIntensityAnalyzer and WordNetLemmatizer
sia = SentimentIntensityAnalyzer()
lemmatizer = WordNetLemmatizer()

# Function to get the WordNet POS tag
def get_wordnet_pos(treebank_tag):
    if treebank_tag.startswith('J'):
        return wordnet.ADJ
    elif treebank_tag.startswith('V'):
        return wordnet.VERB
    elif treebank_tag.startswith('N'):
        return wordnet.NOUN
    elif treebank_tag.startswith('R'):
        return wordnet.ADV
    else:
        return wordnet.NOUN

def normalize_words(words):
    """Lemmatize words and remove punctuation."""
    normalized_words = []
    pos_tags = nltk.pos_tag(words)

    # Process each word based on its POS tag
    for word, tag in pos_tags:
        wn_tag = get_wordnet_pos(tag)
        lemmatized_word = lemmatizer.lemmatize(word, wn_tag).lower()
        normalized_words.append(lemmatized_word)

    # Remove punctuation
    normalized_words = [word for word in normalized_words if word.isalpha()]

    return normalized_words

# Function to classify words based on sentiment
def classify_word(word):
    sentiment = sia.polarity_scores(word)['compound']
    if sentiment > 0.1:
        return 'Positive'
    elif sentiment < -0.1:
        return 'Negative'
    else:
        return 'Neutral'

# Function to classify and count words in 'selftext'
def classify_and_count_words(df):
    df['selftext_split'] = df['selftext'].apply(lambda text: text.split())
    df['selftext_normalized'] = df['selftext_split'].apply(lambda words: normalize_words(words))
    df['selftext_classified'] = df['selftext_normalized'].apply(
        lambda words: [classify_word(word) for word in words]
    )
    return df

# Apply classification to the dataframe
classified_df = classify_and_count_words(final_labeled_df)

# Function to get top N words based on sentiment for each severity group
def get_top_words(group_df, sentiment_label, top_n=10):
    # Extract all words classified with the specified sentiment label
    words = [word for sublist in group_df['selftext_classified'] for word, label in zip(group_df['selftext_normalized'].explode(), group_df['selftext_classified'].explode()) if label == sentiment_label]

    # Count occurrences of each word
    word_counts = Counter(words)

    # Return the most common words
    return word_counts.most_common(top_n)

# Define severity groups
severity_groups = final_labeled_df['predicted_perceived_severity_mapped'].unique()

# Display top 10 negative words for each group
for group in severity_groups:
    group_df = final_labeled_df[final_labeled_df['predicted_perceived_severity_mapped'] == group]
    top_negative_words = get_top_words(group_df, sentiment_label='Negative', top_n=10)
    print(f"Top 10 Negative Words for {group}:")
    for word, count in top_negative_words:
        print(f"{word}: {count}")
    print("\n")

```

Figure 4.18. Code for presenting presents Top Negative Words by perceived severity groups

4.6 Summary

In summary, Chapter 4 offers a comprehensive overview of the implementation process for analyzing health-related data, detailing the methodologies and tools employed. It starts with an overview of the different tools and computational resources used, including GPU-enabled cloud environments for handling large datasets and optimizing model performance. The chapter covers essential steps such as data preprocessing, which includes cleaning and normalization, tokenization for text preparation, and label combination and data augmentation to improve dataset effectiveness. Additionally, it addresses dataset splitting and evaluation to guarantee thorough training and testing of the models.

In data annotation, the chapter describes both human-led sample annotation and the use of automated text classification and generation models. Human annotators reviewed a subset of data to provide accurate labels, while text classification models were trained using TensorFlow and Hugging Face Transformers to predict health belief model components. Additionally, state-of-the-art text generation models, including LLAMA3 and GPT-4o, were employed to generate and structure text data for further analysis.

Finally, The data analysis section focuses on the Health Belief Model (HBM), examining components such as perceived severity, susceptibility, benefits, barriers, and cues to action across various mental health issues. Emotional content analysis was performed using sentiment counts and identification of negative words to understand the emotional tone of the data.

In the next chapter, we present the evaluation of the models and methods implemented, focusing on their effectiveness in predicting health belief model components and analyzing emotional content. This evaluation will assess performance metrics, accuracy, and overall results to determine the success of the implemented strategies.

Chapter 5. Evaluation

This chapter concentrates on evaluating the performance of the implemented models. It details the results from the model training and testing phases, including quantitative metrics such as accuracy, precision, recall, and F1-score. This section also includes a detailed analysis of the results, comparing the effectiveness of different models and techniques used. Together, this chapter provides an in-depth look at how the research methodologies were put into practice and evaluates their success in achieving the research objectives. The insights gained from the implementation and results evaluation are crucial for understanding the practical impact of the research and informing future work in this area.

5.1 Data Annotation

This section covers the data annotation process, including both manual annotation and the development of text classification and generation models.

5.1.1 Human-Led Sample Annotation

The Table 5.1 below presents the initial results of a comparison between the annotations provided by two annotators.

Root Cause	Number of data rows	Perceived susceptibility (Sentiment)	Perceived severity	Perceived benefits	Perceived barriers	Cue to Action	Self-efficacy
Drug and Alcohol	200	79%	66%	74%	79%	76%	89%
Early Life	200	88%	56%	74%	61%	78%	87%
Personality	200	78%	68%	81%	76%	75%	89%
Trauma and Stress	200	81%	63%	76%	72%	82%	92%
Total	800	82%	63%	76%	72%	78%	89%

Table 5.1. The initial results of a comparison between the annotations

As illustrated in the table, the most closely aligned results are those pertaining to perceived susceptibility (81.5%) and self-efficacy (89.25%). Following this are perceived benefits, perceived barriers, and cue to action (76.25%, 72%, and 77.75%, respectively). The lowest match is perceived severity (63.25%), which is primarily attributable to the presence of multiple labels per text. The sample labels are presented in the Table 5.2 below.

Raw Text	Label from Data Team	Label from Psychology Team
Just wondering what everyone's go to things are that work for anxiety/panic attacks. For me I usually pace around, drink green tea, ginger ale, water, sometimes i take an anti-nausea pill, sometimes a benzo if it's really bad/or it's been going on for a while, calming movies/music, doctors talking about anxiety, cutting back on coffee/weed/alcohol. How about you?	Panic Attacks	Anxiety, Panic Attacks
I don't know what to do anymore I can't sleep I can't eat so trying to find someone on here to talk too is my last hope I struggle with depression sometimes I'm sitting there crying for the whole day or night I struggle with nightmares and that comes from being molested when I was only 8 i have anxiety and panic attacks by the minute I can't even finish typing I'm sorry 😞	Depression	Depression, Anxiety, Panic Attacks

Table 5.2. The sample labels from the two annotators

Given that one individual may experience a range of mental health issues, our objective was to ensure the text was as comprehensive as possible. Consequently, we opted to include multiple labels, despite the potential challenges this might present in training a model for perceived severity classification. This is due to the fact that a large number of labels and an imbalance in the data may lead to difficulties in accurately training the model.

The final set of labels with value counts is presented in the Table 5.3 below:

Column	Label	Value Count	Label	Value Count
Root Cause	Drug and Alcohol	200	Personality	200
	Early life	200	Trauma and Stress	200
Perceived susceptibility (Sentiment)	Negative	292	Positive	64
	Neutral	244		
Perceived severity	Depression	288	Anxiety, Loneliness	1
	Anxiety	164	Inflicting pain onto others	1
	Loneliness	77	PTBS, Social Anxiety	1
	Depression, Suicide Attempt	31	Depression, Family conflict	1
	Relationship Anxiety	26	Relationship Anxiety, Suicide Attempts	1
	etc.			
Perceived benefits	Not Mentioned	456	Feeling heard and understood	17
	Finding support	155	Making progress	16
	Getting access to treatment	124	Feeling relieved	11
	Improving quality of life	18	Do not want to bother others	3
Perceived barriers	Not Mentioned	443	Insecurity about seriousness	14
	Feeling unheard and misunderstood 107	107	Facing financial barriers	13
	Feeling helpless	63	Feeling stigmatized	13
	Feeling hopeless	60	Lack of support	11
	Feeling worthless	32	Lack of resources	5
	Do not know how to deal with the situation	17	Do not think they need help	3
	Do not want to bother others	17	Covid Situation	2

Cue to Action	No Action	519	Sharing their situation	65
	Looking for information and resources	105	Benefiting others	23
	Seeking support and treatment	87	Physical Activity	1
Self-efficacy	Troubled Mindset	492	Empowered Mindset	130
	Overcome Mindset	151	Denial Mindset	27

Table 5.3. The final set of labels with value counts

5.1.2 Text Classification Models

This section delves into the text classification models used in the study, outlining the process from model selection through training to evaluation. Each step is crucial in refining the models for accurate and reliable classification of health belief components within the dataset.

Model Selection

Table 5.4 below summarizes the performance metrics of the different models used in the initial testing of the study.

No.	Column to classified	Model	Data Size	Text Preprocessing	Label Combination	Data Augmentation	Max length	Length Cut	Accuracy
1	Root Cause	Distilbert	800				512	Cut ending	74%
2	Root Cause	Distilbert	800	Yes			512	Cut ending	81%
3	Root Cause	Distilbert	800	Yes			512	Cut middle	64%
4	Root Cause	Roberta	800	Yes			256	Cut ending	78%
5	Root Cause	Distilbert	1200	Yes		Yes	512	Cut ending	86%
6	Self-efficacy	Distilbert	800	Raw Text			512	Cut ending	59%
7	Self-efficacy	Distilbert	800	Yes			512	Cut ending	61%
8	Self-efficacy	Distilbert	800	Yes			512	Cut middle	62%
9	Self-efficacy	Roberta	800	Yes			256	Cut middle	64%
10	Self-efficacy	Distilbert	1912	Yes		Yes	512	Cut ending	89%

Table 5.4. The performance metrics of various models employed for the first testing

As illustrated in the table, the models exhibiting the highest degree of accuracy are DistilBERT with data preprocessing and data augmentation. The model demonstrated an accuracy of 86% in root cause classification and 89% in self-efficacy classification. Given that the root cause and self-efficacy datasets comprise only four labels per column, the step label combination was omitted.

The final flow, derived from the results of two columns of testing, was subsequently applied to the remaining five columns.

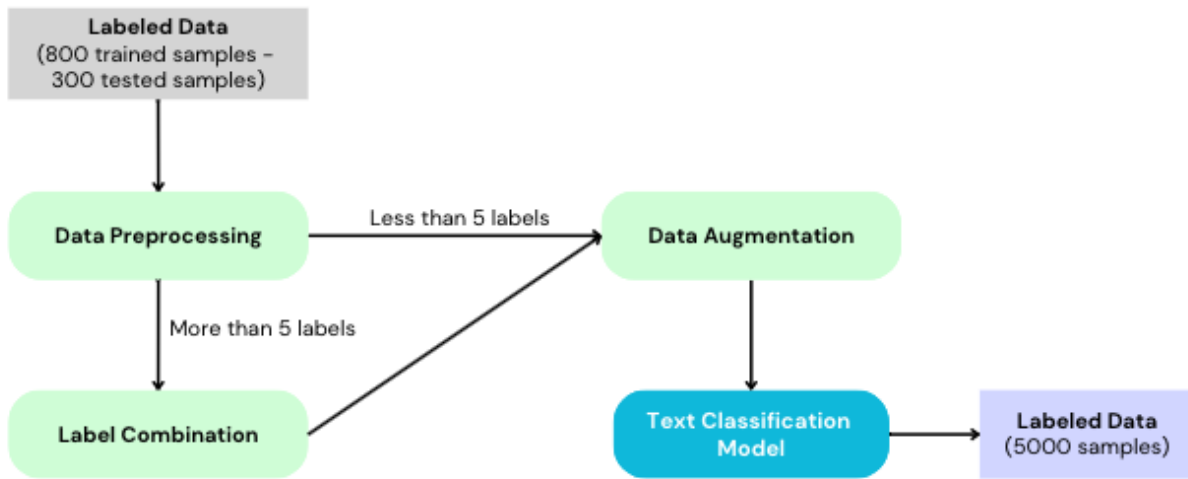


Figure 5.1. The process flow for enhancing and preparing labeled data for text classification

The diagram in Figure 5.1 illustrates the process flow for enhancing and preparing labeled data for text classification. Initially, the labeled dataset comprising 800 training samples and 300 test samples undergoes data preprocessing to clean and normalize the data. Following this, a label combination step is performed if the dataset contains more than five labels, ensuring that infrequent labels are aggregated to reduce data imbalance.

If the data includes fewer than five labels, it proceeds directly to data augmentation to increase the sample size and improve model robustness. The augmented data is then used to train the text classification model, resulting in a refined labeled dataset with 5,000 samples, ready for further analysis. This flow was systematically applied across multiple columns to ensure consistency and accuracy in the final dataset.

Model Training

The results of the models are depicted in the Table 5.5 below. The models demonstrated favorable outcomes, with a precision of over 79%, an F1 score of approximately 80-90%, and a recall rate of 80-90%.

Column	Label	Accuracy	Precision	Recall	F1-score	Support
Root Cause	Drug and Alcohol	86%	90%	99%	94%	71
	Early life		90%	83%	86%	72
	Personality		82%	83%	83%	72
	Trauma and Stress		84%	80%	82%	65
Perceived susceptibility (Sentiment)	Negative	82%	94%	71%	81%	106
	Neutral		72%	95%	82%	91
	Positive		88%	81%	84%	26
Perceived severity	Depression	79%	73%	60%	66%	72
	Anxiety		92%	88%	90%	75
	Loneliness		86%	94%	90%	65
	Others		64%	74%	68%	68
Perceived benefits	Not Mentioned	91%	85%	86%	86%	80
	Finding support		92%	95%	93%	103
	Getting access to treatment		100%	91%	95%	75
	Others		85%	86%	86%	80
Perceived barriers	Not Mentioned	83%	82%	71%	76%	91
	Feeling unheard and misunderstood		93%	92%	93%	74
	Feeling helpless		83%	83%	83%	36
	Feeling hopeless		88%	79%	84%	29
	Others		74%	91%	82%	74
Cue to Action	No Action	87%	93%	77%	84%	108
	Looking for information and resources		92%	93%	93%	61
	Seeking support and treatment		85%	91%	88%	55
	Sharing their situation		71%	95%	81%	37
	Others		86%	100%	92%	12
Self-efficacy	Troubled Mindset	89%	92%	82%	87%	104
	Overcome Mindset		85%	91%	88%	93
	Empowered Mindset		90%	99%	94%	72
	Denial Mindset		100%	87%	93%	15

Table 5.5. The performance metrics of final models for each column

Model Evaluation

These models were constructed to predict the values of 5,000 rows of data, and 296 randomly selected rows were used for evaluation purposes. The Table 5.6 below presents the degree of accuracy observed for each model.

Root Cause	Perceived susceptibility (Sentiment)	Perceived severity	Perceived benefits	Perceived barriers	Cue to Action	Self-efficacy
84%	83%	47%	81%	75%	82%	83%

Table 5.6. The degree of accuracy observed for each model for 5000 samples

The models demonstrated strong performance in predicting most components, with accuracy rates ranging from 75% to 84% for categories such as Root Cause, Perceived Susceptibility, Perceived Benefits, and Self-Efficacy. However, the accuracy for predicting Perceived Severity was notably lower, at 47%. This decrease in accuracy can be attributed to the complexity of Perceived Severity, which involves multiple labels, making it more challenging for the model to classify this aspect accurately. This indicates a need for further refinement in handling multi-label classifications within this component. Overall, the results highlight the models' effectiveness in capturing key health-related constructs, though certain areas, such as Perceived Severity, may require further refinement.

5.1.3 Text Generation Models

Table 5.7 below is an example for the result from text generation models.

selftext	analysis
No matter which way, whether sadness, hate, fear, pain, tiredness, anger, despair, clueless,..... No matter where there is only depth and abyss... No matter how often I build a bridge to go further (feel good) it's always a "matter" of time until I feel the fall... Everytime I think that I am as down as I could ever be that I could not be anymore broken or whatsoever time goes until I fall deeper and again and always and forever. No matter how deep/much I feel to be cut, there is more meat that gets cut even deeper..... Nothing ever cuts through. I never get true cut of (die). It only cuts near the edge...	{'Root cause': 'Personality', 'Sentiment': 'Negative', 'Perceived severity': 'Depression', 'Perceived benefits': 'Not Mentioned', 'Perceived barriers': 'Not Mentioned', 'Cue to Action': 'No Action', 'Self-efficacy': 'Troubled Mindset'}
The past two years of my life have essentially been periods of hallucinating/changes in perceptions of reality, and the spaces between them. Since last October i've been trying to get help, i've been passed from place to place and i'm waiting on a letter from the new place which will no doubt pass me off somewhere else. The hallucinations are okay honestly, I know they're not real, it's not happening. Changes of how I perceive reality are a bit harder. I've spent the past 4 days thinking that because I can hear things others cant, it must be part of a different reality and I'm acting like a bridge between this reality and a reality where my trauma didn't happen. I won't go into the trauma as its irrelevant and honestly I just don't want to. The thoughts creep in, "imagine if", "what if", "it might be", "it probably is", "I must be the only person on this earth that knows this and can experience this" The thought of being a bridge between two worlds isnt in itself awful, but when I start to plan how i'd get to that other reality it gets a bit dark. Im seeing sense currently, not sure how long for. Could be in the matrix again tomorrow, could be in The Truman Show. I just want to know if anyone has experience in mitigating these changes of perception. Thank you for any help or advice :)	{'Root cause': 'Personality', 'Sentiment': 'Neutral', 'Perceived severity': 'Hallucinations', 'Perceived benefits': 'Finding support', 'Perceived barriers': 'Lack of resources', 'Cue to Action': 'Seeking support and treatment', 'Self-efficacy': 'Overcome Mindset'}

Table 5.7. Examples of the results from text generation models

The results were then compared to the human-annotated model and are summarized in the Table 5.8 below.

Model	Root Cause	Perceived susceptibility (Sentiment)	Perceived severity	Perceived benefits	Perceived barriers	Cue to Action	Self-efficacy
LLAMA3	27%	54%	11%	38%	16%	19%	13%
GPT 4o	27%	54%	33%	48%	16%	36%	57%
GPT 4o Mini	40%	65%	22%	40%	20%	31%	61%
GPT 4o Mini Fine-Tuning	68%	78%	32%	70%	57%	73%	73%
GPT 4o Fine-Tuning	71%	77%	42%	70%	62%	77%	74%
DistilBERT Classification	84%	83%	47%	81%	75%	82%	83%

Table 5.8. The result comparison between text generation models

As illustrated in the table, the GPT4o and GPT4o mini fine-tuning model demonstrates the most optimal results across text generation models, with Root Cause at 71% and 68%, Perceived Susceptibility (Sentiment) at 77% and 78%, Perceived Severity at 42% and 32%, Perceived Benefits at 70%, Perceived Barriers at 62% and 57%, Cue to Action at 77% and 73%, and Self-Efficacy at 74% and 73%. This demonstrates the capacity to comprehend text in a nuanced manner within the context of human writing. Nevertheless, when compared to DistilBERT Classification, the GPT4o Fine-tuning model exhibits a lower degree of accuracy, ranging from 5-13% across all columns. Consequently, the final design entails the utilization of DistilBERT Classification for labeling in conjunction with GPT4o mini, with the objective of providing support for extraction.

The column with the lowest accuracy across all model predictions is Perceived Severity, largely due to the complexity of managing multiple labels. To overcome this issue, we propose using GPT to extract relevant keywords, which will then undergo human review to refine the final

labels. Additionally, we recommend an alternative approach that involves predicting the labels, extracting keywords related to mental health issues, and combining these keywords to produce the final prediction for Perceived Severity. Although this method requires extra time for manual verification, it achieves a higher accuracy rate of 81% and ensures greater relevance in the data, thus enhancing its utility for analysis. This was implemented with a simple prompt below.

```
system_prompt = """You are an experienced psychiatrist. Analyze the following text:
```

```
    Health issue: Which mental health issue is mentioned in the text?
    Answers could be: Depression, Anxiety, Suicide Attempt, Hallucinations,
    Suicidal Thoughts, Loneliness, Panic Attacks, Alcoholism, Drug Addiction,
    Bipolar, Trauma, Stress, etc.
```

```
    Keywords: Keywords from text that relates to mental health issues.
```

```
    Respond with a valid JSON object containing Health issue and Keywords
    as keys and your analysis as values. Keep each value concise, preferably
    not more than 5 words."""
```

The Table 5.9 below illustrates examples of the predicted Perceived Severity after manual review, including the Perceived Severity labels and keywords generated by GPT-4o.

selftext	Human Annotated Perceived severity	Gpt4o Perceived severity	Gpt4o Keywords	Manual Checked Gpt4o Perceived severity
I've been feeling miserable for about 4 months. I (20F) been in a 5 year relationship with my boyfriend (21M) he's been extremely supportive all this months, although, I been seeing how my depression is progressively worse now that I live with him. Within those 4 months I got r1p3d and I now have HPV which have affected me in every aspect of my life. I no longer want to be an emotional and economic burden for my boyfriend.	Depression	Depression	miserable, supportive, progressively worse	Depression

I dont care anymore. I dont care about school, or my friends, or family. I cant care about stuff anymore... Its like i've lost what makes me human... Instead of living im just existing... I dont want to deal with it anymore. I dont want to keep living like this. Always just sad and tired. I honestly cant remember the last time i was truly happy, or the last time i had a day where i didnt feel like shit. I've been going to therapy for a while now and its still just keeps getting worse. Like a tumor that just grows and grows until it eventually kills you. Ive had countless fights with my parents about it. Ive tried talking with them but its like they refuse to understand. They think im just lazy and a failure... Even after going to therapy, and even though they've been with me to a few sessions, they still dont want to understand. Every time ive tried to talk with them it just ends in a big fight. Im just better off dead. Maybe its selfish and cruel, but its the truth. Im just another meaningless life amongst billions of others. I wrote the letter years ago, i know how, where, who id call. Ive had the thought for years at this point and the tiny piece of hope i had left is gone. I would finally stop being a burden. My family would finally fighting because of me. It would probably hurt for a little while, but they'd get over it eventually...	Suicidal Thoughts	Suicidal Thoughts	lost, sad, tired, better off dead, burden	Suicidal Thoughts
I got left on by my girlfriend that i waited for 40 days just to know that she want to cut me off in the end..the worst is that i predicted it because i overthink every single thing..my last relationship is also the same a girl left me because she say im just too negative to be around with..yeah ik..im deppresing to be with..sigh yet i feel like i keep falling for someone that always show care to me..i always have suicidal thoughts but honestly only my mom that keep me going..but im afraid if i reach my limit soon...everyone treat me like shit,my brother stole my money,my gf dumped me and ghosted me without a reason and life responbilities is getting to me....Im sorry for this stupid rant i just have nobody to share this with sorry again	Depression, Suicidal Thoughts	Suicidal Thoughts	depressing, suicidal thoughts, loneliness	Depression , Suicidal Thoughts
I have anger issues, but when I start getting angry and I begin to count I start feeling more and more panicked. The same thing happens when I try and take deep breaths. The act of controlling my own breathing freaks me out and I start panicking. Please help me learn to calm down.	Anger, Panic Attacks	Panic Attacks	anger issues, panicked, deep breaths	Anger, Panic Attacks

Table 5.9. Examples of the predicted Perceived Severity after manual review

5.2 Data Analysis Results

This section presents the outcomes of the data analysis, focusing on the application of the Health Belief Model and emotional content analysis. The results provide insights into the perceived severity, susceptibility, benefits, barriers, and emotional tones across different mental health issues.

5.2.1 Health Belief Model Analysis

The Health Belief Model (HBM) Analysis explores how components like perceived severity, susceptibility, and self-efficacy are reflected in the dataset, revealing patterns that inform our understanding of health behaviors and perceptions in mental health contexts.

Perceived Severity and Root Cause

As illustrated in the Figure 5.2 below, the largest proportion of root causes for mental health problems mentioned on Reddit was personality, accounting for 50.1%. Personality traits like chronic negativity or perfectionism can increase vulnerability to mental health issues. This was followed by trauma and stress at 27.4%, as experiences such as abuse or chronic stress can leave deep emotional scars. Early life experiences, accounting for 13.5%, also play a significant role, as neglect or instability during formative years can disrupt emotional development. Finally, drug and alcohol use, at 9.0%, can alter brain chemistry and worsen mental health disorders.

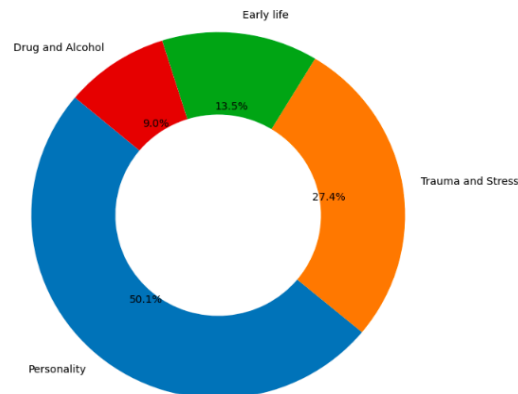


Figure 5.2. Distribution of Root Causes

pattern was observed in discussions about drug and alcohol use, with anxiety (164 posts), depression (140 posts), and suicidal thoughts (96 posts) leading the way. These posts might involve stories of how substance abuse exacerbates these mental health issues. However, loneliness and panic attacks ranked fourth and fifth in personality-related posts, whereas alcoholism (43 posts) and drug addiction (37 posts) took the fourth and fifth spots in drug and alcohol-related discussions.

In contrast, trauma and stress were most frequently associated with loneliness, depression, suicidal thoughts, anxiety, and stress, with 485, 341, 298, 254 and 24 posts, respectively. These discussions likely included narratives about how traumatic events and chronic stress lead to a deep sense of isolation and hopelessness. For early life experiences, depression was the most mentioned issue with 217 posts, followed by suicidal thoughts (186 posts), anxiety (132 posts), loneliness (107 posts), and trauma (55 posts). Here, users may have shared how difficult childhood experiences, such as neglect or abuse, have led to long-lasting mental health challenges.



Figure 65.



Figure 5.4. Word Cloud for perceived severity related to each root cause

Perceived susceptibility and Self-efficacy

The largest proportion of perceptions about mental health issues was neutral, at 48.3%, suggesting that most individuals had a balanced or uncertain view of their mental health. Negative perceptions followed at 45.1%, indicating that a significant number of users viewed their mental health in a more pessimistic light, while only 6.6% expressed a positive outlook. In terms of self-efficacy, the majority of individuals felt they had a troubled mindset (57.6%), reflecting significant challenges in managing their mental health. This was followed by individuals who felt they had overcome obstacles (22.2%) and those who felt empowered to handle their issues (19.3%). A small proportion, 0.9%, reported a denial mindset, suggesting they did not acknowledge or address their mental health challenges.

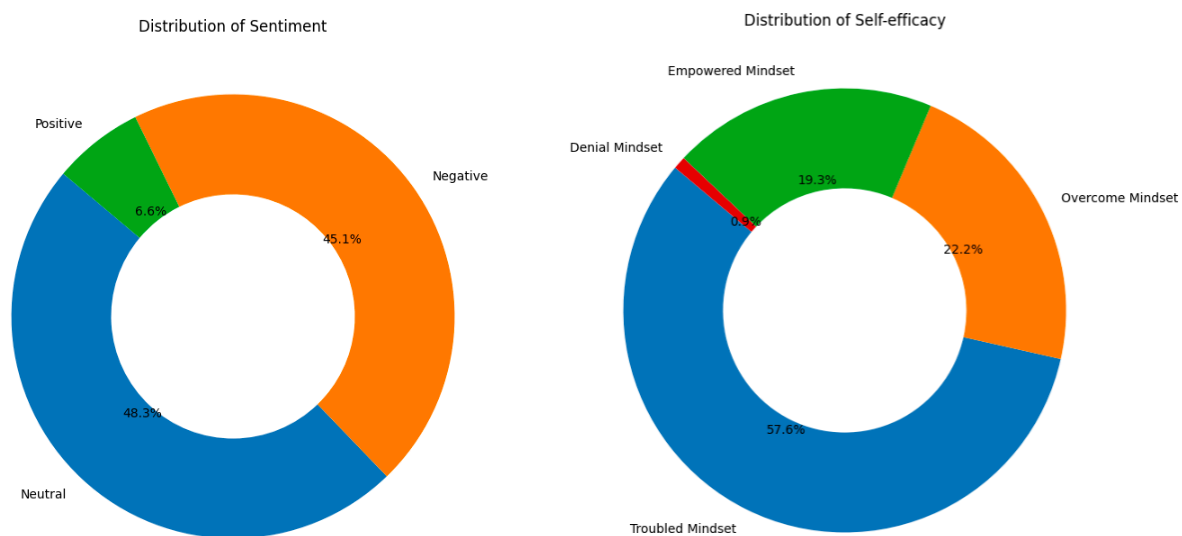


Figure 5.5. Distribution of Sentiment and Self-Efficacy

Examining the relationship between perceived susceptibility and self-efficacy revealed a notable correlation: users with positive and neutral mindsets were more likely to exhibit an empowered or overcome self-efficacy, whereas those with a negative mindset were strongly associated with a troubled mindset. Specifically, positive sentiments were primarily linked with empowerment (65.3%) and also with an overcome mindset (25.1%). In contrast, negative sentiments were

closely associated with a troubled mindset (91.7%). Neutral sentiments were more evenly distributed among overcome (37.3%), troubled (32.4%), and empowered (29%) mindsets.

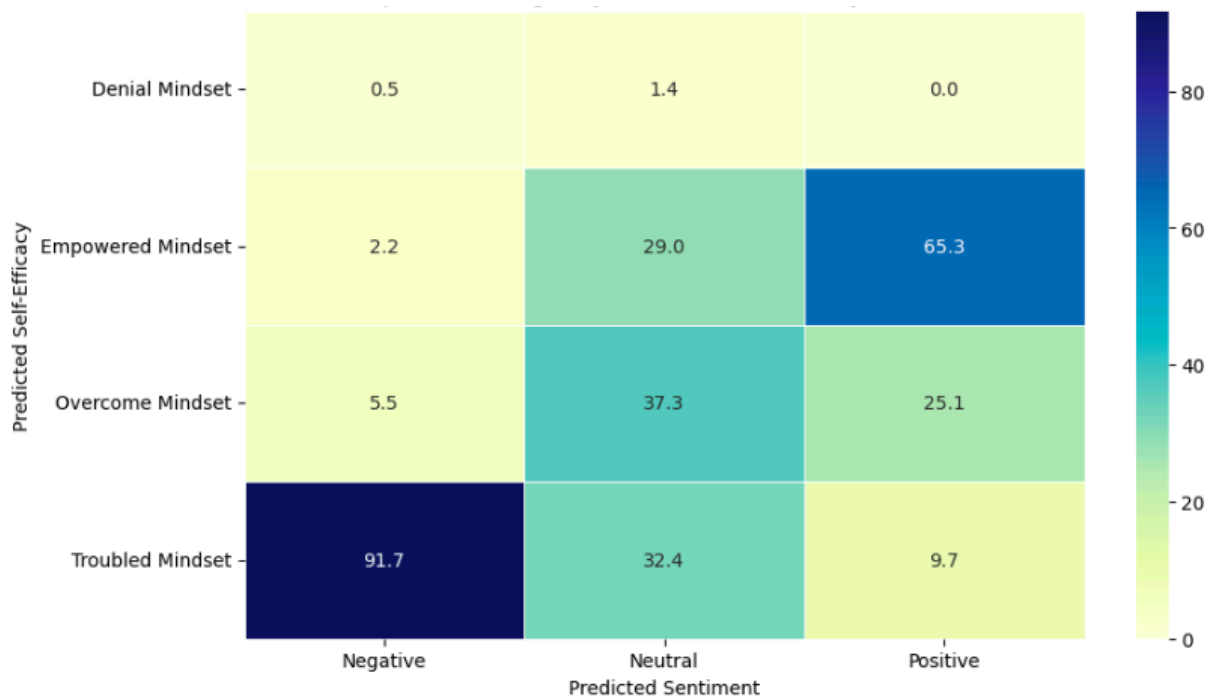


Figure 5.6. Heatmap of Percentages by Predicted Sentiment and Self-Efficacy

One notable case was when the perceived susceptibility was negative, but the mindset was empowered (50 out of 5000 posts). This situation often revealed a blend of negative and positive sentiments, as well as a mix of empowered and troubled mindsets. For instance, the statement below demonstrated a profound negative sentiment characterized by feelings of emptiness, distress, and hopelessness in response to an unexpected challenge. Despite this, the speaker's mindset was notably empowered—they actively sought professional help and showed a commitment to addressing their emotional crisis, despite their turmoil.

“Well this last Sunday I was hit with a bombshell and I just feel empty inside now. For context I am a father of a 9 mo old baby and while I'm not married my partner lives with me. She is the love of my life. We just got our relationship back on track and we were talking of marriage again. Life was great again until...”

This last Sunday I got home from work and I found out my partner is pregnant with a second child. She was sitting in the tub crying and I'll be honest... i didnt take the news well. I didn't become violent or anything, but it sure wasnt a good response. She is conflicted about it as am I. I asked her what she wanted to do and she flat out told me that abortion wasn't an option. She started crying just looking at the planned parenthood website. I told her that I'll support any decision she makes but she fears I'll resent her for keeping the baby. We are both in a deep state of depression and it's hurting us all over again. If I lose her or my son... I'm done.

The thing is I don't even think i have a right to feel the way i do. It's her body, it's her choice. I had my fun and now I need to nut up or leave.

I don't know if I want to raise a second child... I don't know if I can handle it... I don't know what to do. I've stopped eating, I've lost sleep, I am contemplating suicide... I need help. I've already called my therapist and I have an appointment scheduled but I just need to talk to someone or hear something other than my parents advice.”

Perceived benefits - Perceived barriers - Cue to Action

Regarding perceived benefits and barriers, a significant proportion of respondents did not mention these factors, with 69% not identifying perceived benefits and 67.3% not recognizing perceived barriers. In the Cue to Action category, the majority reported no action taken (63.1%). Among those who did identify benefits, finding support or accessing treatment were noted by 21% and 5.4%, respectively, with other benefits, such as feeling relieved, improving quality of life, and making progress, accounting for 4.5%.



Figure 5.7. Distribution of Perceived benefits - Perceived barriers - Cue to Action

For perceived barriers, the most common issues were feeling unheard and misunderstood (11.5%), feeling helpless (4.5%), and feeling hopeless (1.9%). Other barriers included not knowing how to deal with the situation, not wanting to bother others, financial constraints, lack of resources and support, feeling worthless, and experiencing stigma. In the Cue to Action category, the most frequently mentioned actions were looking for information and resources (16.3%), seeking support and treatment (12.1%), and sharing their situation (6.3%), with other actions like benefiting others and engaging in physical activities representing 2.4%.

Analysis of the Health Belief Model Across Top Mental Health Issues

In this section, we analyzed the top five mental health issues based on perceived severity to explore the entire health belief model. This included examining root causes, perceived susceptibility, perceived benefits, perceived barriers, cues to action, and self-efficacy to gain deeper insights into how these factors interact and influence overall mental health perceptions.

Anxiety

The analysis of anxiety revealed that the primary root cause was related to personality (65%), followed by trauma and stress (18.2%). Personality factors, such as perfectionism or low self-esteem, can heighten the risk of developing anxiety. Sentiment analysis showed that only 8.9% of posts were positive, while a significant 70.8% were neutral and 20.2% were negative. In terms of perceived benefits, 42.4% of individuals had sought support, and 6% had already accessed treatment. Perceived barriers were largely not mentioned, accounting for 74% of the cases. Regarding cues to action, 31.1% of individuals had looked for information and resources, while 16.8% had sought support and treatment. Self-efficacy levels indicated that 32.9% felt empowered and 31.1% felt they had overcome their challenges, which was higher than average.

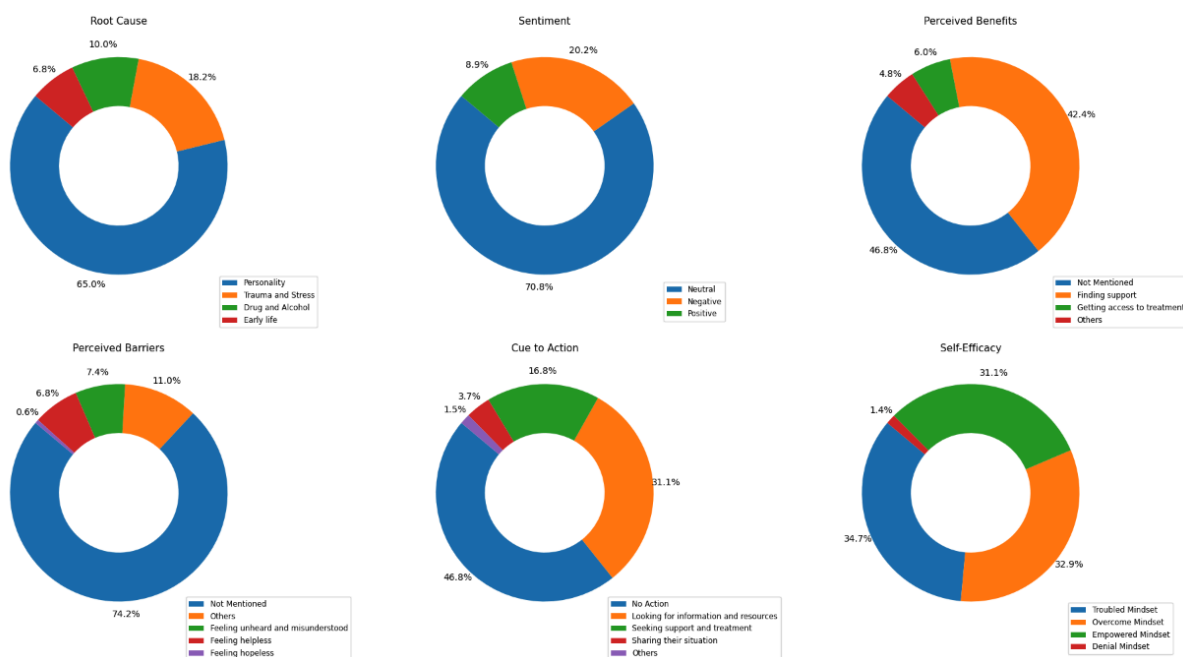


Figure 5.8. The Health Belief Model in Anxiety Group

In conclusion, individuals dealing with anxiety were primarily influenced by personality-related issues, faced significant barriers, and were often prompted to act by seeking information and support. Despite these challenges, a notable proportion felt empowered or had made progress in overcoming their anxiety.

Depression

The next mental health issue analyzed was depression. The primary root cause was related to personality (55%), followed by trauma and stress (24.9%). Sentiment analysis revealed that only 4.9% of posts were positive, while a significant 54.6% were negative and 41% were neutral. Regarding perceived benefits, 75.5% did not mention any, and only 14.4% found support. Seeking resource recommendations can offer valuable information, validation, coping strategies, social connections, and professional help. Perceived barriers were also largely not mentioned, accounting for 67.5% of cases, followed by others and feeling unheard and misunderstood (14.6% and 11.9%, respectively). Regarding cues to action, 69.3% of individuals took no action, with only 11.6% looking for information and resources, and 9.5% seeking support and treatment. Self-efficacy levels showed that 67.1% of individuals were troubled, while 18.9% had overcome their challenges, and 13.3% felt empowered.

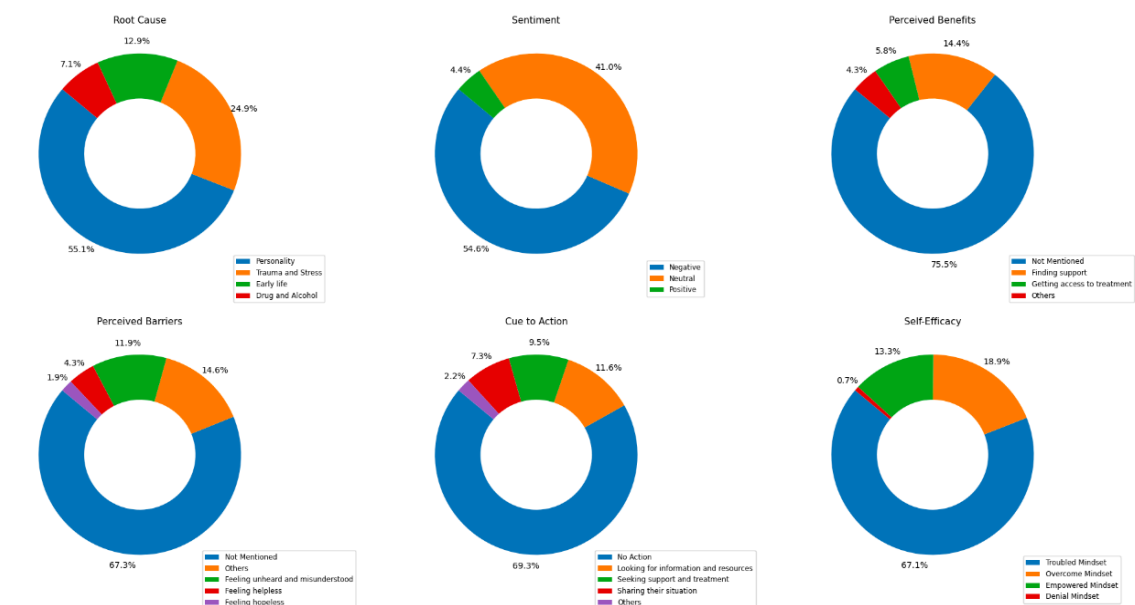


Figure 5.9 The Health Belief Model in Depression Group

The analysis highlights a significant level of inaction and feelings of being unheard among those dealing with depression, with a strong association with personality-related issues and a predominant sense of being troubled. This suggests a critical need for more proactive support systems and better communication strategies to reach and empower individuals with depression.

Anxiety and Depression

The next group examined consisted of individuals experiencing both **anxiety and depression**. As shown in the charts, the trends were similar to those in the anxiety-only group across all six components. The root causes were predominantly linked to personality (55.9%) and trauma and stress (19.6%). Sentiment was mostly neutral (69.6%). In terms of perceived benefits, 29.4% of individuals sought support, and 16.7% had accessed treatment. Perceived barriers were largely not mentioned (67.6%). Regarding cues to action, 30.4% of individuals looked for information and resources, while 18.6% sought support and treatment. Self-efficacy levels indicated that 31.1% felt empowered and 29.4% felt they had overcome their challenges, which was higher than average.

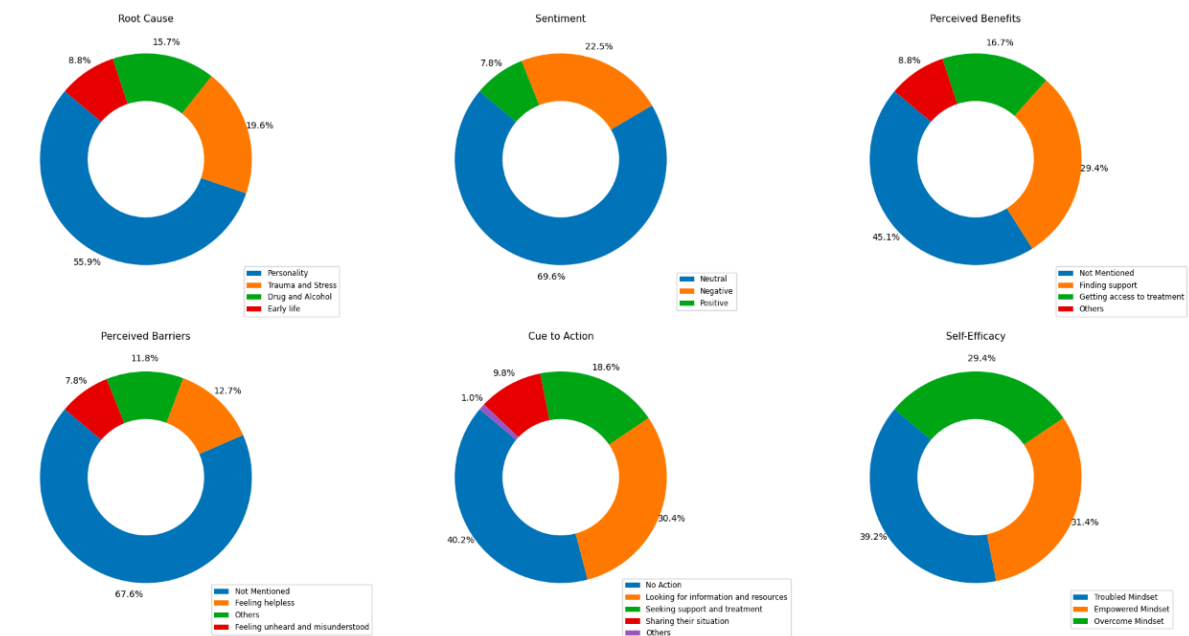


Figure 5.10. The Health Belief Model in Anxiety and Depression Group

Despite dealing with both anxiety and depression, individuals in this group showed a proactive approach similar to those with anxiety alone, with relatively high levels of empowerment and action-oriented behavior. This suggests that even in the presence of comorbid mental health issues, many individuals are actively seeking ways to manage and overcome their challenges.

Suicidal Thoughts

For individuals experiencing suicidal thoughts, the root causes were mainly linked to personality issues and trauma or stress. The sentiment around these issues was predominantly negative (70.9%), with only 2.6% of posts showing positive sentiments. However, the example below illustrates a case where, despite severe mental health struggles - including suicidal thoughts, self-harm, and panic attacks - the individual maintains a positive and proactive mindset. They actively seek help, explore treatment options, and remain open to therapy and medication, despite challenges like financial constraints and uncertainty about where to start. This demonstrates their determination to find support and navigate the path to recovery.

“I (19m) have been struggling with my mental health for quite a few years now (suicidal thoughts, self-harm and panic attacks/breakdowns) but the last few months have been especially bad. I've decided I want to try get help but I don't know where to start. A psychologist, a psychiatrist or a therapist? I don't know who to see or how. I'm open to the thought of medication, most of my family is already on either antidepressants or anxiety meds, but, talking to them about this isn't really an option. I don't have much money and I struggle enough to book a hair appointment on my own let alone figure out this whole process. Any advice would be really appreciated.”

A significant 87.8% of individuals did not mention any perceived benefits, and just 8.3% reported finding support. In terms of barriers, 59.9% were not specified, followed by 17.9% categorized as "others", and 14.4% involving feelings of being unheard and misunderstood. The majority took no action (82.2%), with only 7.8% seeking support or treatment. Self-efficacy levels showed that 81.9% felt troubled, while only 9.6% had overcome their challenges, and a mere 0.7% felt empowered. This data highlights the severe struggle and isolation faced by

individuals with suicidal thoughts, indicating a critical need for focused interventions and support mechanisms.

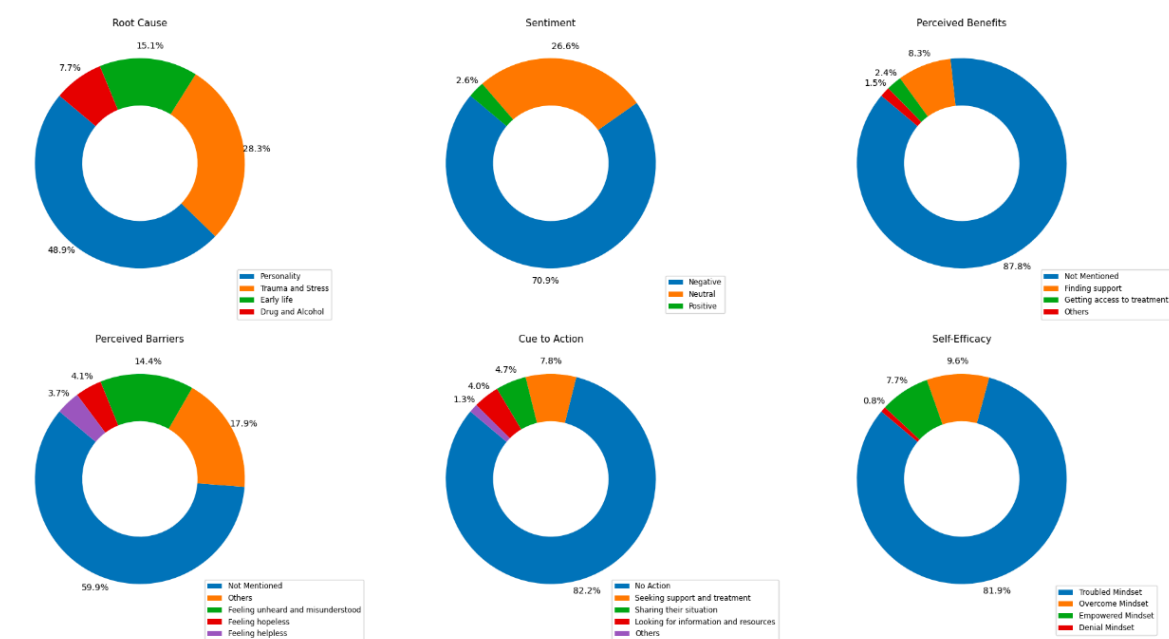


Figure 5.11. The Health Belief Model in Suicidal Thoughts Group.

While both anxiety and depression were heavily influenced by personality-related issues, individuals with anxiety were more likely to seek out support and feel empowered in managing their condition. Those who experienced both anxiety and depression exhibited similar proactive behaviors, with relatively high levels of empowerment and active efforts to find information and resources. In contrast, individuals with depression alone tended to experience more inaction, felt unheard, and were less likely to feel empowered or make progress, indicating a greater need for intervention and support systems tailored specifically to those dealing with depression. For those who struggled with suicidal thoughts, the situation was even more dire—marked by overwhelmingly negative sentiment, significant inaction, and a deeply troubled mindset. These individuals were the least likely to seek support or feel empowered, underscoring the critical need for targeted interventions to address their extreme vulnerability.

Loneliness

Loneliness was often described in the texts as being significantly isolated by a lack of connection with family and friends. The primary causes were trauma and stress (61.5%) and personality-related factors (23.9%). The sentiment expressed was relatively balanced, with 8.5% positive, 42.4% neutral, and 49.1% negative. Despite this, perceived benefits were largely unmentioned (83.4%), and only 9.9% reported finding support. Barriers were similarly underreported (65.5%), with 17.2% citing "other" barriers and 15.7% feeling unheard and misunderstood. In terms of cues to action, 67.7% did not take any action, while 10.7% shared their situation and 10% looked for information and resources. Self-efficacy was predominantly troubled (63%), followed by overcome (20.7%), and empowered (14.7%), with only 1.6% feeling empowered.

Compared to anxiety, where individuals were more proactive and sought support more frequently, those dealing with loneliness exhibited less engagement and fewer efforts to seek help or resources.

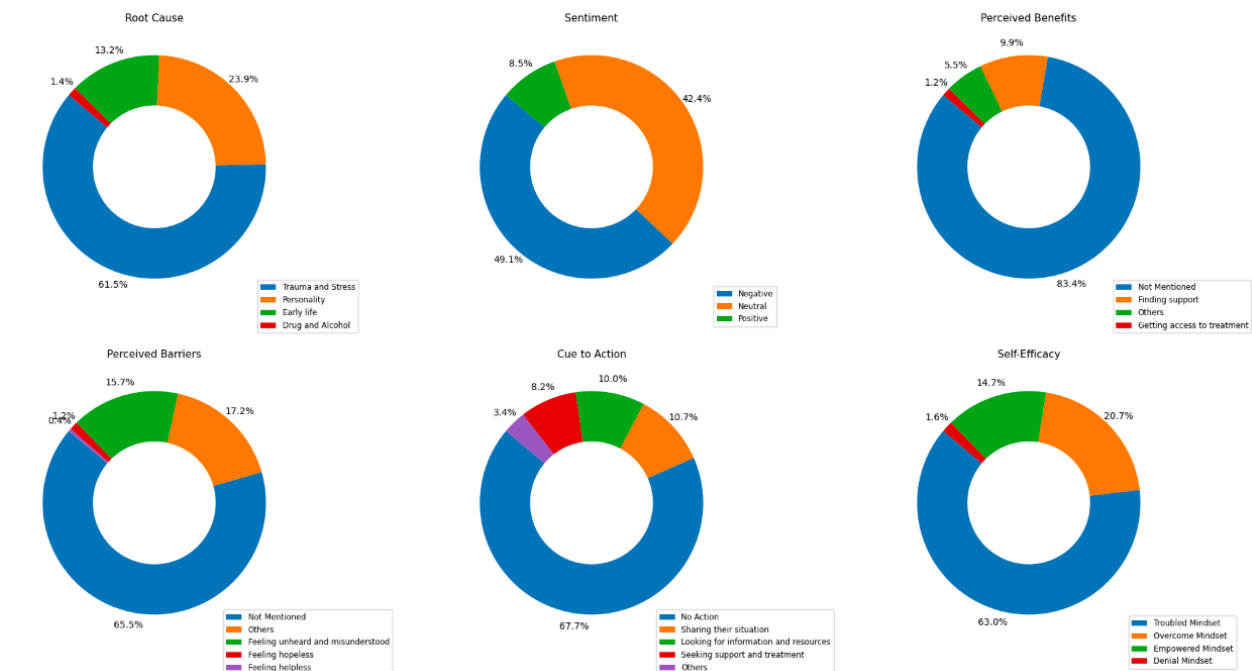


Figure 5.12. The Health Belief Model in Loneliness Group

Trauma or Stress

The combined analysis of trauma and stress (90 posts on trauma and 65 posts on stress) revealed that the primary root cause was related to personality (45.2%), followed by early life experiences (29%) and trauma (21.3%). Sentiment analysis showed that 10.3% of posts were positive, while 52.3% were neutral and 37.4% were negative. In terms of perceived benefits, 60% of individuals did not mention any benefits, whereas 23.2% reported having sought support. Perceived barriers were largely unmentioned, accounting for 67.1% of the cases. Regarding cues to action, 57.4% did not take any action, 16.8% looked for information and resources, and 14.8% sought support and treatment. Self-efficacy levels indicated that 27.7% felt empowered, 23.9% felt they had overcome their challenges, and 48.4% felt troubled. The combined insights showed that while there was a substantial amount of neutral and negative sentiment, individuals were less proactive in seeking support or acknowledging benefits.

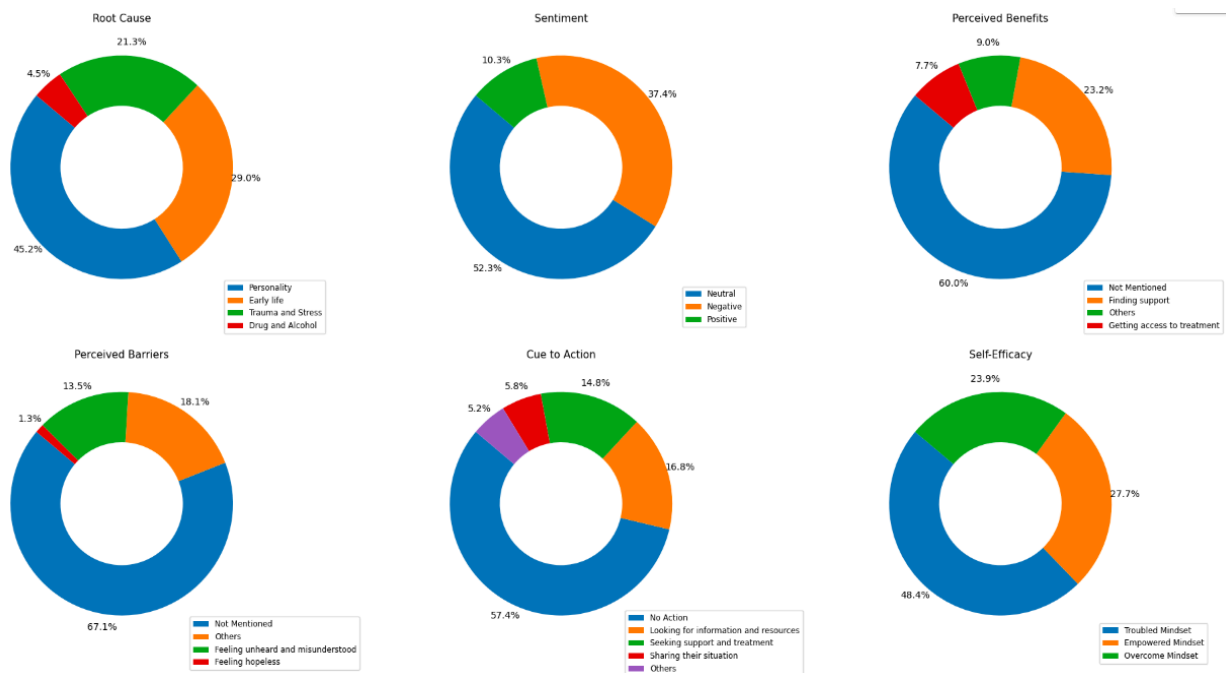


Figure 5.13. The Health Belief Model in Trauma or Stress Group

Alcoholism or Drug Addiction

The combined analysis of alcoholism and drug addiction (35 posts in alcoholism and 25 posts in drug addiction) revealed that the primary root cause was related mainly to drug and alcohol 73.3%, followed by personality (11.7%) and early life (10%). Sentiment analysis showed that 10% of posts were positive, while 43.3% were neutral and 46.7% were negative. In terms of perceived benefits, 65% of individuals did not mention any benefits, whereas 18.3% reported having sought support. Perceived barriers were largely unmentioned, accounting for 75% of the cases. Regarding cues to action, 55% did not take any action, 16.7% looked for information and resources, and 15% sought support and treatment. Self-efficacy levels indicated that 26.7% felt empowered, 25% felt they had overcome their challenges, and 48.3% felt troubled. The analysis of alcoholism and drug addiction indicated that while substance use was the predominant root cause, individuals displayed a mix of sentiment and self-efficacy levels. A significant portion did not mention benefits or take action, highlighting the need for enhanced support and resources to address substance use challenges effectively.

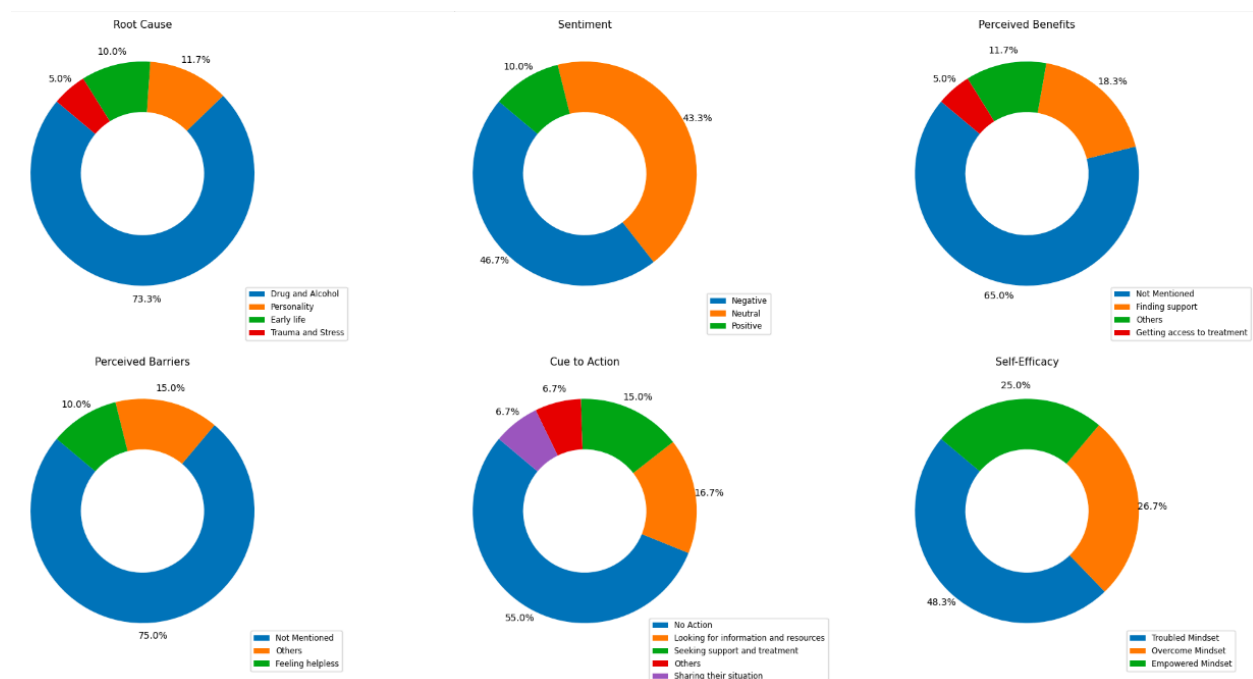


Figure 5.14. The Health Belief Model in Alcoholism or Drug Addiction Group

Others

The combined analysis of miscellaneous conditions (such as bipolar disorder, obsessive compulsive disorder, hallucinations, panic attacks and more) revealed that the primary root cause was predominantly related to personality (52.3%), followed by early life experiences (19.1%), substance use (14.4%), and trauma and stress (14.1%). Sentiment analysis showed that 8.7% of posts were positive, 57.3% were neutral, and 34% were negative. In terms of perceived benefits, 58.5% of individuals did not mention any benefits, while 26.6% reported having sought support. Perceived barriers were largely unmentioned, accounting for 68.3% of the cases. Regarding cues to action, 53.4% did not take any action, 20.9% looked for information and resources, and 16.3% sought support and treatment. Self-efficacy levels indicated that 26.8% felt empowered, 27.6% felt they had overcome their challenges, and 45.2% felt troubled. The analysis of various mental health conditions showed a predominant focus on personality-related causes, with a significant proportion of individuals feeling troubled and a high rate of inaction.

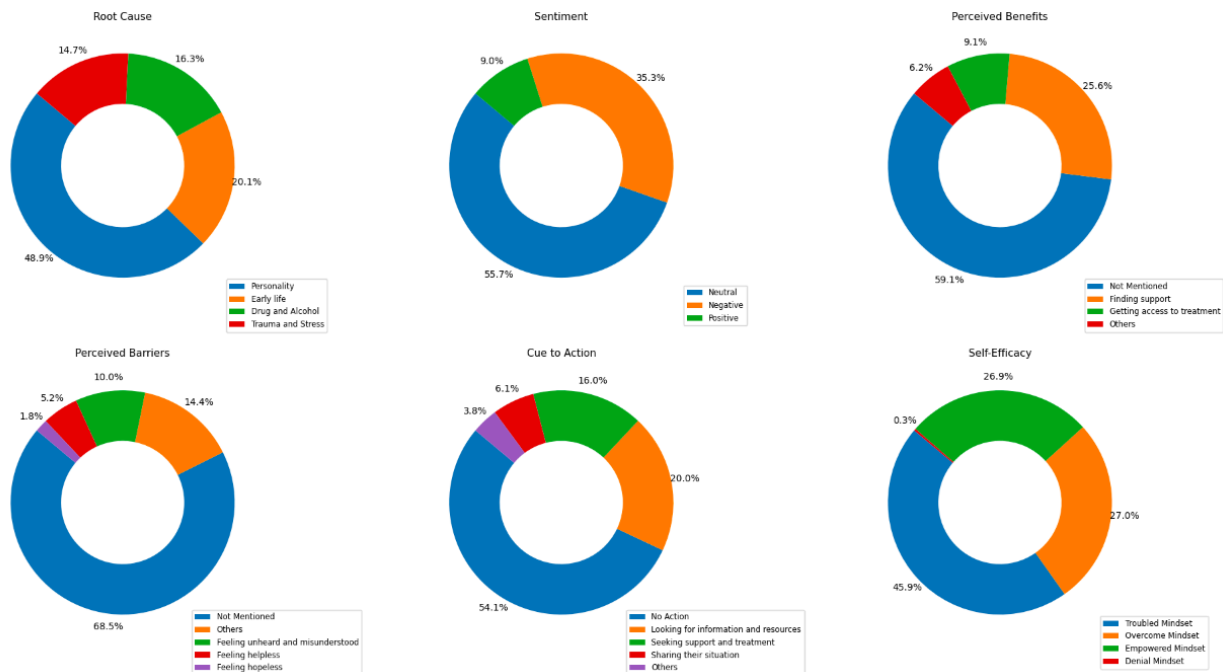


Figure 5.15. The Health Belief Model in Other Group

In conclusion, the analysis across the eight mental health issues revealed several key patterns and insights. Anxiety and Depression were recognized as the most prevalent issues, both heavily influenced by personality-related factors, with anxiety showing higher proactive behaviors and empowerment compared to depression, which displayed more inaction and a sense of being unheard. Suicidal Thoughts were marked by overwhelmingly negative sentiment, a high level of inaction, and a predominance of troubled mindsets, emphasizing the urgent need for targeted support. The combined category of Anxiety and Depression showed similar trends to individual anxiety cases, indicating overlapping issues.

Furthermore, Loneliness was primarily associated with trauma and stress, featuring a significant proportion of neutral sentiments and high levels of inaction. Trauma or Stress showed a complex relationship with root causes, including personality and early life experiences, and demonstrated moderate levels of self-efficacy. Alcoholism or Drug Addiction had a high prevalence of negative sentiment and a significant number of unmentioned barriers, highlighting a critical need for improved intervention strategies. The Others category, encompassing various conditions like bipolar disorder, OCD, panic attacks and more was mostly influenced by personality and early life factors, with a notable proportion feeling troubled and facing barriers to support.

Overall, these findings suggest that while personality-related issues are a major root cause across conditions, each issue has unique characteristics that need customized support and intervention strategies to effectively meet the varied needs of individuals.

5.2.2 Emotional Content Analysis

The Emotional Content Analysis section examines the emotional tone and sentiment within the dataset, categorizing and comparing word sentiment across different perceived severity groups. This analysis highlights the emotional landscape associated with different levels of perceived severity in mental health issues, offering a deeper understanding of the emotional drivers behind health-related behaviors.

Word Sentiment Counts by Perceived Severity Group

As illustrated in the chart, neutral sentiment predominated across all Perceived Severity Groups, ranging from 89% to 91%. In contrast, the proportions of positive and negative sentiments varied between the different groups.

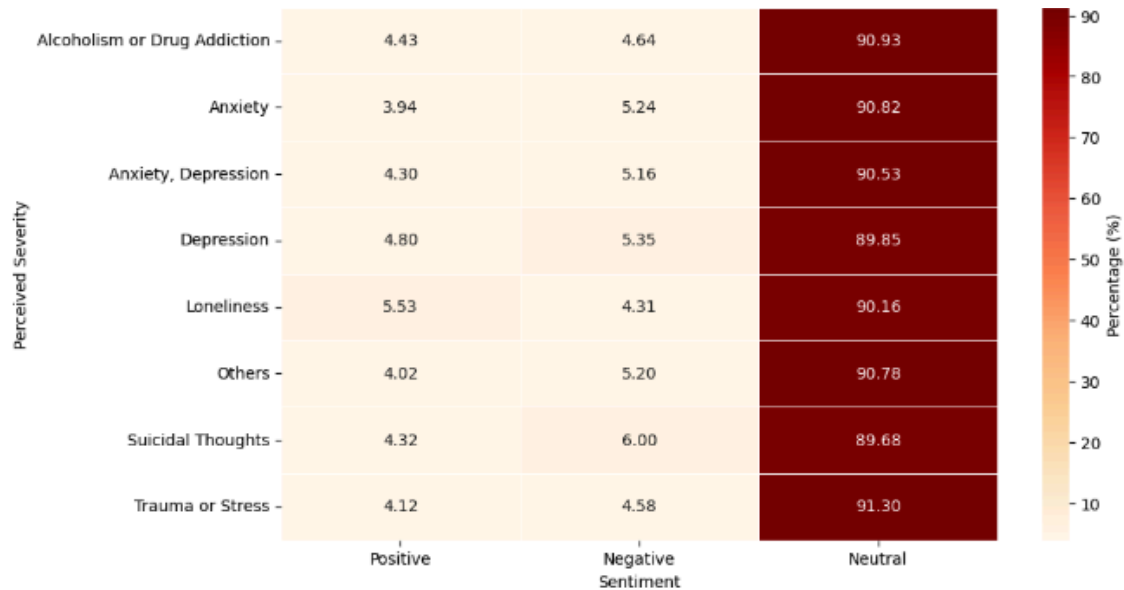


Figure 5.16. Heatmap of Percentages of Word Sentiment by Predicted Severity

A closer examination of the comparison between positive and negative words (excluding neutral) revealed that the only group with a higher percentage of positive words than negative was Loneliness, with 56.22% positive versus 43.78% negative. In contrast, the groups with the highest proportions of negative words were Suicidal Thoughts (58.12%), Anxiety (57.11%), and Others (56.37%). The remaining groups exhibited a range of negative word percentages between 51% and 55%. This analysis underscored that while Loneliness was relatively more positive, other mental health issues were predominantly characterized by negative sentiments.

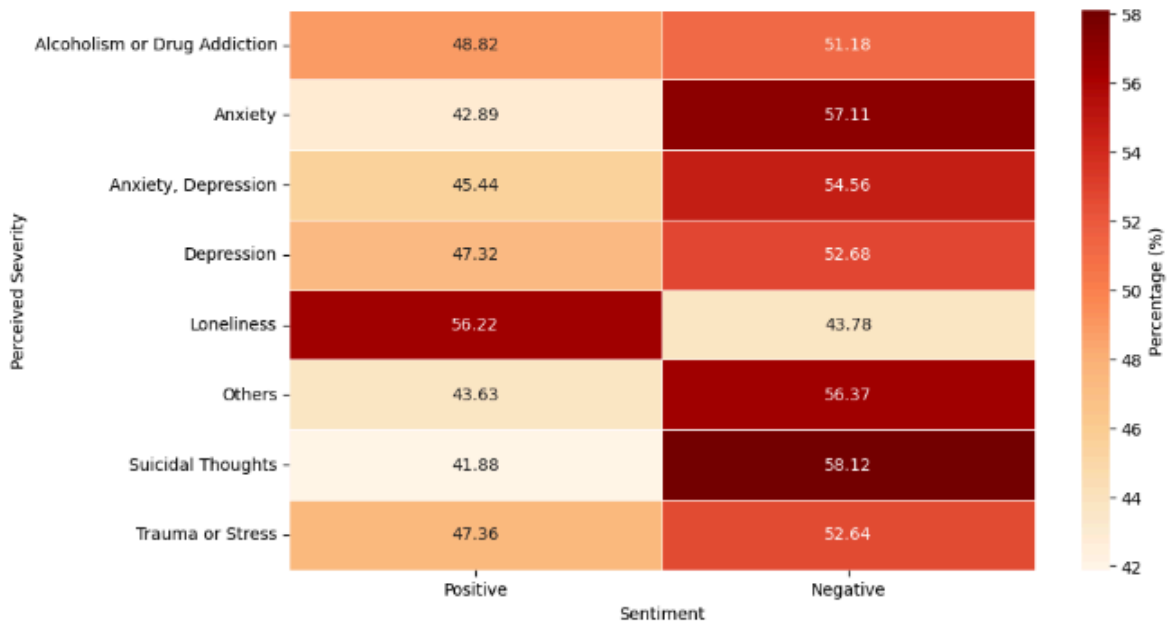


Figure 5.17. Heatmap of Percentages of Word Sentiment by Predicted Severity (Excluding Neutral)

Top Negative Words by Perceived Severity Group

This section delves into the most frequently used negative words within each perceived severity group. By identifying and analyzing the top 10 negative words for each group, we seek to gain insights into the specific concerns and emotional challenges associated with different mental health issues.

The heat map shows that "no" and "bad" were consistently among the top 10 negative words across all groups. "No" ranked as the top negative word in the Depression, Others, Loneliness, Alcoholism or Drug Addiction, and Trauma or Stress categories. It was second in Suicidal Thoughts and third in Anxiety. Although "bad" generally ranked lower, it frequently appeared in the top 5 across most groups.

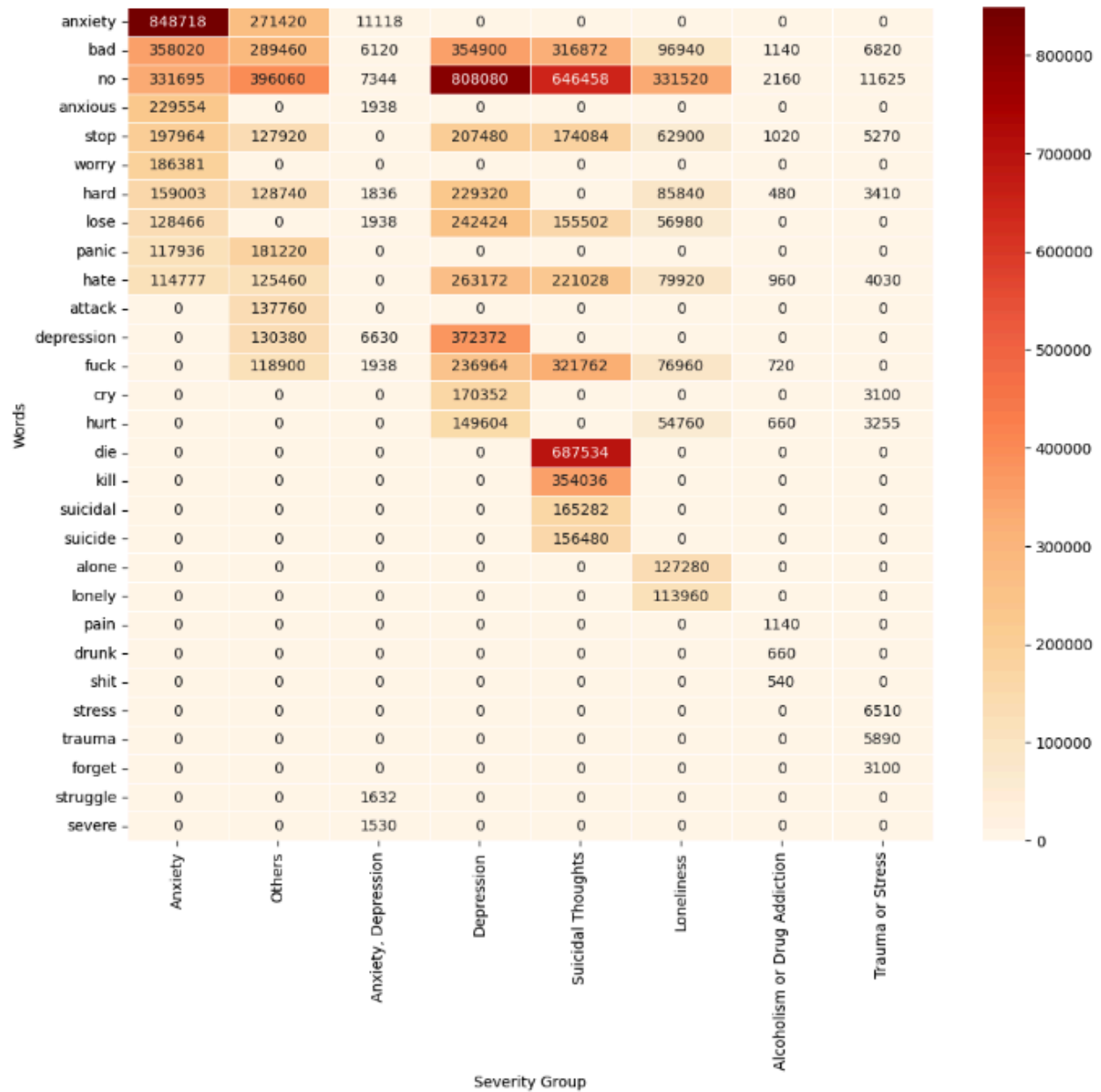


Figure 5.18. Heatmap of Top 10 Negative Words by Predicted Severity Group

A deeper analysis of each group reveals that, aside from "no" and "bad," words like "anxiety," "anxious," "stop," and "worried" were prominent in the Anxiety group. In the Depression group, "depression," "hate," "lose," and "fuck" were the most common negative words. In the Anxiety and Depression groups, there was a mix of words, with "anxiety" and "depression" being top

negative terms. For Suicidal Thoughts, the most frequent words also included "die," "kill," "suicidal," and "suicide."

In the Loneliness group, "alone," "lonely," "hard," and "hate" were top words, while in the Alcoholism or Drug Addiction group, "stop" and "pain" were most common. In the Trauma or Stress group, "stress," "trauma," and "stop" were prominent, and in the Others category, the top words were "anxiety," "hard," "stop," and "hate."

In summary, "no" and "bad" were the most consistent negative words across all groups, while each specific group had additional negative words that reflected their unique emotional struggles.

5.2.3 Summary of Health Belief Model and Emotional Content Analyses

This section presents a consolidated overview of the key findings from the Health Belief Model (HBM) Analysis and Emotional Content Analysis. The Table 5.10 below highlights the main components examined, such as perceived severity, susceptibility, self-efficacy, and emotional tone, summarizing the patterns and relationships observed across the dataset. This integrated analysis offers a thorough understanding of health-related behaviors, perceptions, and emotional content within the context of mental health issues.

Issue	Anxiety	Depression	Anxiety & Depression	Suicidal Thoughts	Loneliness	Trauma/ Stress	Alcohol/ Drug Addiction	Others
Root Causes	65% Personality, 18.2% Trauma/ Stress	55% Personality, 24.9% Trauma/ Stress	55.9% Personality, 19.6% Trauma/ Stress	Personality, Trauma/ Stress	61.5% Trauma/ Stress, 23.9% Personality	45.2% Personality, 29% Early Life, 21.3% Trauma	73.3% Drug/ Alcohol, 11.7% Personality	52.3% Personality, 19.1% Early Life
Sentiment Analysis	8.9% Positive, 70.8% Neutral, 20.2% Negative	4.9% Positive, 54.6% Negative, 41% Neutral	69.6% Neutral	2.6% Positive, 70.9% Negative	8.5% Positive, 42.4% Neutral, 8.5% Negative	10.3% Positive, 52.3% Neutral, 37.4% Negative	10% Positive, 43.3% Neutral, 46.7% Negative	8.7% Positive, 57.3% Neutral, 34% Negative

Perceived Benefits	42.4% Sought Support, 6% Accessed Treatment	14.4% Found Support, 75.5% Not Mentioned	29.4% Sought Support, 16.7% Accessed Treatment	8.3% Found Support, 87.8% Not Mentioned	9.9% Found Support, 83.4% Not Mentioned	23.2% Sought Support, 60% Not Mentioned	18.3% Sought Support, 65% Not Mentioned	26.6% Sought Support, 58.5% Not Mentioned
Perceived Barriers	74% Not Mentioned	67.5% Not Mentioned, 14.6% Other, 11.9% Unheard	67.6% Not Mentioned	59.9% Not Specified, 17.9% Other, 14.4% Unheard	65.5% Not Mentioned, 17.2% Other, 15.7% Unheard	67.1% Not Mentioned	75% Not Mentioned	68.3% Not Mentioned
Cues to Action	31.1% Sought Info/Resources, 16.8% Sought Support/Treatment	69.3% No Action, 11.6% Sought Info/Resources, 9.5% Sought Support/Treatment	30.4% Sought Info/Resources, 18.6% Sought Support/Treatment	82.2% No Action, 7.8% Sought Support/Treatment	67.7% No Action, 10.7% Shared Situation, 10% Sought Info/Resources	57.4% No Action, 16.8% Sought Info/Resources, 14.8% Sought Support/Treatment	55% No Action, 16.7% Sought Info/Resources, 15% Sought Support/Treatment	53.4% No Action, 20.9% Sought Info/Resources, 16.3% Sought Support/Treatment
Self-Efficacy	32.9% Empowered, 31.1% Overcome	67.1% Troubled, 18.9% Overcome, 13.3% Empowered	31.1% Empowered, 29.4% Overcome	81.9% Troubled, 9.6% Overcome, 0.7% Empowered	63% Troubled, 20.7% Overcome, 14.7% Empowered	27.7% Empowered, 23.9% Overcome, 48.4% Troubled	26.7% Empowered, 25% Overcome, 48.3% Troubled	26.8% Empowered, 27.6% Overcome, 45.2% Troubled
% Word Sentiment (Except Neutral)	57% Negative, 43% Positive	53% Negative, 47% Positive	55% Negative, 45% Positive	58% Negative, 42% Positive	44% Negative, 56% Positive	53% Negative, 47% Positive	51% Negative, 49% Positive	56% Negative, 44% Positive
Top Negative Words	"anxiety," "worried," "stop," "bad," "no"	"depression," "hate," "lose," "bad," "no"	"anxiety," "depression," "worried," "stop," "bad"	"die," "kill," "suicidal," "suicide," "no"	"alone," "lonely," "hard," "hate," "no"	"stress," "trauma," "stop," "bad," "no"	"pain," "stop," "addiction," "bad," "no"	"anxiety," "hard," "stop," "bad," "no"

Table 5.10. Overview of the key findings from the Health Belief Model

5.3 Summary

The models demonstrated strong performance across several components, achieving accuracy rates between 75% and 84% for Root Cause, Perceived Susceptibility, Perceived Benefits, and Self-Efficacy. However, the accuracy for predicting Perceived Severity was notably lower at 47%, largely due to its complex multi-label classification. Among the models, GPT4o and GPT4o mini fine-tuning models showed the best results in text generation tasks, with accuracy rates from 70% to 78%. Despite these strengths, GPT4o fine-tuning models had lower accuracy compared to DistilBERT Classification, which was 5-13% more accurate across all components. Therefore, it is recommended to use DistilBERT Classification for labeling, in conjunction with GPT4o mini for enhanced text generation and extraction support. To address the low accuracy in Perceived Severity, we propose leveraging GPT for keyword extraction followed by human review to improve label precision.

The data analysis reveals distinct patterns across various mental health issues. Anxiety is predominantly influenced by personality traits, exhibiting moderate negative sentiment along with moderate self-efficacy and proactivity. In contrast, Depression is marked by high negative sentiment and significant inaction, with personality factors playing a major role, leading many individuals to feel troubled and unheard. The combination of Anxiety and Depression shows similar trends to Anxiety alone, with moderate negative sentiment and considerable self-efficacy. Suicidal Thoughts are characterized by the highest negative sentiment and extreme inaction, highlighting an urgent need for targeted interventions. Loneliness, associated with trauma and stress, displays a balanced sentiment but substantial inaction; individuals feel troubled but are less proactive. Trauma and stress present a mix of sentiment with moderate negative levels, underscoring the need for more proactive support. Alcohol and drug addiction are marked by high negative sentiment and significant inaction, with substance use being a major contributing factor. The "Others" category encompasses a variety of conditions influenced by personality and early life experiences, showing moderate negative sentiment and high levels of inaction. This summary captures the root causes, sentiment, perceived benefits and barriers, cues to action, self-efficacy, and notable negative words associated with each mental health issue.

Chapter 6. Discussion and Future Works

In this section, we discuss the key findings of our research, addressing their implications and potential limitations. We also outline areas for improvement and propose directions for future work, highlighting opportunities to expand upon our current findings and explore new applications or methodologies.

6.1 Discussion

The evaluation of computational techniques for analyzing Reddit datasets within the Health Belief Model framework provides valuable insights into the strengths and limitations of current approaches in understanding mental health discussions online. This research aligns with the growing trend of using social media data, particularly from Reddit, to study mental health issues, as highlighted by [Boettcher \(2021\)](#) in their scoping review of depression and anxiety studies using Reddit data. The strong performance in predicting components such as Root Cause, Perceived Susceptibility, Perceived Benefits, and Self-Efficacy (accuracy rates between 75% and 84%) demonstrates the potential of these techniques. However, the challenge in predicting Perceived Severity (47% accuracy) underscores the complexity of multi-label classification in mental health contexts. This difficulty echoes findings from other studies that have attempted to classify mental health states of Reddit users, suggesting a common challenge in the field ([Boettcher 2021](#)).

The proposed alternative approach combining label prediction with keyword extraction for mental health issues aligns with recent advancements in natural language processing (NLP) techniques applied to Reddit data. This success suggests that hybrid approaches, such as the one proposed in this study, could indeed enhance both accuracy and relevance in mental health text analysis. The comparison of model performance between GPT-4o, GPT-4o mini, and DistilBERT Classification provides interesting insights into the strengths of different model architectures. The superior classification accuracy of DistilBERT aligns with findings from other studies that have successfully applied BERT-based models to mental health text classification tasks. The

suggestion to combine DistilBERT for labeling and GPT-4o mini for text generation represents an innovative approach that could leverage the strengths of both model types.

The data analysis identifies distinct patterns in mental health discussions: Anxiety is linked to moderate negative sentiment and self-efficacy, while Depression shows high negative sentiment and inaction, emphasizing a need for empathetic interventions. Suicidal Thoughts reveal the highest negative sentiment and inaction, highlighting the urgency for targeted support. Loneliness, related to trauma and stress, shows balanced sentiment but significant inaction, suggesting barriers in seeking help. These findings align with [Castilla-Puentes and others \(2022\)](#), which emphasize the importance of tailored interventions for different mental health conditions.

The ethical considerations raised regarding privacy and consent when analyzing Reddit posts on mental health are crucial and reflect ongoing discussions in the field. While the platform is public, users often share sensitive information, expecting some anonymity. Researchers must anonymize data and ensure no personally identifiable details are used. Given the vulnerability of this population, care should be taken to avoid causing harm or stigmatization by the analysis.

Overall, the analysis highlights that while some mental health conditions share common themes of negative sentiment and inaction, each issue also has unique characteristics that must be addressed. The findings indicate that successful interventions must be multifaceted, addressing both emotional and practical aspects of these conditions. This nuanced understanding of various mental health issues can aid in developing more effective and personalized support strategies. In addition, when analyzing Reddit posts on mental health, privacy and consent are key concerns. While the platform is public, users often share sensitive information, expecting some anonymity. Researchers must anonymize data and ensure no personally identifiable details are used. Given the vulnerability of this population, care should be taken to avoid causing harm or stigmatization.

6.2 Answer to Research Questions

RQ1: What computational techniques can be utilized to effectively analyze large-scale Reddit datasets within the context of the Health Belief Model?

Effective computational techniques for analyzing large-scale Reddit datasets include fine-tuning models such as GPT-4o and leveraging classification algorithms like DistilBERT. These techniques facilitate nuanced text generation and accurate classification of health-related constructs within the Health Belief Model framework. Combining these approaches enhances both the understanding and categorization of complex health discussions.

RQ2: How can these techniques be applied to ensure accuracy and efficiency in the analysis of health-related discussions on Reddit?

To ensure both accuracy and efficiency, it is advisable to use DistilBERT for initial labeling due to its superior classification performance, alongside GPT4o mini for generating and refining text. This hybrid approach capitalizes on the strengths of each model, improving the overall analysis of health-related discussions. Additionally, incorporating keyword extraction and human review for components like Perceived Severity can further enhance accuracy.

RQ3: What are the key challenges in applying computational methods to Reddit data, and how can they be addressed?

Key challenges include the complexity of multi-label classification, particularly for components like Perceived Severity, and the need for nuanced text comprehension. These challenges can be addressed by using a combination of models, such as DistilBERT for classification and GPT-4o for text generation, and by implementing keyword extraction followed by human review to improve label precision.

RQ4: How do health beliefs influence the nature and content of public health discussions on Reddit?

Health beliefs significantly shape the nature and content of public health discussions on Reddit. They determine how users interpret and discuss health issues, influencing the focus and tone of the conversations. For instance, beliefs about susceptibility or benefits can drive the intensity and direction of discussions, impacting overall engagement and sentiment.

RQ5: What role does emotional content play in shaping these discussions and subsequent public health behaviors?

Emotional content plays a crucial role in shaping public health discussions and behaviors on Reddit. Emotional expressions, presented in word sentiment, can amplify the impact of health beliefs, influencing how users engage with content and each other. Emotional content often drives higher engagement levels and can affect behavior changes by resonating more deeply with individuals' experiences and concerns.

RQ6: Can patterns be identified in how different health beliefs and emotional expressions correlate with engagement levels and behavior changes on Reddit?

Patterns can indeed be identified in the correlation between health beliefs, emotional expressions, engagement levels, and behavior changes on Reddit. Health beliefs and emotional content often interact to influence user engagement and behavior. For example, discussions framed with strong emotional expressions and certain health beliefs may lead to higher engagement and observable behavior changes, reflecting how these factors interplay in shaping public health discourse.

6.3 Limitation

This thesis acknowledges several limitations. The quality of Reddit data is inconsistent, often containing noise and irrelevant information that can affect analysis accuracy. Findings may not be generalizable to other social media platforms or offline discussions due to Reddit's unique user base and culture. Inferring health beliefs from text-based discussions within the Health

Belief Model framework involves subjective judgments, which may not fully capture the complexity of human beliefs and emotions. Furthermore, human-labeled data used for training models can be inherently biased, reflecting the subjective perspectives of the annotators and potentially influencing the model's predictions.

Challenges in analyzing emotional content arise from language nuances, such as sarcasm and context-specific meanings, which can lead to inaccuracies in sentiment analysis. Moreover, the analysis might not completely account for temporal dynamics, as health beliefs and emotions can evolve over time. Computational techniques used may also face constraints in processing power and scalability, limiting the ability to handle extremely large datasets. Ethical considerations regarding user privacy and consent are crucial, even with publicly available Reddit data.

Transformer-based models, while effective in natural language processing, face challenges such as handling rare or out-of-vocabulary words and domain adaptation. Although generation models excel at understanding general text contexts, they face challenges in specialized domains such as the Health Belief Model. These models perform well with broad, general content but may struggle with the complexities and nuances inherent in domain-specific frameworks where the ideas are more intricate. This discrepancy highlights the need for enhanced domain adaptation techniques and more precise models to effectively capture and analyze specialized health-related concepts.

Addressing these challenges requires incorporating mechanisms for rare word handling and developing techniques for adapting models to specific domains. There are also opportunities for improvement, including developing more efficient and scalable transformer models using methods like model distillation and parameter sharing. Exploring novel architectures and training methodologies tailored to specific tasks and domains can advance the field. Additionally, improving the interpretability and explainability of transformer models can enhance their trustworthiness and real-world applicability. Ethical considerations, including bias, fairness, and misinformation, must be addressed to ensure responsible AI development and deployment, maximizing benefits while mitigating risks.

6.4 Future Works

Based on the findings and limitations of this thesis, several potential avenues for future research and development are suggested. One key direction is the enhancement of domain-specific models tailored to complex frameworks like the Health Belief Model. Future work should focus on developing and fine-tuning models that are specifically designed to handle the nuances and intricacies of specialized health-related concepts. This includes creating domain-adapted versions of transformer models and incorporating domain-specific knowledge and vocabulary to improve model performance and accuracy in these contexts.

Another important area for future research is the improved handling of rare or out-of-vocabulary words. Advanced techniques such as subword tokenization, adaptive embeddings, or hybrid approaches that combine pre-trained language models with domain-specific data could enhance the model's robustness and generalization capabilities. Additionally, integrating temporal dynamics into the analysis could provide a more comprehensive understanding of how health beliefs and emotional content evolve over time. Future studies might explore incorporating temporal features into models to track and analyze changes in beliefs and sentiments more accurately.

Scalability and efficiency improvements are also crucial for managing extremely large datasets. Research into more efficient transformer architectures, such as those employing model distillation, parameter sharing, or pruning, can help reduce computational complexity while maintaining high performance. This will enable the handling of larger and more sophisticated datasets, facilitating more in-depth research.

Expanding research to include multimodal data sources, such as images and videos, could further enrich the analysis of health-related content. Incorporating visual and auditory elements alongside text can provide a more holistic view of how health beliefs and emotional expressions are conveyed and perceived.

Addressing ethical and bias considerations remains a priority. Future research should aim to develop frameworks and guidelines for responsible AI deployment to mitigate risks associated with bias and fairness. Ensuring that models are trained and evaluated with ethical implications in mind, and implementing measures to detect and correct biases, is essential for responsible AI use.

Enhancing the interpretability and explainability of transformer models is another important area for future work. Developing techniques that make model predictions more transparent and understandable can increase their trustworthiness and practical utility. Lastly, expanding research to include other social media platforms and offline data sources can offer a broader perspective and validate findings across different contexts. By comparing results from Reddit with data from other platforms, researchers can assess the generalizability of the models and insights, leading to more accurate and comprehensive applications of AI in understanding and improving public health discussions.

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Disclaimer: *AI tools were used in the preparation of this thesis for purposes such as rephrasing and grammar correction. All content, analysis, and conclusions are the result of the author's original work.*

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